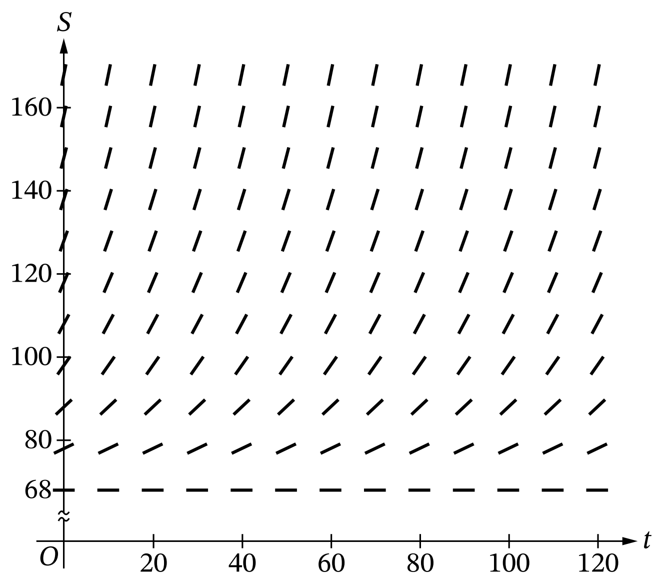
7-1-3

1. For $0 \le t \le 120$, the temperature of a bowl of soup at time t is modeled by the function S that satisfies the differential equation $\frac{dS}{dt} = -\frac{1}{20}(S - 68)$, where $S(t)$ is measured in degrees Fahrenheit ($^{\circ}\mathrm{F}$) and t is measured in minutes. The temperature of the bowl of soup is always greater than $68^{\circ}\mathrm{F}$ and is $148^{\circ}\mathrm{F}$ at time $t = 0$.

Part A

Explain why the following could not be a slope field for the differential equation $\frac{dS}{dt} = -\frac{1}{20}(S - 68)$.



Part B

Use the line tangent to the graph of S at $t = 0$ to approximate the temperature of the soup at time $t = 5$ minutes.

Part C

According to the model, is the approximation found in part B an overestimate or an underestimate of the temperature of the soup at time $t = 5$ minutes? Give a reason for your answer.

Part D

Using separation of variables, find the particular solution $y = S(t)$ to the differential equation $\frac{dS}{dt} = -\frac{1}{20}(S - 68)$ with initial condition $S(0) = 148$.

Part A Point 1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

7-1-3

- **P1** can be earned for a response of "because there are no negative slopes" or "because the slopes are positive."
- **P1** can be earned by using a local argument at any point (t, S) with $0 \leq t \leq 120$ and $S > 68$, rather than a global argument.

Model Solution:

The slopes of the line segments in a slope field for the given differential equation should all be negative because for $S > 68$, $\frac{dS}{dt} < 0$. However, the slopes of the line segments in the slope field given are all nonnegative.

0	1 ✓

The response accurately includes the criteria below.

- Explanation

Part B Point 2

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response can earn **P2** with $\left.\frac{dS}{dt}\right|_{(0,\,148)} = -4$, "slope is -4," or equivalent.
- A response that presents a linear approximation with a slope of -4 also earns **P2**. For example, a response of $148 - 4(5)$ earns **P2**.
- A response that declares the slope equal to any nonzero value $k \neq -4$ does not earn **P2**.

Model Solution:

$$\left.\frac{dS}{dt}\right|_{(0,148)} = -\frac{1}{20}(148 - 68) = -4$$

0	1 ✓

The response accurately includes the criteria below.

- Slope of tangent line

Part B Point 3

Select a point value to view scoring criteria, solutions, and examples or to score the response.

7-1-3

Scoring Notes:

- A response that declares the slope equal to any nonzero value $k \neq -4$:
  - Earns **P3** if there is evidence that the given differential equation was used to calculate k and that k was used correctly in a tangent line approximation of the form $148 + k(5)$. Subsequent errors in simplification will not be considered in scoring for **P3**.
  - Does not earn **P3** if k is simply declared to be a slope without a connection to the differential equation.
- **P3** cannot be earned for an approximation at any value of t other than 5.
- A response does not have to present the tangent line equation but must clearly demonstrate its use at $t = 5$ in finding the requested approximation in order to earn **P3**.
- A response of $148 - 4(5)$ earns **P3**. Subsequent errors in simplification will not be considered in scoring for **P3**.

Model Solution:

$S(5) \approx 148 - 4(5) = 128$

The temperature of the soup at $t = 5$ minutes is approximately $128^\circ\mathrm{F}$.

✓

0	1

The response accurately includes the criteria below.

- Tangent line approximation

Part C Point 4

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P4**, a response must include one of the following: $\frac{d^2S}{dt^2}$, $-\frac{1}{20}\frac{dS}{dt}$, $\frac{1}{400}(S-68)$, “the second derivative,” or equivalent.
  - “S' is increasing (or decreasing)” or a reference to concavity without a connection to the second derivative is not sufficient to earn **P4**.

Model Solution:

Because $S(t) > 68$, $\frac{d^2S}{dt^2} = -\frac{1}{20}\frac{dS}{dt} = \left(-\frac{1}{20}\right)\left(-\frac{1}{20}\right)(S-68) = \frac{1}{400}(S-68) > 0$.

✓

0	1

7-1-3

The response accurately includes the criteria below.

- Considers $\frac{d^2S}{dt^2}$

Part C Point 5

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To be eligible to earn **P5**, a response must have earned **P4**.
- Responses do not need to say “concave up” to earn **P5**. A response of “underestimate because the second derivative is positive” is sufficient to earn **P5**.

Model Solution:

Because the second derivative is always positive, the tangent line approximation is an underestimate of the temperature predicted by the model.

0	1

The response accurately includes the criteria below.

- Answer with reason

Part D Point 6

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response with no separation of variables does not earn **P6**.
- A response earns **P6** if the separation of variables results in an equation of the form $A\frac{dS}{S-68} = B\ dt$, where A and B are nonzero constants.
- Special case: If the response begins with a separation of the form $A\frac{dS}{S+68} = B\ dt$, where A and B are correct coefficients, it earns **P6**.

Model Solution:

$$\frac{dS}{S-68} = -\frac{1}{20}dt$$

7-1-3

0	1

The response accurately includes the criteria below.

- Separate variables

Part D Point 7

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response with no separation of variables does not earn **P7**.
- A response is eligible to earn **P7** if the separation of variables resulted in an equation of the form $A\frac{dS}{S-68} = B\ dt$, where A and B are nonzero constants. A response that presents consistent antiderivatives of the form $A\ln(S-68)$ (with either parentheses or absolute value) and Bt earns **P7**.
- Special case: If the response began with a separation of the form $A\frac{dS}{S+68} = B\ dt$, where A and B are correct coefficients, it earns **P7** with two consistent antiderivatives.
- A response with no constant of integration is eligible to earn **P7**.

Model Solution:

$$\ln|S-68| = -\frac{t}{20} + C$$

0	1

The response accurately includes the criteria below.

- Finds antiderivatives

Part D Point 8

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response with no separation of variables does not earn **P8**.
- A response with no constant of integration does not earn **P8**.
- A response is eligible for **P8** if it has earned P6 and has one of the two antiderivatives correct OR if it has earned **P7**.

7-1-3

- An eligible response earns **P8** by correctly including the constant of integration in an equation and substituting 0 for t and 148 for S.

Model Solution:

$$\ln|148-68| = -\frac{0}{20} + C \Rightarrow \ln 80 = C$$

$$\ln(S-68) = -\frac{t}{20} + \ln 80$$

0	1

The response accurately includes the criteria below.

- Constant of integration and uses initial condition

Part D Point 9

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response is eligible for **P9** only if it has earned **P6**, **P7**, and **P8**.
- A response earns **P9** only for an answer of $S(t) = 68 + 80e^{-t/20}$ or equivalent.

Model Solution:

$$S - 68 = e^{-t/20+\ln 80} = 80e^{-t/20}$$

$$S(t) = 68 + 80e^{-t/20}$$

0	1

The response accurately includes the criteria below.

- Solves for $S(t)$

7-1-3

2. If $P(t)$ is the size of a population at time t, which of the following differential equations describes linear growth in the size of the population?

(A) $\dfrac{dP}{dt} = 200$

(B) $\dfrac{dP}{dt} = 200t$

(C) $\dfrac{dP}{dt} = 100t^2$

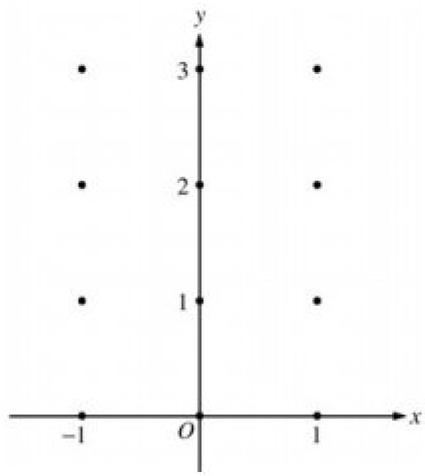
(D) $\dfrac{dP}{dt} = 200P$

(E) $\dfrac{dP}{dt} = 100P^2$

7-1-3

3. Consider the differential equation $\frac{dy}{dx} = x^4(y-2)$.

(a) On the axes provided, sketch a slope field for the given differential equation at the twelve points indicated.



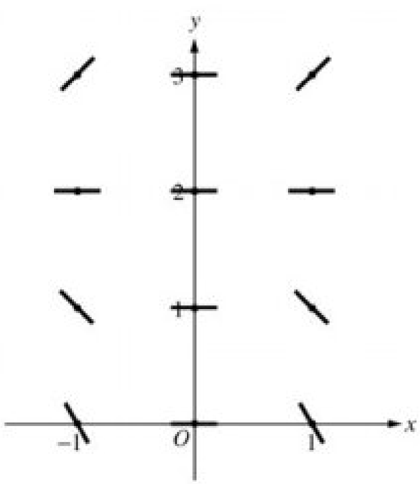
(b) While the slope field in part (a) is drawn at only twelve points, it is defined at every point in the *xy*-plane. Describe all points in the *xy*-plane for which the slopes are negative.

(c) Find the particular solution $y = f(x)$ to the given differential equation with the initial condition $f(0) = 0$.

Part A

1 point is earned for zero slope at each point (x, y) where $x = 0$ or $y = 2$

1 point is earned for positive slope at each point (x, y) where $x \neq 0$ and $y > 2$ **OR** negative slope at each point (x, y) where $x \neq 0$ and $y < 2$



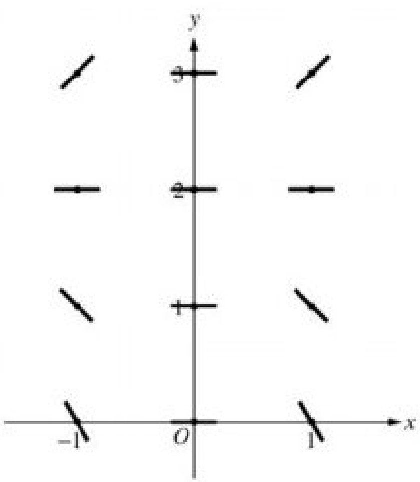
7-1-3

0	1	2

The student response earns all of the following points:

1 point is earned for zero slope at each point (x, y) where $x = 0$ or $y = 2$

1 point is earned for positive slope at each point (x, y) where $x \neq 0$ and $y > 2$ **OR** negative slope at each point (x, y) where $x \neq 0$ and $y < 2$



Part B

1 point is earned for the description

Slopes are negative at points (x, y) where $x \neq 0$ and $y < 2$.

0	1

The student response earns all of the following points:

1 point is earned for the description

Slopes are negative at points (x, y) where $x \neq 0$ and $y < 2$.

Part C

1 point is earned if separates variables

2 points are earned for the antiderivatives

7-1-3

1 point is earned for constant of integration

1 point is earned if uses initial condition

1 point is earned if solves for y, 0/1 if y is not exponential

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables

Solution:

$$\frac{1}{y-2}dy = x^4dx$$
$$\ln\left|y-2\right| = \frac{1}{5}x^5 + C$$
$$\left|y-2\right| = e^Ce^{\frac{1}{5}x^5}$$
$$y-2 = Ke^{\frac{1}{5}x^5},\ K = \pm e^C$$
$$-2 = Ke^0 = K$$
$$y = 2 - 2e^{\frac{1}{5}x^5}$$

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned if separates variables

2 points are earned for the antiderivatives

1 point is earned for constant of integration

1 point is earned if uses initial condition

1 point is earned is solves for y, 0/1 if y is not exponential

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables

Solution:

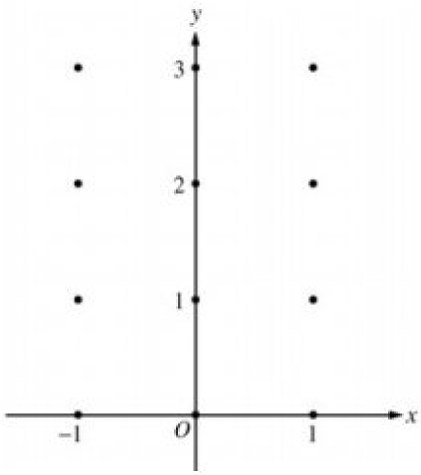
7-1-3

$$\frac{1}{y-2}dy = x^4dx$$
$$\ln\left|y-2\right| = \frac{1}{5}x^5 + C$$
$$\left|y-2\right| = e^Ce^{\frac{1}{5}x^5}$$
$$y-2 = Ke^{\frac{1}{5}x^5},\ K = \pm e^C$$
$$-2 = Ke^0 = K$$
$$y = 2 - 2e^{\frac{1}{5}x^5}$$

7-1-3

4. Consider the differential equation $\frac{dy}{dx} = x^4 (y - 2)$.

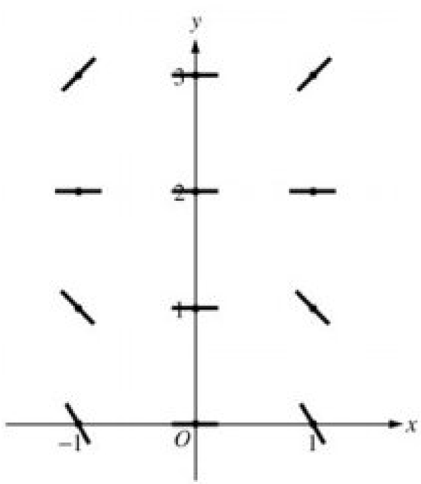
On the axes provided, sketch a slope field for the given differential equation at the twelve points indicated.



Part A

1 point is earned for zero slope at each point (x , y) where x =0 or y =2

1 point is earned for positive slope at each point (x,y) where x $\neq$ 0 and y >2 **OR** negative slope at each point (x,y) where x $\neq$ 0 and y<2



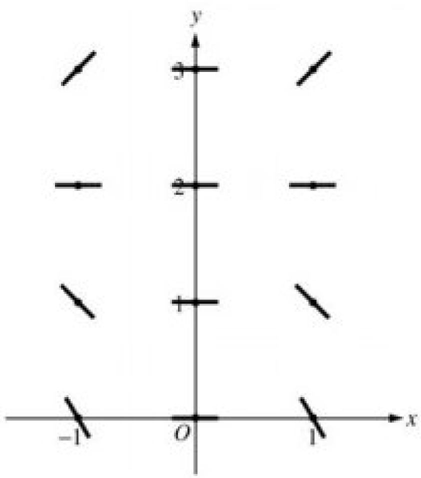
0	1	2 ✓

The student response earns none of the following points:

1 point is earned for zero slope at each point (x , y) where x =0 or y =2

7-1-3

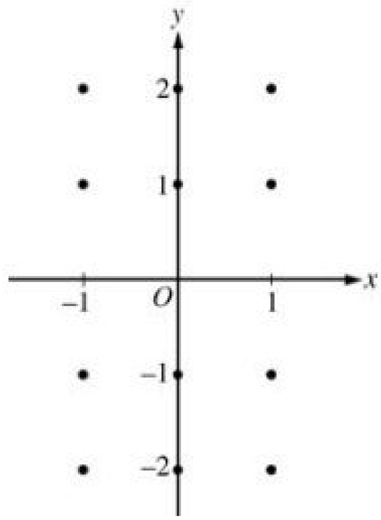
1 point is earned for positive slope at each point (x,y) where x $\neq$ 0 and y >2 **OR** negative slope at each point (x,y) where x $\neq$ 0 and y<2



7-1-3

5. Consider the differential equation $\frac{dy}{dx} = -\frac{2x}{y}$.

(a) On the axes provided, sketch a slope field for the given differential equation at the twelve points indicated.



(b) Let $y = f(x)$ be the particular solution to the differential equation with the initial condition $f(1) = -1$. Write an equation for the line tangent to the graph of f at $(1, -1)$ and use it to approximate $f(1.1)$.

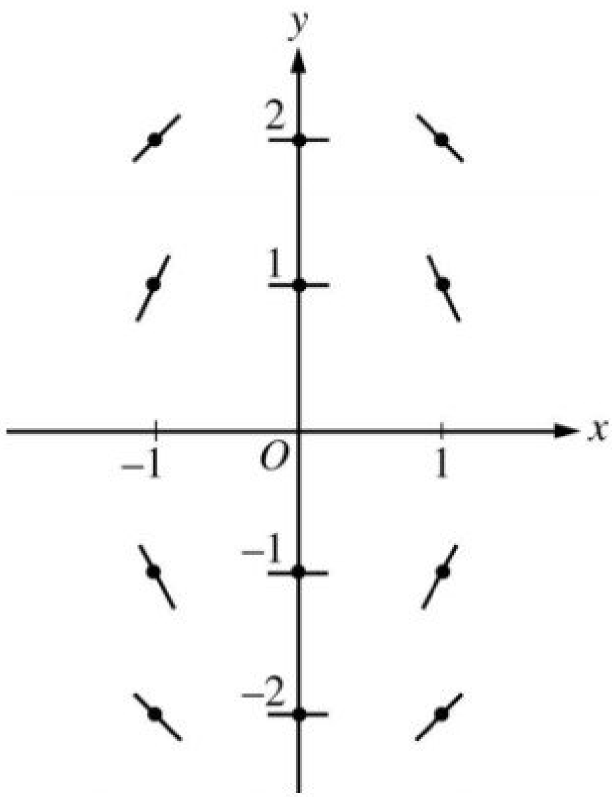
(c) Find the particular solution $y = f(x)$ to the given differential equation with the initial condition $f(1) = -1$.

Part A

1 point is earned for the zero slopes

1 point is earned for the nonzero slopes

7-1-3



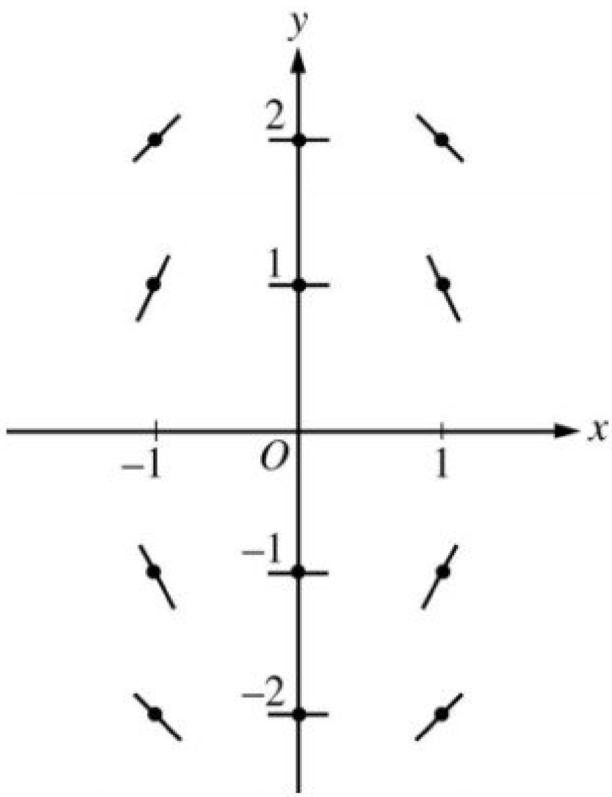
0	1	2

The student response earns all of the following points:

1 point is earned for the zero slopes

1 point is earned for the nonzero slopes

7-1-3



Part B

1 point is earned for the equation of the tangent line

1 point is earned for the approximation for $f(1.1)$

The line tangent to f at (1, -1) is $y + 1 = 2(x - 1)$. Thus, $f(1.1)$ is approximately -0.8.

0	1	2

The student response earns all of the following points:

1 point is earned for the equation of the tangent line

1 point is earned for the approximation for $f(1.1)$

The line tangent to f at (1, -1) is $y + 1 = 2(x - 1)$. Thus, $f(1.1)$ is approximately -0.8.

Part C

1 point is earned for the separates variables

7-1-3

1 point is earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for uses initial condition

1 point is earned for solves for y

$$\frac{dy}{dx} = -\frac{2x}{y}$$
$$ydy = -2xdx$$
$$\frac{y^2}{2} = -x^2 + C$$
$$\frac{1}{2} = -1 + C;\ C = \frac{3}{2}$$
$$y^2 = -2x^2 + 3$$

Since the particular solution goes through (1, -1), y must be negative. Thus the particular solution is $y = -\sqrt{3 - 2x^2}$.

0	1	2	3	4	5

The student response earns all of the following points:

1 point is earned for the separates variables

1 point is earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for uses initial condition

1 point is earned for solves for y

$$\frac{dy}{dx} = -\frac{2x}{y}$$
$$ydy = -2xdx$$
$$\frac{y^2}{2} = -x^2 + C$$
$$\frac{1}{2} = -1 + C;\ C = \frac{3}{2}$$
$$y^2 = -2x^2 + 3$$

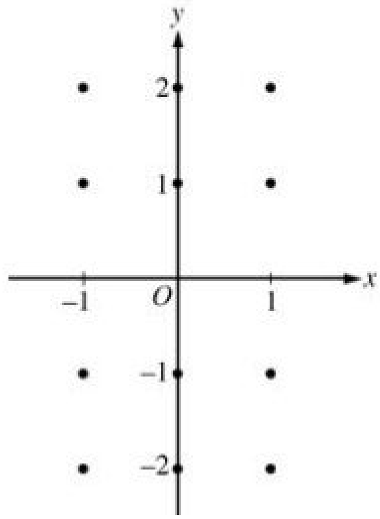
Since the particular solution goes through (1, -1), y must be negative. Thus the particular solution is $y = -\sqrt{3 - 2x^2}$.

7-1-3

7-1-3

6. Consider the differential equation $\dfrac{dy}{dx} = -\dfrac{2x}{y}$.

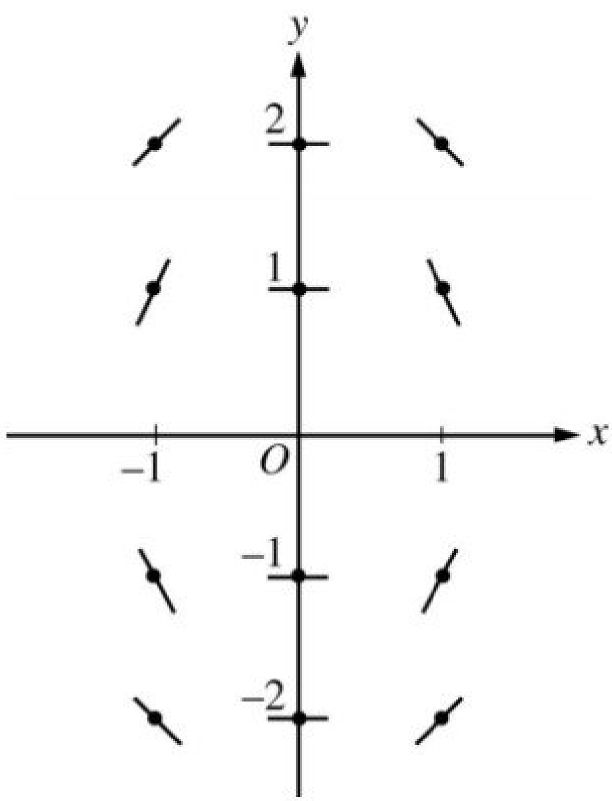
On the axes provided, sketch a slope field for the given differential equation at the twelve points indicated.



Part A

1 point is earned for the zero slopes

1 point is earned for the nonzero slopes



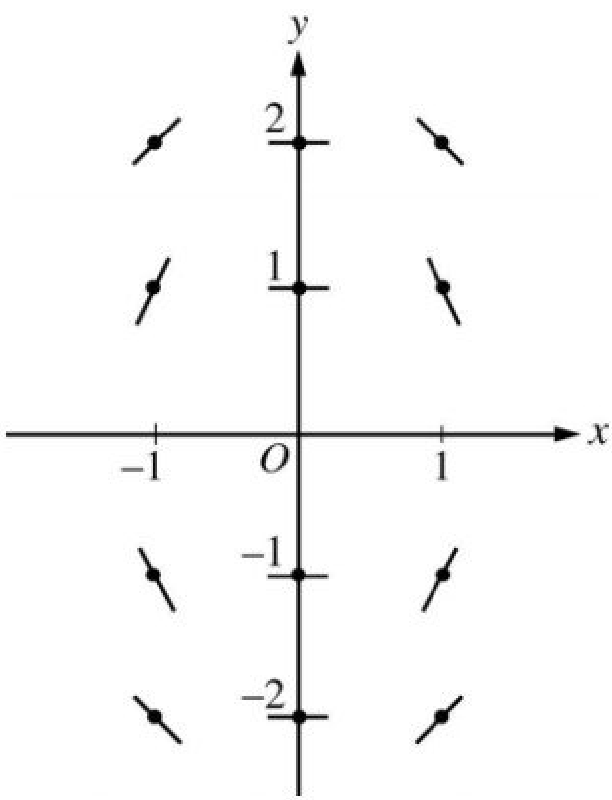
7-1-3

0	1	2

The student response earns all of the following points:

1 point is earned for the zero slopes

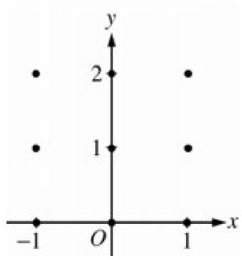
1 point is earned for the nonzero slopes



7-1-3

7. Consider the differential equation $\frac{dy}{dx} = \frac{1}{2}x + y - 1$.

(a) On the axes provided, sketch a slope field for the given differential equation at the nine points indicated.



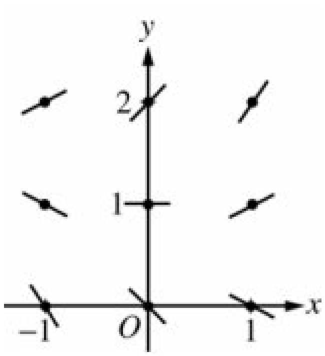
(b) Find $\frac{d^2y}{dx^2}$ in terms of x and y. Describe the region in the xy-plane in which all solution curves to the differential equation are concave up.

(c) Let $y = f(x)$ be a particular solution to the differential equation with the initial condition $f(0) = 1$. Does f have a relative minimum, a relative maximum, or neither at $x = 0$? Justify your answer.

(d) Find the values of the constants m and b, for which $y = mx + b$ is a solution to the differential equation.

Part A

2 points are earned for sign of each slope at each point and relative steepness of slope lines in rows and columns

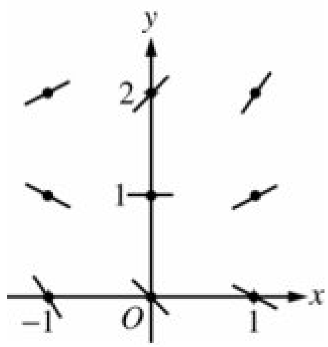


0	1	2

The student response earns all of the following points:

2 points are earned for sign of each slope at each point and relative steepness of slope lines in rows and columns

7-1-3



Part B

2 points are earned for $\frac{d^2y}{dx^2}$

1 point is earned for the description

$$\frac{d^2y}{dx^2} = \frac{1}{2} + \frac{dy}{dx} = \frac{1}{2}x + y - \frac{1}{2}$$

Solution curves will be concave up on the half-plane above the line

$y = -\frac{1}{2}x + \frac{1}{2}$.

0	1	2	3

The student response earns all of the following points:

2 points are earned for $\frac{d^2y}{dx^2}$

1 point is earned for the description

$$\frac{d^2y}{dx^2} = \frac{1}{2} + \frac{dy}{dx} = \frac{1}{2}x + y - \frac{1}{2}$$

Solution curves will be concave up on the half-plane above the line

$y = -\frac{1}{2}x + \frac{1}{2}$.

Part C

1 point is earned for the answer

1 point is earned for the justification

7-1-3

$$\frac{dy}{dx}\Big|_{(0,1)} = 0 + 1 - 1 = 0 \text{ and } \frac{d^2y}{dx^2}\Big|_{(0,1)} = 0 + 1 - \tfrac{1}{2} > 0$$

Thus, f has a relative minimum at (0,1).

0	1	2

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

$$\frac{dy}{dx}\Big|_{(0,1)} = 0 + 1 - 1 = 0 \text{ and } \frac{d^2y}{dx^2}\Big|_{(0,1)} = 0 + 1 - \tfrac{1}{2} > 0$$

Thus, f has a relative minimum at (0,1).

Part D

1 point is earned for the value of m

1 point is earned for the value of b

Substituting $y = mx + b$ into the differential equation:

$m = \frac{1}{2}x + (mx + b) - 1 = (m + \frac{1}{2})x + (b - 1)$

Then $0 = m + \frac{1}{2}$ and $m = b - 1$: $m = -\frac{1}{2}$ and $b = \frac{1}{2}$.

0	1	2

The student response earns all of the following points:

1 point is earned for the value of m

1 point is earned for the value of b

Substituting $y = mx + b$ into the differential equation:

7-1-3

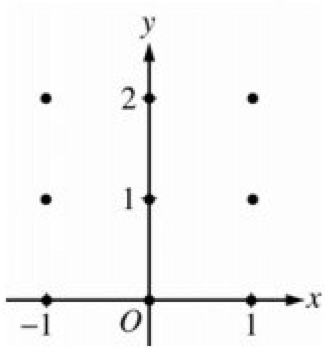
$m = \frac{1}{2}x + (mx + b) - 1 = (m + \frac{1}{2})x + (b - 1)$

Then $0 = m + \frac{1}{2}$ and $m = b - 1$: $m = -\frac{1}{2}$ and $b = \frac{1}{2}$.

7-1-3

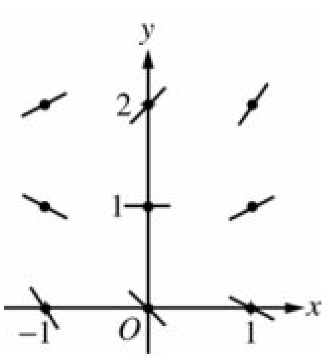
Consider the differential equation $\frac{dy}{dx} = \frac{1}{2}x + y - 1$

8. On the axes provided, sketch a slope field for the given differential equation at the nine points indicated. (Note: Use the axes provided in the exam booklet.)



Part A

2 points are earned for sign of each slope at each point and relative steepness of slope lines in rows and columns

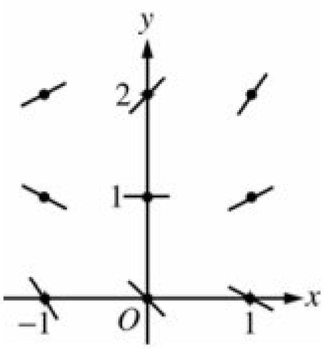


0	1	2

The student response earns all of the following points:

2 points are earned for sign of each slope at each point and relative steepness of slope lines in rows and columns

7-1-3



9. Find the values of the constants m and b, for which $y=mx+b$ is a solution to the differential equation.

Part D

1 point is earned for the value of m

1 point is earned for the value of b

Substituting $y=mx+b$ into the differential equation:

$$m = \frac{1}{2}x + (mx + b) - 1 = \left(m + \frac{1}{2}\right)x + (b - 1)$$

Then $0 = m + \frac{1}{2}$ and $m = b - 1 : m = -\frac{1}{2}$ and $b = \frac{1}{2}$

0	1	2

The student response earns all of the following points:

1 point is earned for the value of m

1 point is earned for the value of b

Substituting $y=mx+b$ into the differential equation:

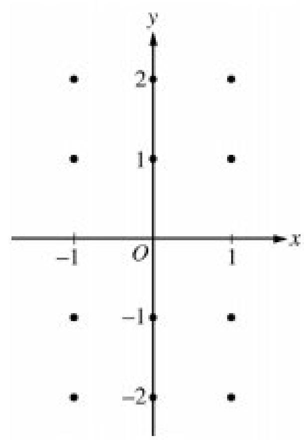
$$m = \frac{1}{2}x + (mx + b) - 1 = \left(m + \frac{1}{2}\right)x + (b - 1)$$

Then $0 = m + \frac{1}{2}$ and $m = b - 1 : m = -\frac{1}{2}$ and $b = \frac{1}{2}$

7-1-3

10. Consider the differential equation $\frac{dy}{dx} = \frac{x+1}{y}$.

(a) On the axes provided, sketch a slope field for the given differential equation at the twelve points indicated, and for $-1 < x < 1$, sketch the solution curve that passes through the point $(0, -1)$.



(b) While the slope field in part (a) is drawn at only twelve points, it is defined at every point in the xy-plane for which $y \neq 0$. Describe all points in the xy-plane, $y \neq 0$, for which $dy/dx = -1$.

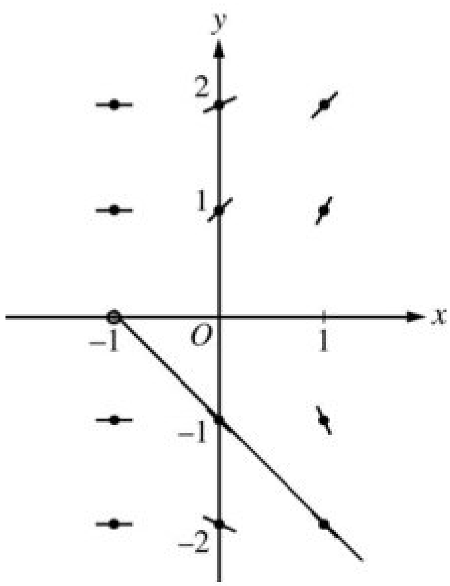
(c) Find the particular solution $y = f(x)$ to the given differential equation with the initial condition $f(0) = -2$.

Part A

1 point is earned for zero slopes

1 point is earned for nonzero slopes

1 point is earned for solution curve through (0, -1)



7-1-3

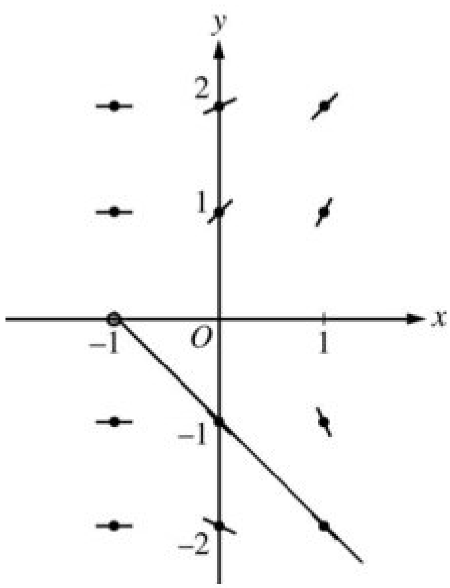
0	1	2	3

The student response earns all of the following points:

1 point is earned for zero slopes

1 point is earned for nonzero slopes

1 point is earned for solution curve through (0, -1)



Part B

1 point is earned description

$$-1 = \frac{x+1}{y} \Rightarrow y = -x - 1$$

$$\frac{dy}{dx} = -1 \text{ for all } (x, y) \text{ with } y = -x - 1 \text{ and } y \neq 0$$

0	1

The student response earns all of the following points:

1 point is earned description

7-1-3

$-1 = \frac{x+1}{y} \Rightarrow y = -x - 1$

$\frac{dy}{dx} = -1$ for all (x, y) with $y = -x - 1$ and $y \neq 0$

Part C

1 point is earned if separates variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

1 point is earned if uses initial condition

1 point is earned if solves for y

Note: max 2/5 [1-1-0-0-0] if no constant of integration

Note: 0/5 if no separation of variables

$$\int y\, dy = \int (x+1) dx$$
$$\frac{y^2}{2} = \frac{x^2}{2} + x + c$$
$$\frac{(-2)^2}{2} = \frac{0^2}{2} + 0 + c \Rightarrow c = 2$$
$$y^2 = x^2 + 2x + 4$$

Since the solution goes through (0, -2), y must be negative. Therefore $y = -\sqrt{x^2 + 2x + 4}$.

0	1	2	3	4	5 ✓

The student response earns all of the following points:

1 point is earned if separates variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

1 point is earned if uses initial condition

7-1-3

1 point is earned if solves for y

Note: max 2/5 [1-1-0-0-0] if no constant of integration

Note: 0/5 if no separation of variables

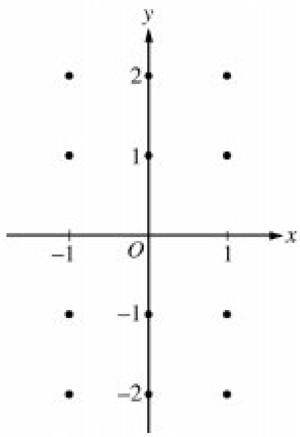
$$\int y\,dy = \int (x+1)dx$$
$$\frac{y^2}{2} = \frac{x^2}{2} + x + c$$
$$\frac{(-2)^2}{2} = \frac{0^2}{2} + 0 + c \Rightarrow c = 2$$
$$y^2 = x^2 + 2x + 4$$

Since the solution goes through (0, -2), y must be negative. Therefore $y = -\sqrt{x^2 + 2x + 4}$.

7-1-3

11. Consider the differential equation dy/dx = x+1/y.

On the axes provided, sketch a slope field for the given differential equation at the twelve points indicated, and for -1<*x*<1, sketch the solution curve that passes through the point (0, -1) .

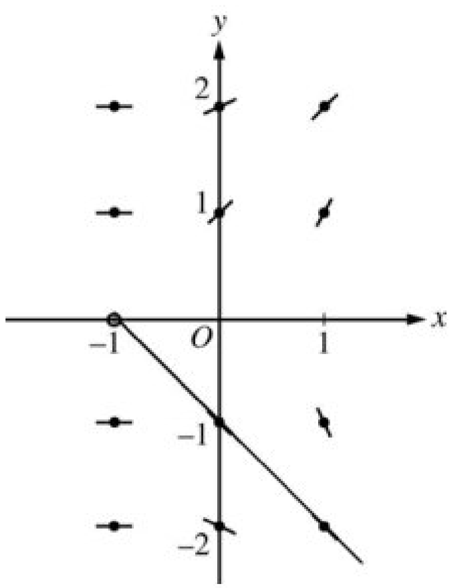


Part A

1 point is earned for zero slopes

1 point is earned for nonzero slopes

1 point is earned for solution curve through (0,-1)



0	1	2	3

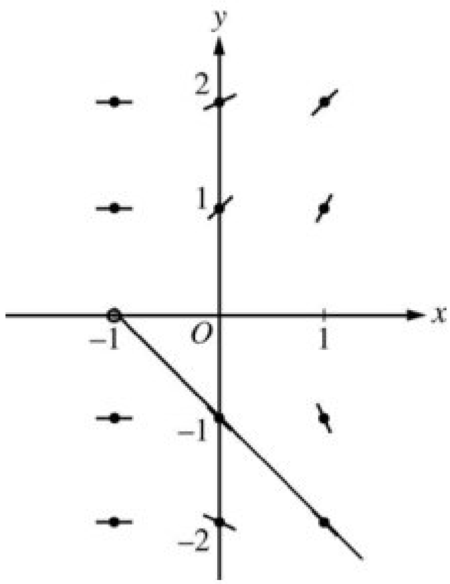
The student response earns all of the following points:

7-1-3

1 point is earned for zero slopes

1 point is earned for nonzero slopes

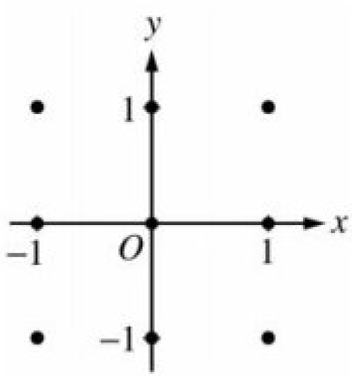
1 point is earned for solution curve through (0,-1)



7-1-3

Consider the differential equation $dy/dx = (y-1)^2 \cos(\pi x)$.

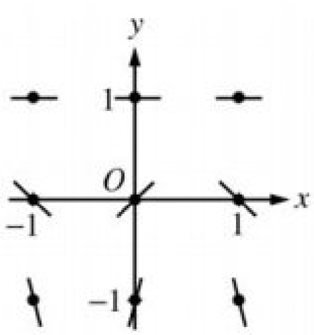
12. On the axes provided, sketch a slope field for the given differential equation at the nine points indicated. (Note: Use the axes provided in the exam booklet.)



Part A

1 point is earned for the zero slopes

1 point is earned for all other slopes



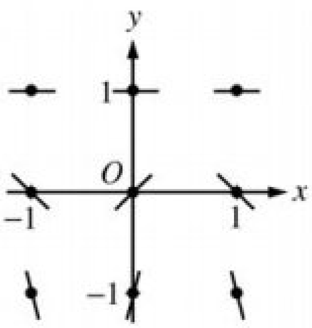
0	1	2

The student response earns all of the following points:

1 point is earned for the zero slopes

1 point is earned for all other slopes

7-1-3



13. There is a horizontal line with equation $y=c$ that satisfies this differential equation. Find the value of c.

Part B

1 point is earned for $c=1$

The line $y=1$ satisfies the differential equation, so $c=1$.

0	1

The student response earns one of the following points:

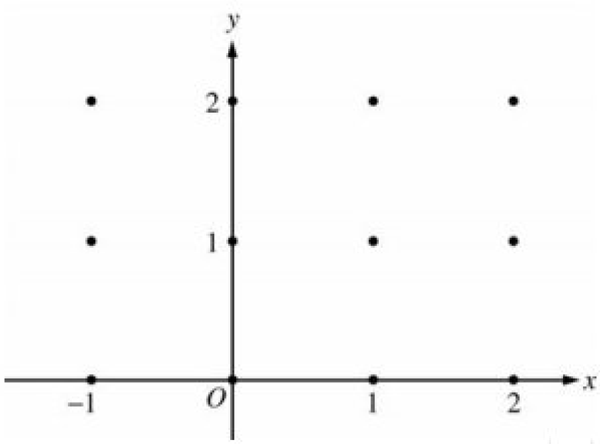
1 point is earned for $c=1$

The line $y=1$ satisfies the differential equation, so $c=1$.

7-1-3

14. Consider the differential equation $dy/dx = -xy^2/2$. Let $y = f(x)$ be the particular solution to this differential equation with the initial condition $f(-1) = 2$.

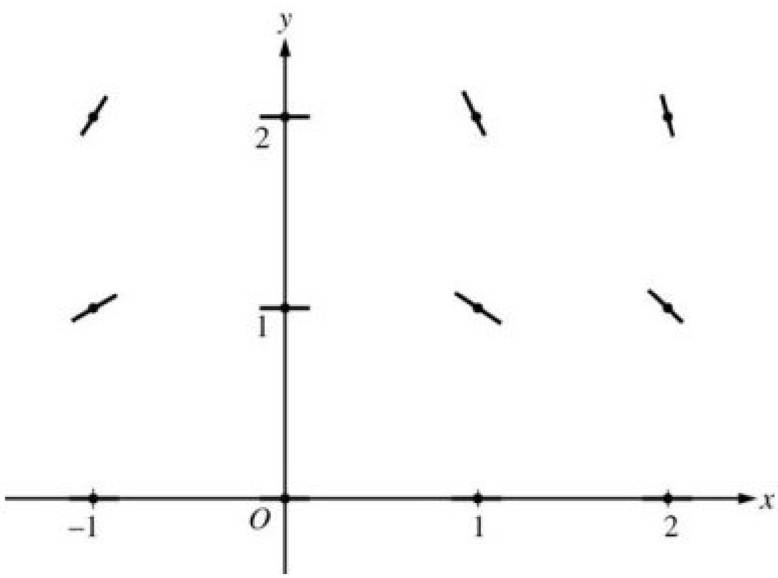
On the axes provided, sketch a slope field for the given differential equation at the twelve points indicated.



Part A

1 point is earned for zero slopes

1 point is earned for nonzero slopes



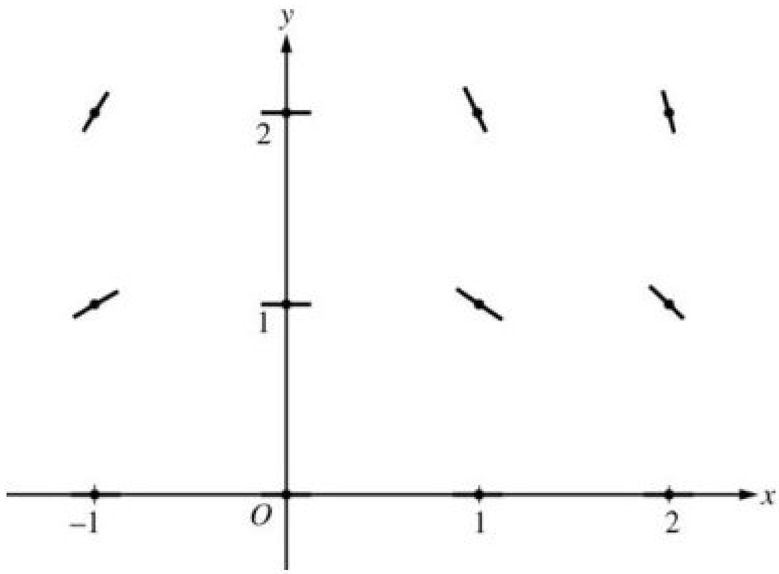
0	1	2

The student response earns all of the following points:

1 point is earned for zero slopes

7-1-3

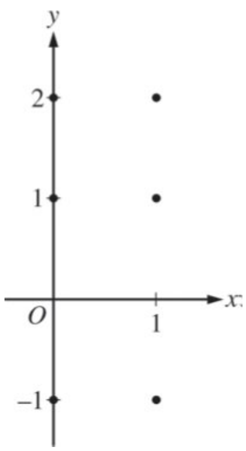
1 point is earned for nonzero slopes



7-1-3

15. Consider the differential equation $\frac{dy}{dx} = 2x - y$

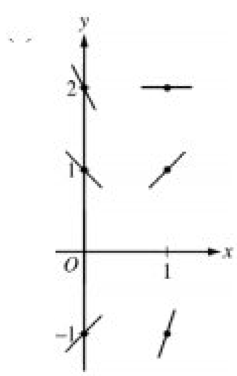
On the axes provided, sketch a slope field for the given differential equation at the six points indicated.



Part A

1 point is earned for slopes where x = 0

1 point is earned for slopes where x = 1



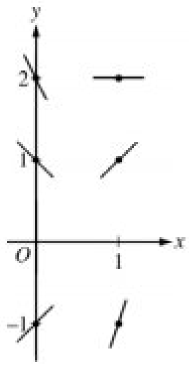
0	1	2

The student response earns all of the following points:

1 point is earned for slopes where x = 0

1 point is earned for slopes where x = 1

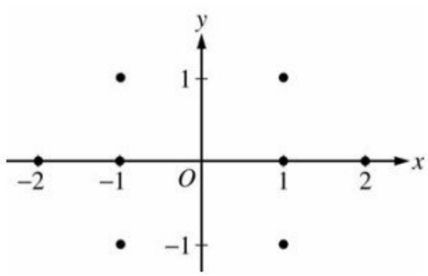
7-1-3



7-1-3

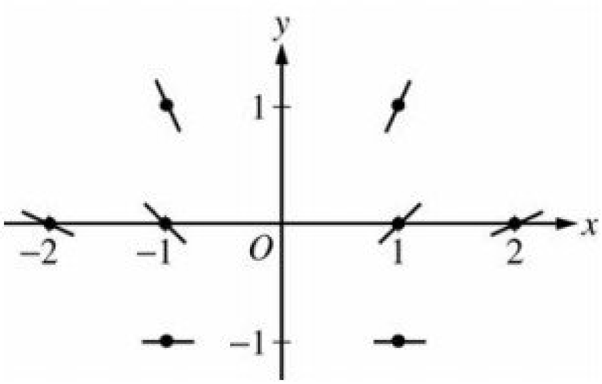
16. Consider the differential equation $\frac{dy}{dx} = \frac{1+y}{x}$, where $x \neq 0$.

On the axes provided, sketch a slope field for the given differential equation at the eight points indicated.



Part A

2 points are earned for the sign of slope at each point and relative steepness of slope lines in rows and columns

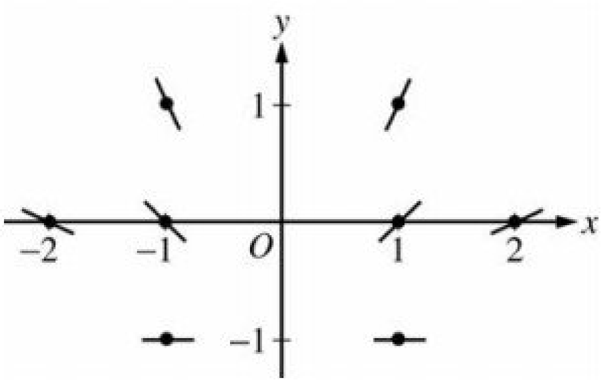


0	1	2

The student response earns all of the following points:

2 points are earned for the sign of slope at each point and relative steepness of slope lines in rows and columns

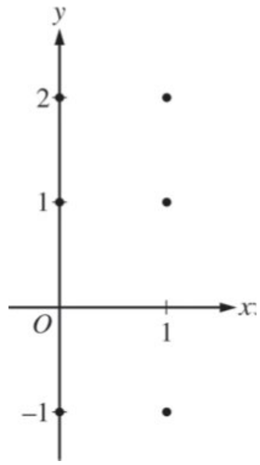
7-1-3



7-1-3

17. Consider the differential equation $\frac{dy}{dx} = 2x - y$.

(a) On the axes provided, sketch a slope field for the given differential equation at the six points indicated.



(b) Find $\frac{d^2y}{dx^2}$ in terms of x and y. Determine the concavity of all solution curves for the given differential equation in Quadrant II. Give a reason for your answer.

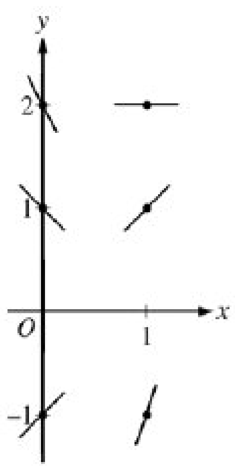
(c) Let $y = f(x)$ be the particular solution to the differential equation with the initial condition $f(2) = 3$. Does f have a relative minimum, a relative maximum, or neither at $x = 2$? Justify your answer.

(d) Find the values of the constants m and b for which $y = mx + b$ is a solution to the differential equation.

Part A

1 point is earned for slopes where $x = 0$

1 point is earned for slopes where $x = 1$



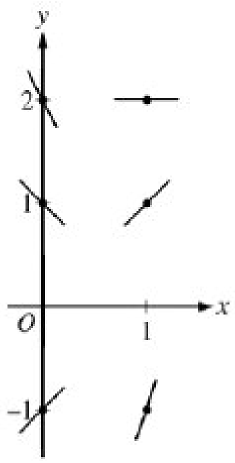
0	1	2

The student response earns all of the following points:

7-1-3

1 point is earned for slopes where $x = 0$

1 point is earned for slopes where $x = 1$



Part B

1 point is earned for $\frac{d^2y}{dx^2}$

1 point is earned for concave up with reason

$$\frac{d^2y}{dx^2} = 2 - \frac{dy}{dx} = 2 - (2x - y) = 2 - 2x + y$$

In Quadrant II, $x < 0$ and $y > 0$, so $2 - 2x + y > 0$. Therefore, all solution curves are concave up in Quadrant II.

0	1	2

The student response earns all of the following points:

1 point is earned for $\frac{d^2y}{dx^2}$

1 point is earned for concave up with reason

$$\frac{d^2y}{dx^2} = 2 - \frac{dy}{dx} = 2 - (2x - y) = 2 - 2x + y$$

In Quadrant II, $x < 0$ and $y > 0$, so $2 - 2x + y > 0$. Therefore, all solution curves are concave up in Quadrant II.

Part C

7-1-3

1 point is earned for considering $\left.\frac{dy}{dx}\right|_{(x,y)=(2,3)}$

1 point is earned for the conclusion with justification

$$\left.\frac{dy}{dx}\right|_{(x,y)=(2,3)} = 2(2) - 3 = 1 \neq 0$$

Therefore, f has neither a relative minimum nor a relative maximum at $x = 2$.

0	1	2

The student response earns all of the following points:

1 point is earned for considering $\left.\frac{dy}{dx}\right|_{(x,y)=(2,3)}$

1 point is earned for the conclusion with justification

$$\left.\frac{dy}{dx}\right|_{(x,y)=(2,3)} = 2(2) - 3 = 1 \neq 0$$

Therefore, f has neither a relative minimum nor a relative maximum at $x = 2$.

Part D

1 point is earned for $\frac{d}{dx}(mx+b) = m$

1 point is earned for $2x - y = m$

1 point is earned for answer

$$y = mx + b \Rightarrow \frac{dy}{dx} = \frac{d}{dx}(mx+b) = m$$
$$2x - y = m$$
$$2x - (mx + b) = m$$
$$(2-m)x - (m+b) = 0$$
$$2 - m = 0 \Rightarrow m = 2$$
$$b = -m \Rightarrow b = -2$$

Therefore, $m = 2$, and $b = -2$

7-1-3

0	1	2	3

The student response earns all of the following points:

1 point is earned for $\frac{d}{dx}(mx+b)=m$

1 point is earned for $2x - y = m$

1 point is earned for answer

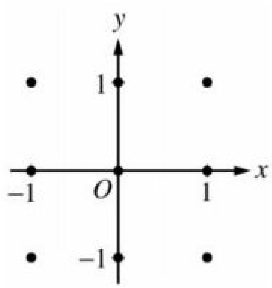
$y=mx+b \Rightarrow \frac{dy}{dx}=\frac{d}{dx}(mx+b)=m$
$2x-y=m$
$2x-(mx+b)=m$
$(2-m)x-(m+b)=0$
$2-m=0 \Rightarrow m=2$
$b=-m \Rightarrow b=-2$

Therefore, $m = 2$, and $b = -2$

7-1-3

18. Consider the differential equation $dy/dx = (y - 1)^2\cos(\pi x)$.

(a) On the axes provided, sketch a slope field for the given differential equation at the nine points indicated.



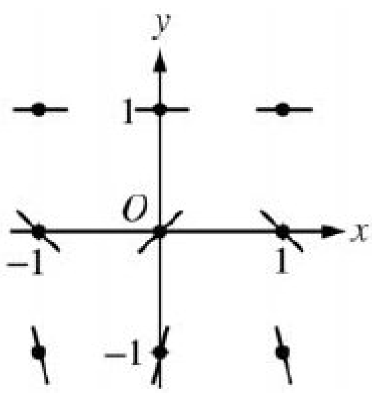
(b) There is a horizontal line with equation $y = c$ that satisfies this differential equation. Find the value of c.

(c) Find the particular solution $y = f(x)$ to the differential equation with the initial condition $f(1) = 0$.

Part A

1 point is earned for the zero slopes

1 point is earned for all other slopes



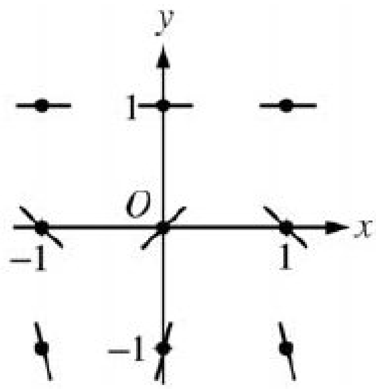
0	1	2

The student response earns all of the following points:

1 point is earned for the zero slopes

1 point is earned for all other slopes

7-1-3



Part B

1 point is earned for $c = 1$

The line $y = 1$ satisfies the differential equation, so $c = 1$.

0	1

The student response earns all of the following points:

1 point is earned for $c = 1$

The line $y = 1$ satisfies the differential equation, so $c = 1$.

Part C

1 point is earned for separating the variables.

2 points are earned for antiderivatives

1 point is earned for constant of integration

1 point is earned if uses initial condition

1 point is earned for answer

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note:0/6 if no separation of variables

7-1-3

$$\frac{1}{(y-1)^2}dy = \cos(\pi x)dx$$
$$-(y-1)^{-1} = \frac{1}{\pi}\sin(\pi x) + C$$
$$\frac{1}{1-y} = \frac{1}{\pi}\sin(\pi x) + C$$
$$1 = \frac{1}{\pi}\sin(\pi) + C = C$$
$$\frac{1}{1-y} = \frac{1}{\pi}\sin(\pi x) + 1$$
$$\frac{\pi}{1-y} = \sin(\pi x) + \pi$$
$$y = 1 - \frac{\pi}{\sin(\pi x)+\pi} \text{ for } -\infty < x < \infty$$

0	1	2	3	4	5	6 ✓

The student response earns all of the following points:

1 point is earned for separating the variables.

2 points are earned for antiderivatives

1 point is earned for constant of integration

1 point is earned if uses initial condition

1 point is earned for answer

Note: max 3/6 [1-2-0-0-0] if no constant of integration

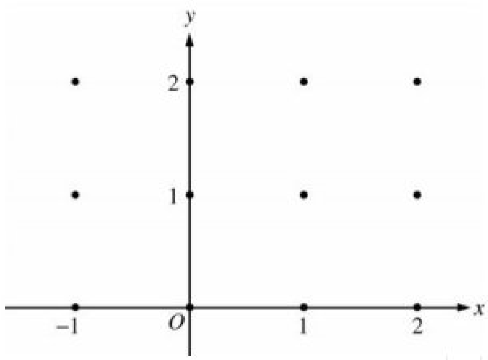
Note:0/6 if no separation of variables

$$\frac{1}{(y-1)^2}dy = \cos(\pi x)dx$$
$$-(y-1)^{-1} = \frac{1}{\pi}\sin(\pi x) + C$$
$$\frac{1}{1-y} = \frac{1}{\pi}\sin(\pi x) + C$$
$$1 = \frac{1}{\pi}\sin(\pi) + C = C$$
$$\frac{1}{1-y} = \frac{1}{\pi}\sin(\pi x) + 1$$
$$\frac{\pi}{1-y} = \sin(\pi x) + \pi$$
$$y = 1 - \frac{\pi}{\sin(\pi x)+\pi} \text{ for } -\infty < x < \infty$$

7-1-3

19. Consider the differential equation $dy/dx = -xy^2/2$. Let $y = f(x)$ be the particular solution to this differential equation with the initial condition $f(-1) = 2$.

(a) On the axes provided, sketch a slope field for the given differential equation at the twelve points indicated.



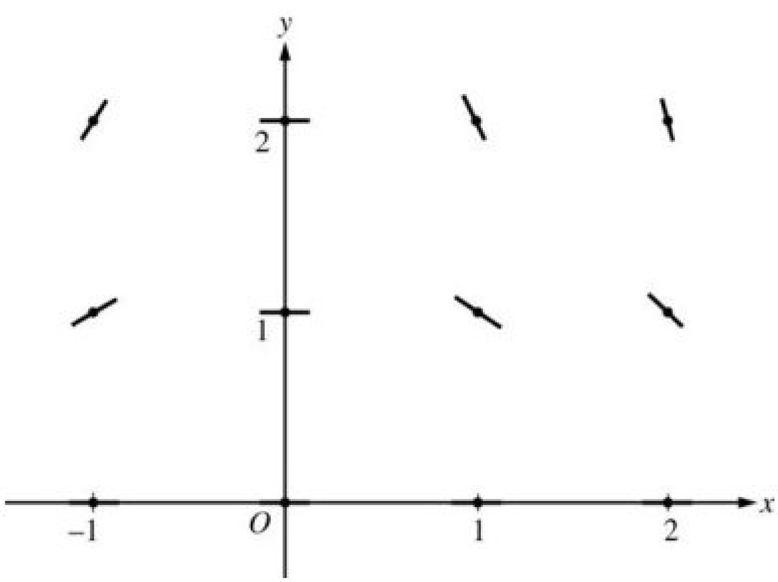
(b) Write an equation for the line tangent to the graph of f at $x = -1$.

(c) Find the solution $y = f(x)$ to the given differential equation with the initial condition $f(-1) = 2$.

Part A

1 point is earned for zero slopes

1 point is earned for nonzero slopes



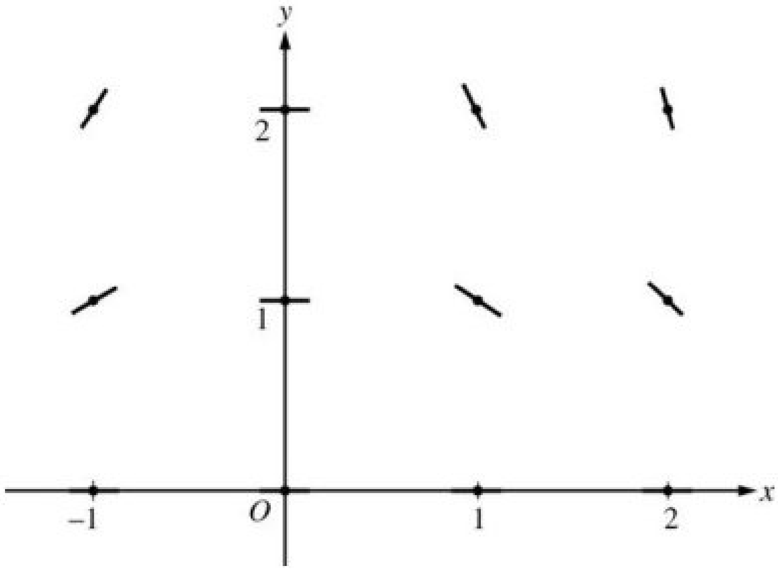
0	1	2

The student response earns all of the following points:

7-1-3

1 point is earned for zero slopes

1 point is earned for nonzero slopes



Part B

1 point is earned for the equation

$$\text{Slope} = \frac{-(-1)4}{2} = 2$$
$$y - 2 = 2(x + 1)$$

0	1

The student response earns all of the following points:

1 point is earned for the equation

$$\text{Slope} = \frac{-(-1)4}{2} = 2$$
$$y - 2 = 2(x + 1)$$

Part C

1 point is earned if separates variables

2 points are earned for antiderivatives

1 point is earned for constant of integration

7-1-3

1 point is earned if uses initial condition

1 point is earned if solves for y

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: max 0/6 if no separation of variables

$$\frac{1}{y^2}dy = -\frac{x}{2}dx$$
$$-\frac{1}{y} = -\frac{x^2}{4} + C$$
$$-\frac{1}{2} = -\frac{1}{4} + C;\ C = -\frac{1}{4}$$
$$y = \frac{1}{\frac{x^2}{4}+\frac{1}{4}} = \frac{4}{x^2+1}$$

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned if separates variables

2 points are earned for antiderivatives

1 point is earned for constant of integration

1 point is earned if uses initial condition

1 point is earned if solves for y

Note: max 3/6 [1-2-0-0-0] if no constant of integration

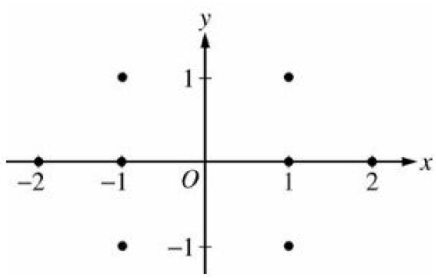
Note: max 0/6 if no separation of variables

$$\frac{1}{y^2}dy = -\frac{x}{2}dx$$
$$-\frac{1}{y} = -\frac{x^2}{4} + C$$
$$-\frac{1}{2} = -\frac{1}{4} + C;\ C = -\frac{1}{4}$$
$$y = \frac{1}{\frac{x^2}{4}+\frac{1}{4}} = \frac{4}{x^2+1}$$

7-1-3

20. Consider the differential equation $\frac{dy}{dx} = \frac{1+y}{x}$, where $x \neq 0$.

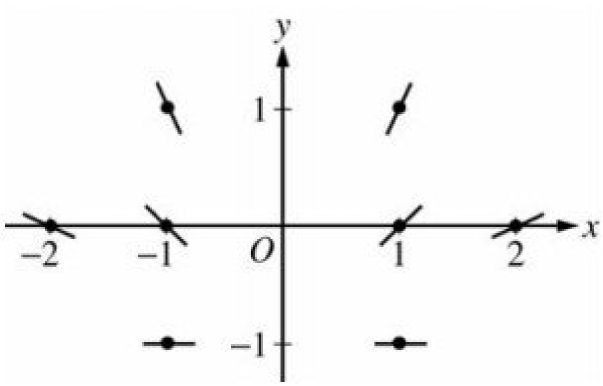
(a) On the axes provided, sketch a slope field for the given differential equation at the eight points indicated.



(b) Find the particular solution $y = f(x)$ to the differential equation with the initial condition $f(-1) = 1$ and state its domain.

Part A

2 points are earned for the sign of slope at each point and relative steepness of slope lines in rows and columns

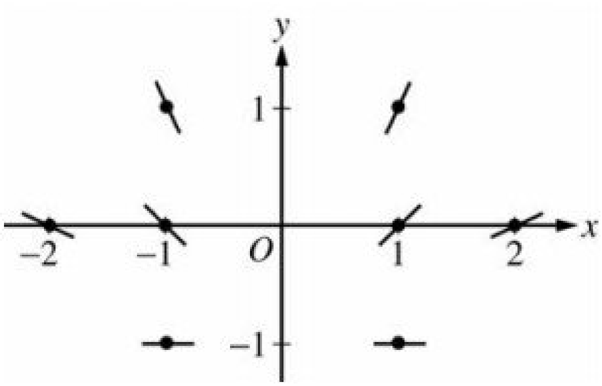


0	1	2

The student response earns all of the following points:

2 points are earned for the sign of slope at each point and relative steepness of slope lines in rows and columns

7-1-3



Part B

1 point is earned for the separates variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned if uses initial condition

1 point is earned if solves for y

1 point is earned for the domain

$\frac{1}{1+y}dy = \frac{1}{x}dx$
$\ln|1+y| = \ln|x| + K$
$|1+y| = e^{\ln|x|+K}$
$1+y = C|x|$
$2 = C$
$1+y = 2|x|$
$y = 2|x| - 1$ and $x < 0$
or
$y = -2x - 1$ and $x < 0$

0	1	2	3	4	5	6	7

The student response earns all of the following points:

7-1-3

1 point is earned for the separates variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned if uses initial condition

1 point is earned if solves for y

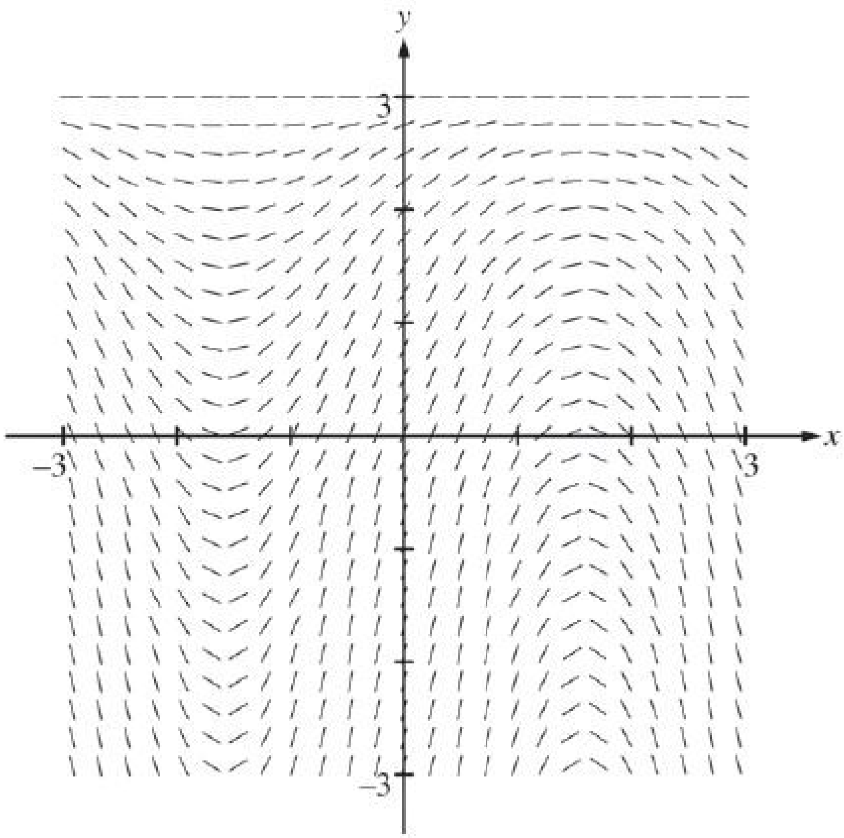
1 point is earned for the domain

$\frac{1}{1+y}dy = \frac{1}{x}dx$
$\ln|1+y| = \ln|x| + K$
$|1+y| = e^{\ln|x|+K}$
$1+y = C|x|$
$2 = C$
$1+y = 2|x|$
$y = 2|x| - 1$ and $x < 0$
or
$y = -2x - 1$ and $x < 0$

7-1-3

21. Consider the differential equation $\frac{dy}{dx} = (3 - y)\cos x$. Let $y = f(x)$ be the particular solution to the differential equation with the initial condition $f(0) = 1$. The function f is defined for all real numbers.

(a) A portion of the slope field of the differential equation is given below. Sketch the solution curve through the point (0,1).



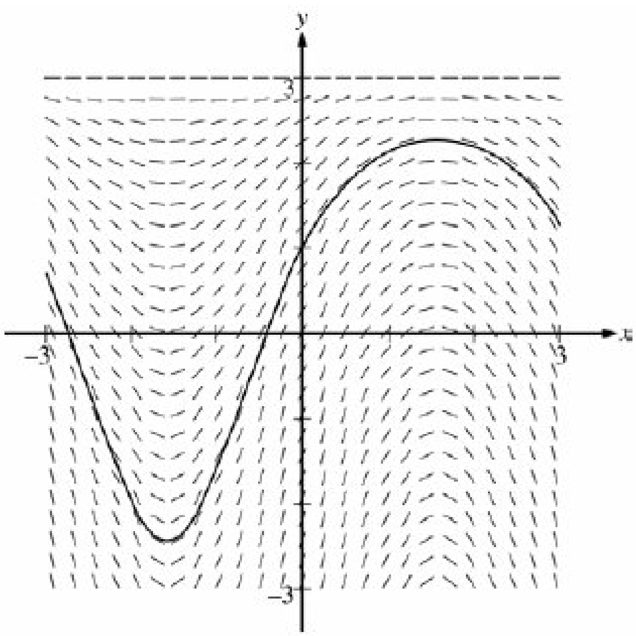
(b) Write an equation for the line tangent to the solution curve in part (a) at the point (0,1). Use the equation to approximate $f(0.2)$.

(c) Find $y = f(x)$, the particular solution to the differential equation with the initial condition $f(0) = 1$.

Part A

1 point is earned for solution curve

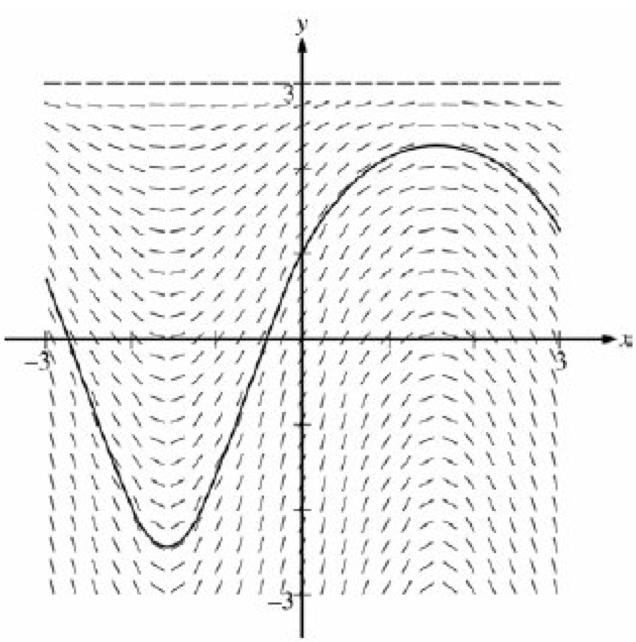
7-1-3



0	1

The student response earns all of the following points:

1 point is earned for solution curve



Part B

1 point is earned for the tangent line equation

1 point is earned for the approximation

7-1-3

$$\left.\frac{dy}{dx}\right|_{(x,y)=(0,1)} = 2\cos 0 = 2$$

An equation for the tangent line is $y = 2x + 1$.

$f(0.2) \approx 2(0.2) + 1 = 1.4$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for the tangent line equation

1 point is earned for the approximation

$$\left.\frac{dy}{dx}\right|_{(x,y)=(0,1)} = 2\cos 0 = 2$$

An equation for the tangent line is $y = 2x + 1$.

$f(0.2) \approx 2(0.2) + 1 = 1.4$

Part C

1 point is earned for the separation of variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

$$\frac{dy}{dx} = (3-y)\cos x$$
$$\int \frac{dy}{3-y} = \int \cos x dx$$
$$-\ln|3-y| = \sin x + C$$
$$-\ln 2 = \sin 0 + C \Rightarrow C = -\ln 2$$
$$-\ln|3-y| = \sin x - \ln 2$$

7-1-3

Because $y(0) = 1$, $y < 3$, so $|3 - y| = 3 - y$

$$3 - y = 2e^{-\sin x}$$
$$y = 3 - 2e^{-\sin x}$$

Note: this solution is valid for all real numbers.

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables

0	1	2	3	4	5	6 ✓

The student response earns all of the following points:

1 point is earned for the separation of variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

$$\frac{dy}{dx} = (3 - y)\cos x$$
$$\int \frac{dy}{3 - y} = \int \cos x dx$$
$$-\ln|3 - y| = \sin x + C$$
$$-\ln 2 = \sin 0 + C \Rightarrow C = -\ln 2$$
$$-\ln|3 - y| = \sin x - \ln 2$$

Because $y(0) = 1$, $y < 3$, so $|3 - y| = 3 - y$

$$3 - y = 2e^{-\sin x}$$
$$y = 3 - 2e^{-\sin x}$$

Note: this solution is valid for all real numbers.

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables

7-1-3

7-1-3

22. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

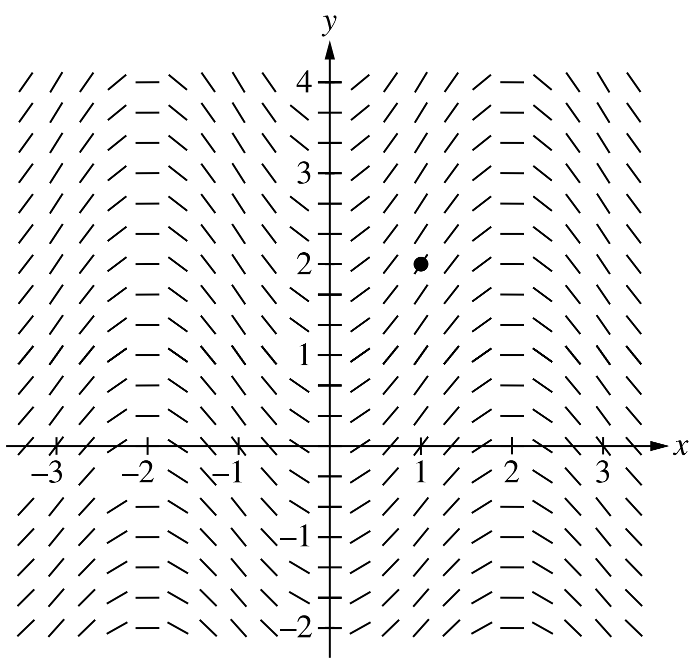
Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the differential equation $\frac{dy}{dx} = \frac{1}{2}\sin\left(\frac{\pi}{2}x\right)\sqrt{y+7}$. Let $y = f(x)$ be the particular solution to the differential equation with the initial condition $f(1) = 2$. The function f is defined for all real numbers.

(a) A portion of the slope field for the differential equation is given. Sketch the solution curve through the point $(1, 2)$.



(b) Write an equation for the line tangent to the solution curve in part (a) at the point $(1, 2)$. Use the equation to approximate $f(0.8)$.

(c) It is known that $f''(x) > 0$ for $-1 \leq x \leq 1$. Is the approximation found in part (b) an overestimate or an underestimate for $f(0.8)$? Give a reason for your answer.

(d) Use separation of variables to find $y = f(x)$, the particular solution to the differential equation $\frac{dy}{dx} = \frac{1}{2}\sin\left(\frac{\pi}{2}x\right)\sqrt{y+7}$ with the initial condition $f(1) = 2$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

7-1-3

The solution curve must pass through the point $(1, 2)$, extend reasonably close to the left and right edges of the square and have no obvious conflicts with the given slope lines.

Only portions of the solution curve within the given slope field are considered.

The solution curve must indicate $f(x) > 0$ for all points on the curve.

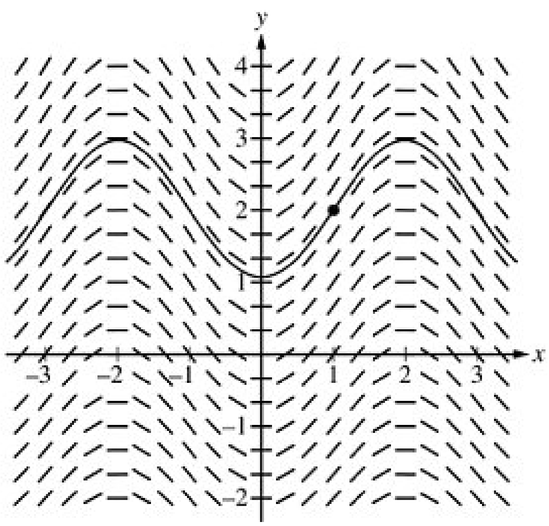
All local maximum/minimum points on the solution curve must occur at horizontal line segments in the slope field**.**

0	1 ✓

The student response accurately includes the criteria below.

☐ Solution curve

Solution:



Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The tangent line equation can be presented in any equivalent form.

An incorrect tangent line equation with a slope of $\frac{3}{2}$ is eligible to earn the second point for a consistent answer.

A response of only $2 + \frac{3}{2}(0.8 - 1)$ earns the second point but not the first.

0	1	2 ✓

7-1-3

The student response accurately includes **both** of the criteria below.

- ☐ Tangent line equation
- ☐ Approximation

Solution:

$$\left.\frac{dy}{dx}\right|_{(x,\ y)=(1,\ 2)} = \frac{1}{2} \cdot 3 \cdot \sin\left(\frac{\pi}{2}\right) = \frac{3}{2}$$

An equation for the tangent line is $y = 2 + \frac{3}{2}(x - 1)$.

$f\,(0.8) \approx 2 + \frac{3}{2}(0.8 - 1) = 1.7$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to

score the response.

The reason must include $f''(x) > 0$, $f'(x)$ is increasing, or $f\,(x)$ is concave up.

0	1 ✓

The student response accurately includes the criteria below.

- ☐ Answer with reason

Solution:

Because $f''(x) > 0, f$ is concave up on $-1 \leq x \leq 1$, the tangent line lies below the graph of $y = f\,(x)$ at $x = 0.8$, and the approximation for $f\,(0.8)$ is an underestimate.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with no separation of variables earns 0 out of 5 points.

A response with no constant of integration can earn at most the first 3 points.

A response is eligible for the fourth point only if it has earned the first point and at least 1 of the 2 antiderivative points.

7-1-3

· Special case: The incorrect separation of $\sqrt{y+7}\ \ dy = \frac{1}{2}\sin\left(\frac{\pi}{2}x\right)\ dx$ does not earn the first point, is only eligible for the antiderivative point for $-\frac{1}{\pi}\cos\left(\frac{\pi}{2}x\right)$, and is eligible for the fourth point.

An eligible response earns the fourth point by correctly including the constant of integration in an equation and substituting 1 for *x* and 2 for *y*.

A response is eligible for the fifth point only if it has earned the first 4 points.

0	1	2	3	4	5

The student response accurately includes **all five** of the criteria below.

- [ ] Separation of variables
- [ ] One correct antiderivative
- [ ] The other correct antiderivative
- [ ] Constant of integration and uses initial condition
- [ ] Solves for *y*

Solution:

$$\int \frac{dy}{\sqrt{y+7}} = \int \frac{1}{2}\sin\left(\frac{\pi}{2}x\right) dx$$

$$2\sqrt{y+7} = -\frac{1}{\pi}\cos\left(\frac{\pi}{2}x\right) + C$$

$$f\,(1) = 2 \ \Rightarrow 2\sqrt{2+7} = -\frac{1}{\pi}\cos\left(\frac{\pi}{2}\cdot 1\right) + C$$

$$\Rightarrow \ 6 = -\frac{1}{\pi}\cos\left(\frac{\pi}{2}\right) + C \Rightarrow C = 6$$

$$\sqrt{y+7} = 3 - \frac{1}{2\pi}\cos\left(\frac{\pi}{2}x\right)$$

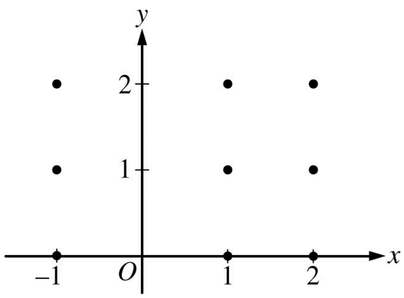
$$y = \left(3 - \frac{1}{2\pi}\cos\left(\frac{\pi}{2}x\right)\right)^2 - 7$$

7-1-3

23. The following are related to this scenario:

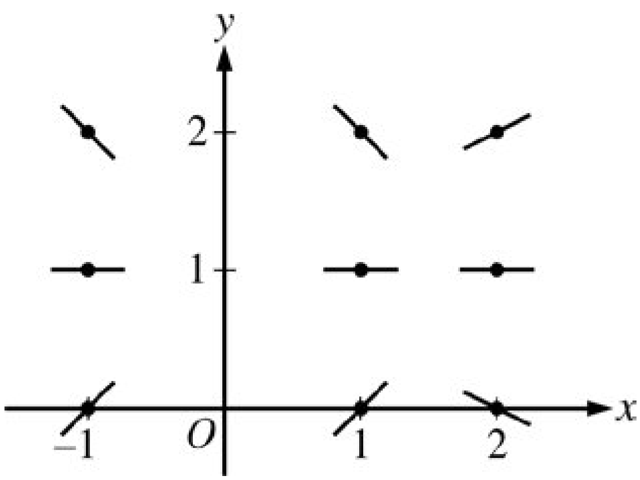
Consider the differential equation $\frac{dy}{dx}=\left(1-\frac{2}{x^2}\right)(y-1)$,where $x\neq0$. Let $y=f(x)$ be the particular solution to the differential equation with initial condition $f(1)=2$.

On the axes provided, sketch a slope field for the given differential equation at the nine points indicated.



Part B

The response can earn up to 2 points:



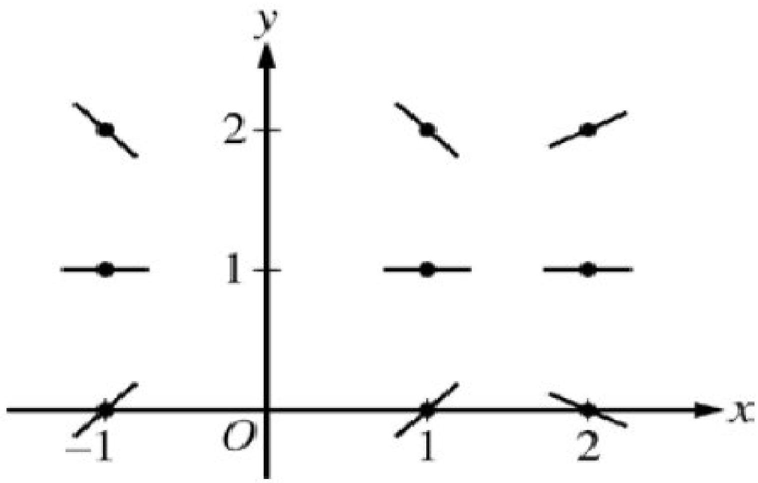
1 point: For zero slope at each point (x, y) where $y=1$

1 point: For the remaining slopes

0	1	2

The response earns both of the following points:

7-1-3



1 point: For zero slope at each point (x, y) where $y = 1$

1 point: For the remaining slopes

24. Which of the following is a solution to the differential equation $y - 4y'' = 0$?

(A) $y = \sin\left(\frac{x}{2}\right)$

(B) $y = \sin(2x)$

(C) $y = e^{x/2}$

(D) $y = e^{2x}$

Answer C

Correct. One way to verify that a function is a solution to a differential equation is to check that the function and its derivatives satisfy the differential equation. The correct derivative must be computed and the algebra correctly done to verify that the differential equation is satisfied.

$y = e^{x/2}$

$y' = \frac{1}{2}e^{x/2}$

$y'' = \frac{1}{4}e^{x/2}$

Substituting these into the differential equation gives $y - 4y'' = e^{x/2} - 4\left(\frac{1}{4}e^{x/2}\right) = e^{x/2} - e^{x/2} = 0$. Since the differential equation is satisfied, $y = e^{x/2}$ is a solution.

7-1-3

25. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

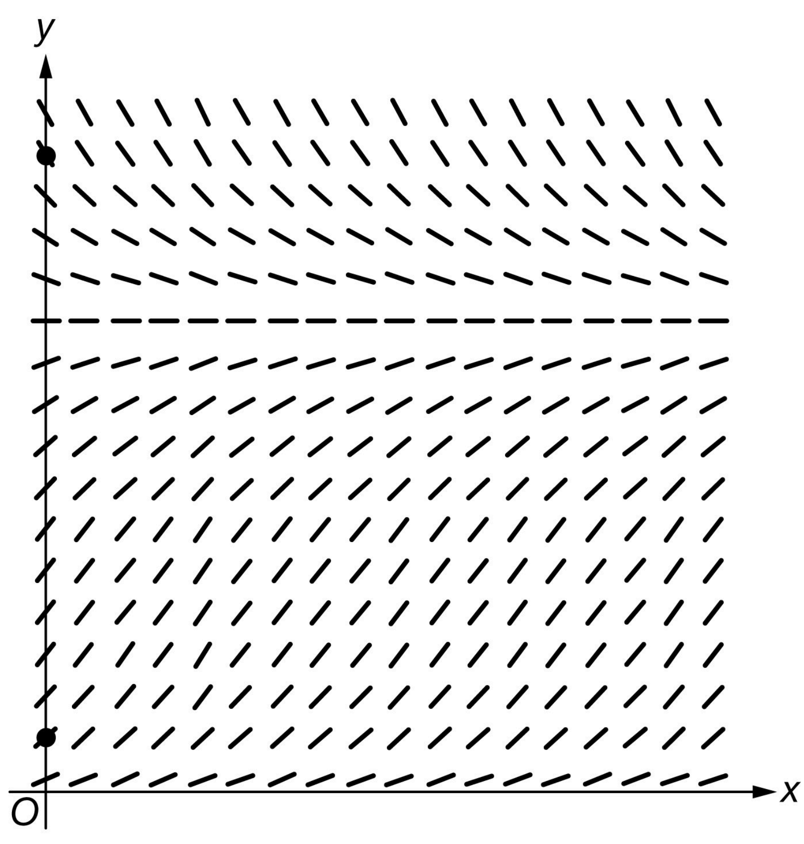
Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the differential equation $\frac{dy}{dx} = \frac{-y}{4}\ln\left(\frac{y}{50}\right)$.

(a) A portion of the slope field for the differential equation is given below. Sketch the solution curves through the two indicated points for $x \geq 0$.



(b) Find $\lim\limits_{y\to 50} \frac{dy}{dx}$.

(c) Let f be the particular solution to the differential equation with initial condition $f(0) = 20$. Find $f'(0)$, and explain why f is increasing for all x.

(d) For $0 < y < 40$, at what value of y does $\frac{dy}{dx}$ attain its maximum value? Justify your answer.

Part A

7-1-3

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Each solution curve must pass through the appropriate point, extend reasonably close to the right edge of the square, and have no obvious conflicts with the given slope lines.

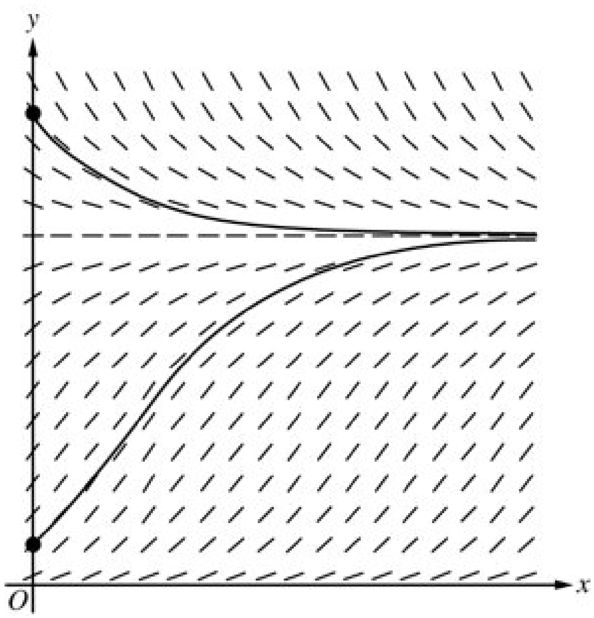
Ignore any portion of a solution curve that extends beyond the square.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Top solution curve
- ☐ Bottom solution curve

Solution:



Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

An answer of 0 earns the point.

The answer of $\frac{-50}{4}\ln\left(\frac{50}{50}\right)$ need not be simplified, but any subsequent simplification must be correct.

Limit notation is not required.

7-1-3

Expressions that equate the limit with the derivative such as $\lim_{y\to 50}\frac{dy}{dx} = \frac{-y}{4}\ln\left(\frac{y}{50}\right) = \frac{-50}{4}\ln\left(\frac{50}{50}\right)$, or other linkage errors, will not earn the point, even in the presence of the correct evaluation.

Expressions with incomplete limit notation, such as $\lim_{y\to 50} = \frac{-50}{4}\ln\left(\frac{50}{50}\right)$, are eligible for the point.

The expression $\lim_{y\to 50}\frac{-50}{4}\ln\left(\frac{50}{50}\right)$ is not incorrect, but the limit must be resolved to earn this point.

0	1 ✓

The student response accurately includes the criteria below.

☐ Limit

Solution:

$$\lim_{y\to 50}\frac{dy}{dx} = \lim_{y\to 50}\frac{-y}{4}\ln\left(\frac{y}{50}\right) = \frac{-50}{4}\ln\left(\frac{50}{50}\right) = 0$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for $-5\ln\left(\frac{2}{5}\right)$ or the equivalent; no supporting work is necessary.

The evaluation does not need to be simplified, and incorrect attempts at simplifying a correct expression will not be read.

For the second point to be earned, the response must:

· Argue that $\frac{dy}{dx}$ is positive for $0 < y < 50$, thus f is increasing for $0 < y < 50$.

· Argue that the particular solution cannot cross the line $y = 50$. (This may be done implicitly by restricting the domain of consideration to $0 < y < 50$.)

Explanations may be graphical but must address both of the above points.

Explanations that appeal only to the sign of $f'(0)$ will not earn this point.

0	1	2 ✓

7-1-3

The student response accurately includes **both** of the criteria below.

- ☐ $f'(0)$
- ☐ Explanation

Solution:

$f'(0) = \left.\frac{dy}{dx}\right|_{(x,\ y)=(0,\ 20)} = \frac{-20}{4}\ln\left(\frac{20}{50}\right) = -5\ln\left(\frac{2}{5}\right)$

Because $\frac{dy}{dx} = \frac{-y}{4}\ln\left(\frac{y}{50}\right) > 0$ for $0 < y < 50$, f is increasing provided $f(x) < 50$.

Note that $y = 50$ is also a solution to the differential equation, so the graph of the particular solution $y = f(x)$ with $f(0) = 20$ is below the line $y = 50$ and cannot cross the line $y = 50$.

Thus, f is increasing for all x.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by correctly applying the product rule in finding $\frac{d^2y}{dx^2}$, with or without using the chain rule. For example, $\frac{d^2y}{dx^2} = -\frac{1}{4} \cdot \ln\left(\frac{y}{50}\right) - \frac{y}{4} \cdot \frac{1}{y}$ would earn the first point.

The second point is earned for correctly applying the chain rule (twice) in finding $\frac{d^2y}{dx^2}$. This point can be earned even if an error was made in applying the product rule that resulted in the first point not being earned. In general, expressions of the form $\frac{d^2y}{dx^2} = c \cdot \ln\left(\frac{y}{50}\right) \cdot \frac{dy}{dx} + g(y) \cdot \frac{dy}{dx}$, where $g(y)$ may be a nonzero constant and c is a nonzero constant, earn this point.

For the first two points, if both the factors 50 and $\frac{1}{50}$ are not present in the response, assume that these factors were canceled. Furthermore, in the presence of the factor 50, assume that a missing factor $\frac{1}{50}$ is an error in the application of the chain rule. Thus, $\frac{d^2y}{dx^2} = -\frac{1}{4} \cdot \frac{dy}{dx} \cdot \ln\left(\frac{y}{50}\right) - \frac{y}{4} \cdot \frac{50}{y} \cdot \frac{dy}{dx}$ would earn the first point but not the second point.

For the first two points, each point is earned with the correct expression and any subsequent errors in simplification come off of the fourth point.

The third point is earned when the declared $\frac{d^2y}{dx^2}$ is set equal to zero and one initial (correct or otherwise) algebraic step towards solving this equation for y is made. However, the declared expression for $\frac{d^2y}{dx^2}$ must contain the variable y and must be distinct from $\frac{dy}{dx} = \frac{-y}{4}\ln\left(\frac{y}{50}\right)$. (This means a score of 0-0-1-0 is possible in this part.)

7-1-3

The fourth point cannot be earned without having earned the first three points.

The fourth point requires a discussion of the behavior of $\frac{d^2y}{dx^2}$ both to the left and right of $y = 50e^{-1}$.

Sign charts, labeled or otherwise, are not read for the fourth point.

0	1	2	3	4

The student response accurately includes **all four** of the criteria below.

- ☐ Product rule
- ☐ Chain rule
- ☐ Attempts to solve $\frac{d^2y}{dx^2} = 0$ for y
- ☐ Answer with justification

Solution:

$$\frac{d^2y}{dx^2} = -\frac{1}{4} \cdot \frac{dy}{dx} \cdot \ln\left(\frac{y}{50}\right) - \frac{y}{4} \cdot \frac{50}{y}\frac{1}{50}\frac{dy}{dx}$$

$$= -\frac{1}{4} \cdot \frac{dy}{dx} \cdot \ln\left(\frac{y}{50}\right) - \frac{y}{4} \cdot \frac{1}{y}\frac{dy}{dx} = -\frac{1}{4} \cdot \frac{dy}{dx} \cdot \left(\ln\left(\frac{y}{50}\right) + 1\right)$$

$$\frac{d^2y}{dx^2} = 0 \Rightarrow \left(\ln\left(\frac{y}{50}\right) + 1\right) = 0 \Rightarrow y = 50e^{-1}$$

Because $\frac{d^2y}{dx^2} > 0$ for $0 < y < 50e^{-1}$ and $\frac{d^2y}{dx^2} < 0$ for $50e^{-1} < y < 40$, $\frac{dy}{dx}$ attains its maximum value at $y = 50e^{-1}$.

26. The rate of change of the volume, V, of water in a tank with respect to time, t, is directly proportional to the square root of the volume. Which of the following is a differential equation that describes this relationship?

(A) $V(t) = k\sqrt{t}$

(B) $V(t) = k\sqrt{V}$

(C) $\frac{dV}{dt} = k\sqrt{t}$

(D) $\frac{dV}{dt} = \frac{k}{\sqrt{V}}$

(E) $\frac{dV}{dt} = k\sqrt{V}$ ✓

7-1-3

27. The rate of change of the volume, V, of water in a tank with respect to time, t, is directly proportional to the square root of the volume. Which of the following is a differential equation that describes this relationship?

(A) $V(t) = k\sqrt{t}$

(B) $V(t) = k\sqrt{V}$

(C) $\frac{dV}{dt} = k\sqrt{t}$

(D) $\frac{dV}{dt} = \frac{k}{\sqrt{V}}$

(E) $\frac{dV}{dt} = k\sqrt{V}$ ✓

28. A rumor spreads among a population of N people at a rate proportional to the product of the number of people who have heard the rumor and the number of people who have not heard the rumor. If p denotes the number of people who have heard the rumor, which of the following differential equations could be used to model this situation with respect to time t, where k is a positive constant?

(A) $\frac{dp}{dt} = kp$

(B) $\frac{dp}{dt} = kp\,(N - p)$ ✓

(C) $\frac{dp}{dt} = kp\,(p - N)$

(D) $\frac{dp}{dt} = kt\,(N - t)$

(E) $\frac{dp}{dt} = kt\,(t - N)$

7-1-3

29. A particle moves along the x-axis with position at time t given by $x(t) = e^{-t}\sin t$ for $0 \le t \le 2\pi$.

Find the value of the constant A for which $x(t)$ satisfies the equation $Ax''(t) + x'(t) + x(t) = 0$ for $0 < t < 2\pi$.

Part B

2 points are earned for: x''(t)

1 point is earned for substitutes x''(t), x'(t), and x(t) into Ax''(t) + x'(t) + x(t)

1 point is earned for the answer

$$x''(t) = -e^{-t}(\cos t - \sin t) + e^{-t}(\sin t - \cos t)$$
$$= -2e^{-t}\cos t$$
$$Ax''(t) + x'(t) + x(t)$$
$$= A\left(-2e^{-t}\cos t\right) + e^{-t}(\cos t - \sin t) + e^{-t}\sin t$$
$$= (-2A + 1)e^{-t}\cos t$$
$$= 0$$
$$Therefore, A = \frac{1}{2}$$

0	1	2	3	4

The student response earns all of the following points:

2 points are earned for: x''(t)

1 point is earned for substitutes x''(t), x'(t), and x(t) into Ax''(t) + x'(t) + x(t)

1 point is earned for the answer

$$x''(t) = -e^{-t}(\cos t - \sin t) + e^{-t}(\sin t - \cos t)$$
$$= -2e^{-t}\cos t$$
$$Ax''(t) + x'(t) + x(t)$$
$$= A\left(-2e^{-t}\cos t\right) + e^{-t}(\cos t - \sin t) + e^{-t}\sin t$$
$$= (-2A + 1)e^{-t}\cos t$$
$$= 0$$
$$Therefore, A = \frac{1}{2}$$

7-1-3

30. Which of the following is the solution to the differential equation $\frac{dy}{dx} = e^{y+x}$ with the initial condition $y(0) = -\ln 4$?

(A) $y = -x - \ln 4$

(B) $y = x - \ln 4$

(C) $y = -\ln(-e^x + 5)$ ✓

(D) $y = -\ln(e^x + 3)$

(E) $y = \ln(e^x + 3)$

31. A curve has slope $2x + 3$ at each point (x,y) on the curve. Which of the following is an equation for this curve if it passes through the point $(1,2)$?

(A) $y = 5x - 3$

(B) $y = x^2 + 1$

(C) $y = x^2 + 3x$

(D) $y = x^2 + 3x - 2$ ✓

(E) $y = 2x^2 + 3x - 3$

32. If $dy/dx=y \sec^2 x$ and $y=5$ when $x=0$, then $y=$

(A) $e^{\tan x} + 4$

(B) $e^{\tan x} + 5$

(C) $5e^{\tan x}$ ✓

(D) $\tan x + 5$

(E) $\tan x + 5e^x$

7-1-3

33. The function $y = e^{3x} - 5x + 7$ is a solution to which of the following differential equations?

(A) $y'' - 3y' - 15 = 0$ ✓

(B) $y'' - 3y' + 15 = 0$

(C) $y'' - y' - 5 = 0$

(D) $y'' - y' + 5 = 0$

Answer A

This option is correct. One way to verify that a function is a solution to a differential equation is to check that the function and its derivatives satisfy the differential equation. The differential equations in this problem involve both y' and y''. The correct derivatives must be computed and the algebra correctly done to verify that the differential equation is satisfied.

$y' = 3e^{3x} - 5$
$y'' = 9e^{3x}$
$y'' - 3y' - 15 = 9e^{3x} - 3(3e^{3x} - 5) - 15$
$= 9e^{3x} - 9e^{3x} + 15 - 15 = 0$

34. Which of the following is a solution to the differential equation $y'' - 4y = 0$?

(A) $y = e^{2x}$ ✓

(B) $y = 2e^{x}$

(C) $y = sin\,(2x)$

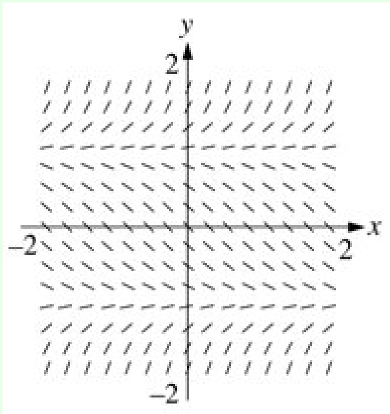
(D) $y = cos\,(2x)$

7-1-3

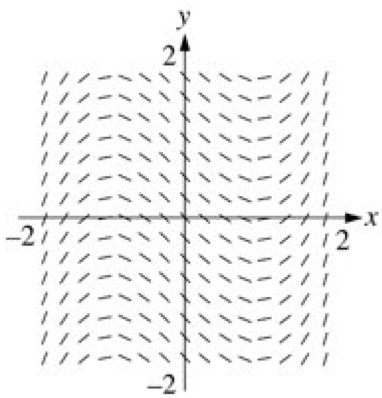
35. Which of the following could be the slope field for the differential equation $\frac{dy}{dx} = y^2 - 1$?

7-1-3

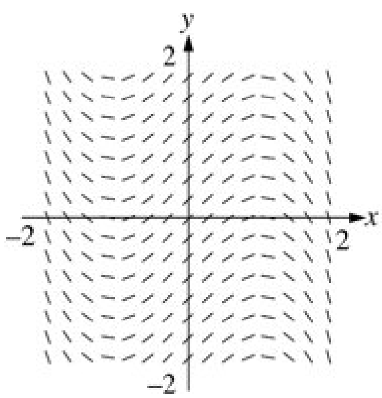
(A)



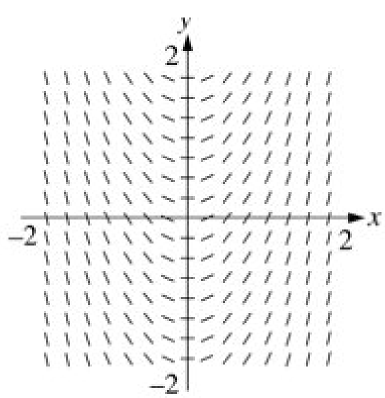
(B)



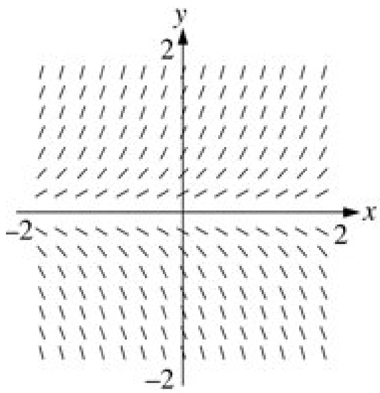
(C)



(D)



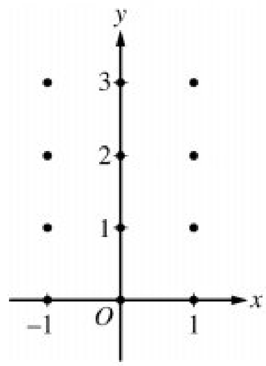
7-1-3

(E) 

7-1-3

36. Consider the differential equation $\frac{dy}{dx} = \frac{x(y-1)}{4}$.

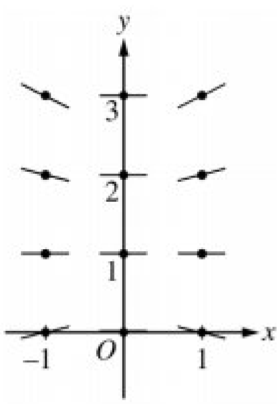
On the axes provided, sketch a slope field for the given differential equation at the twelve points indicated.



General

1 point is earned for: zero slopes

1 point is earned for: other slopes



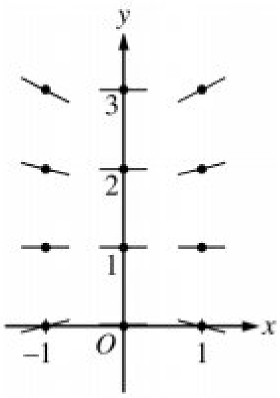
0	1	2

The student response earns all of the following points:

1 point is earned for: zero slopes

1 point is earned for: other slopes

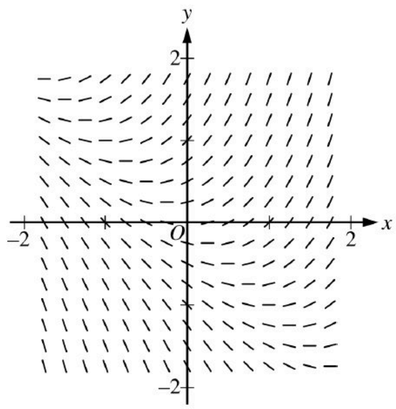
7-1-3



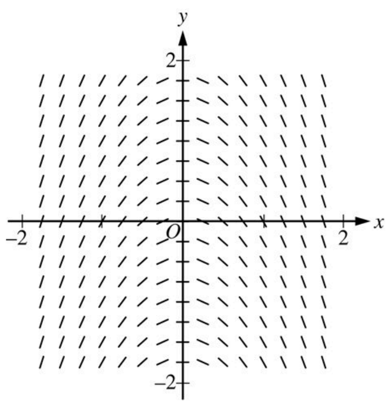
7-1-3

37. Which of the following could be a slope field for the differential equation $\frac{dy}{dx} = x^2 + y$?

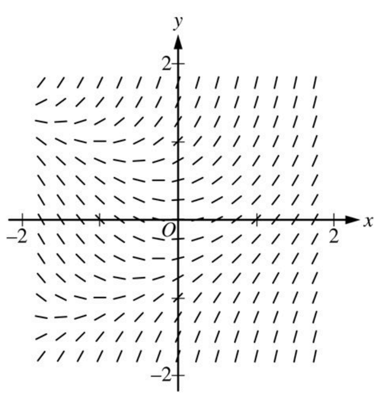
(A)



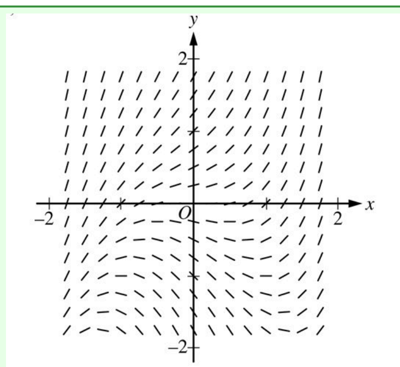
(B)



(C)



(D)

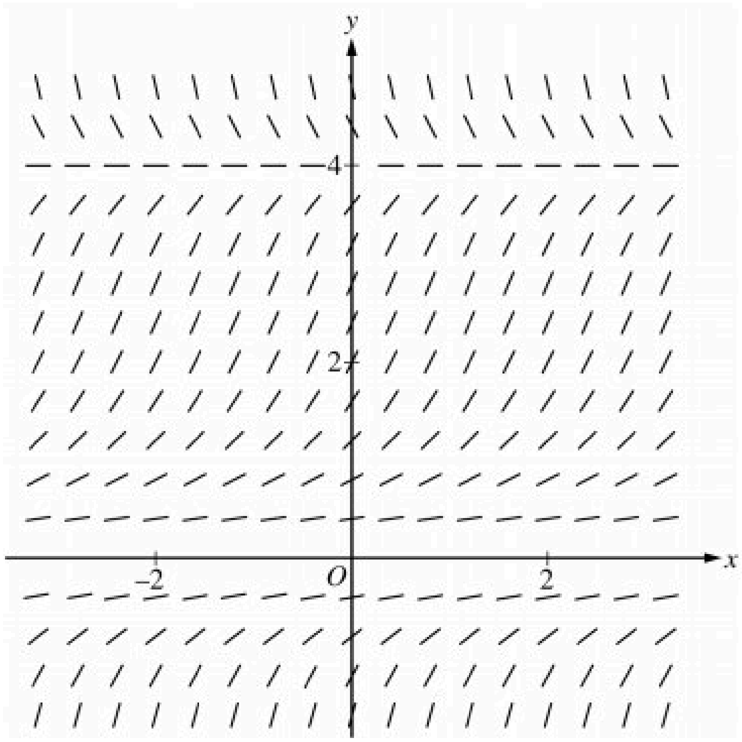


7-1-3

Answer D

This option is correct. The line segments in the slope field have slopes given by $\frac{dy}{dx} = x^2 + y$ at the point (x, y). In Quadrants I and II, all slopes must be positive or zero since $y > 0$ in those quadrants and $x^2 \geq 0$. This is the only option in which that condition is true.

38.



Shown is a slope field for which of the following differential equations?

(A) $\frac{dy}{dx} = \frac{x(4-y)}{4}$

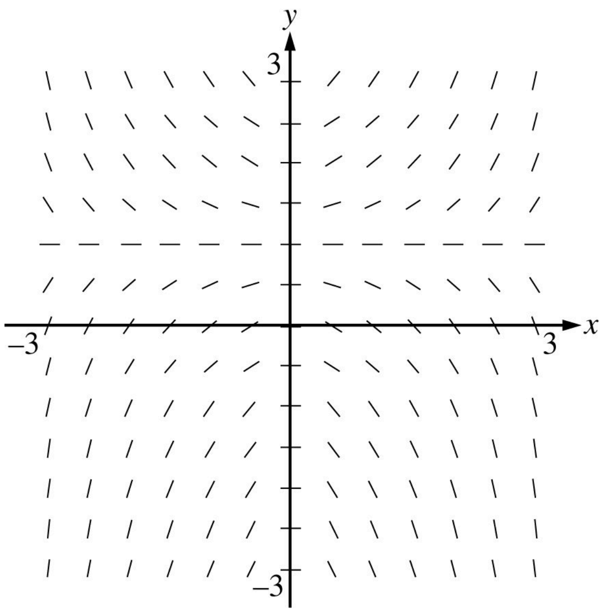
(B) $\frac{dy}{dx} = \frac{y(4-y)}{4}$

(C) $\frac{dy}{dx} = \frac{xy(4-y)}{4}$

(D) $\frac{dy}{dx} = \frac{y^2(4-y)}{4}$

7-1-3

39.



Shown above is a slope field for which of the following differential equations?

(A) $\dfrac{dy}{dx} = xy - x$

(B) $\dfrac{dy}{dx} = xy + x$

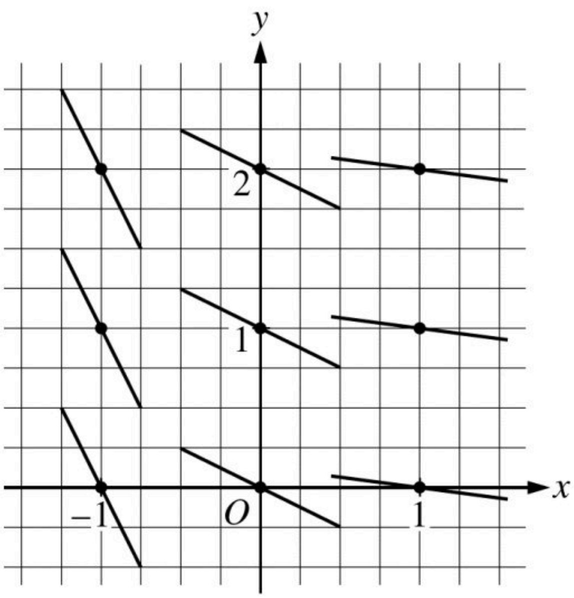
(C) $\dfrac{dy}{dx} = y - x^2$

(D) $\dfrac{dy}{dx} = (y-1)x^2$

(E) $\dfrac{dy}{dx} = (y-1)^3$

7-1-3

40.



A slope field for a differential equation is shown in the figure above. If $y = f(x)$ is the particular solution to the differential equation through the point (-1,2) and $h(x) = 3x \cdot f(x)$, then $h'(-1) =$

(A) -6

(B) -2

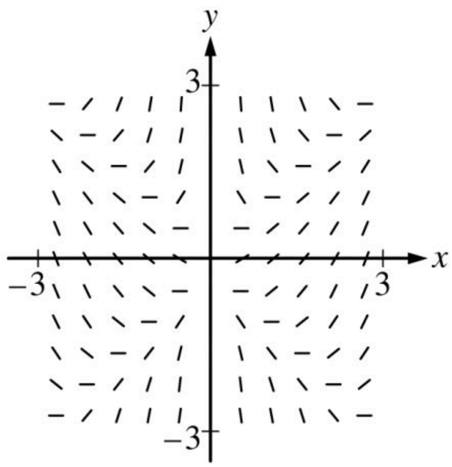
(C) 0

(D) 1

(E) 12 ✓

7-1-3

41.



Shown above is a slope field for which of the following differential equations?

(A) $\frac{dy}{dx} = \frac{x^2 - y^2}{x}$

(B) $\frac{dy}{dx} = \frac{x^2 - y^2}{y}$

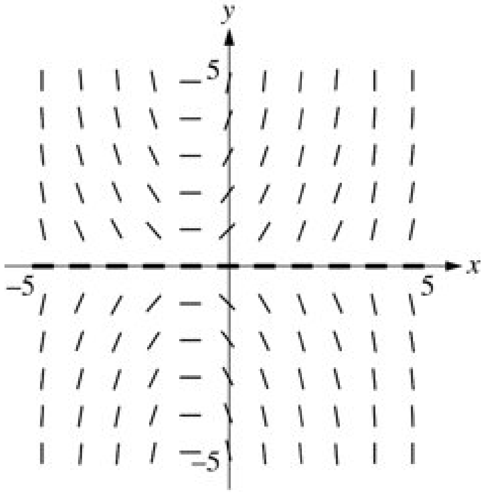
(C) $\frac{dy}{dx} = x^2 - y^2$

(D) $\frac{dy}{dx} = \frac{x^2 + y^2}{y}$

(E) $\frac{dy}{dx} = x^2 + y^2$

7-1-3

42.



Shown above is a slope field for which of the following differential equations?

(A) $\dfrac{dy}{dx} = xy$

(B) $\dfrac{dy}{dx} = xy - y$

(C) $\dfrac{dy}{dx} = xy + y$

(D) $\dfrac{dy}{dx} = xy + x$

(E) $\dfrac{dy}{dx} = (x+1)^3$

43. Which of the following is a solution to the differential equation $xy' - 2y = 2$?

(A) $y = x^2 - 2$

(B) $y = x^2 + 2x$

(C) $y = 4x^2 - 1$

(D) $y = 2x^3 - 1$

Answer C

Correct. One way to verify that a function is a solution to a differential equation is to check that the function and its derivative satisfy the differential equation for all values of x.

$y = 4x^2 - 1$

$y' = 8x$

$xy' - 2y = x(8x) - 2\left(4x^2 - 1\right) = 8x^2 - 8x^2 + 2 = 2$

Therefore, y and y' satisfy the differential equation for all values of x, so this y is a solution.

7-1-3

44. Which of the following is a solution to the differential equation $xy' - 3y = 6$?

(A) $y = x^2 + 2$

(B) $y = 3x^2 - 2$

(C) $y = 5x^3 - 2$

(D) $y = 7x^3 + 2$

Answer C

Correct. One way to verify that a function is a solution to a differential equation is to check that the function and its derivative satisfy the differential equation.

$y = 5x^3 - 2$

$y' = 15x^2$

$xy' - 3y = x\left(15x^2\right) - 3\left(5x^3 - 2\right) = 15x^3 - 15x^3 + 6 = 6$

Therefore, y and y' satisfy the differential equation, so this y is a solution.

45. Of the following, which are solutions to the differential equation $4y + y'' = 0$?

I. $y = 3\sin(2x)$
II. $y = 5\cos(2x)$
III. $y = e^{2x}$

(A) I only

(B) II only

(C) III only

(D) I and II only

Answer D

Correct. A solution to a differential equation can be verified by substituting the function and its derivative(s) into the equation. For $y = 3\sin(2x)$, the derivatives are $y' = 6\cos(2x)$ and $y'' = -12\sin(2x)$. It follows that $4y + y'' = 4(3\sin(2x)) + (-12\sin(2x)) = 0$, so the differential equation is satisfied. For $y = 5\cos(2x)$, the derivatives are $y' = -10\sin(2x)$ and $y'' = -20\cos(2x)$. It follows that $4y + y'' = 4(5\cos(2x)) + (-20\cos(2x)) = 0$, so the differential equation is satisfied. For $y = e^{2x}$, the derivatives are $y' = 2e^{2x}$ and $y'' = 4e^{2x}$. It follows that $4y + y'' = 4e^{2x} + 4e^{2x} = 8e^{2x} \neq 0$, so the differential equation is not satisfied.

7-1-3

46. Of the following, which is not a solution to the differential equation $y''' + 4y' = 0$?

(A) $y = 10$

(B) $y = 4e^{-2x}$

(C) $y = 3\sin(2x)$

(D) $y = 2\cos(2x) + 4$

Answer B

Correct. For $y = 4e^{-2x}$, the derivatives are $y' = -8e^{-2x}$, $y'' = 16e^{-2x}$, and $y''' = -32e^{-2x}$. This leads to the following.
$y''' + 4y' = -32e^{-2x} + 4\left(-8e^{-2x}\right) = -64e^{-2x} \neq 0$
So the differential equation is not satisfied.

47. For what value of k, if any, is $y = e^{2x} + ke^{-3x}$ a solution to the differential equation $4y - y'' = 10e^{-3x}$?

(A) -2

(B) $\frac{10}{3}$

(C) 10

(D) There is no such value of k.

Answer A

Correct. One way to verify that a function is a solution to a differential equation is to check that the function and its derivatives satisfy the differential equation. The differential equation in this problem involves both y and y''.
$y = e^{2x} + ke^{-3x}y' = 2e^{2x} - 3ke^{-3x}y'' = 4e^{2x} + 9ke^{-3x}4y - y'' = 4e^{2x} + 4ke^{-3x} - 4e^{2x} - 9ke^{-3x} = -5ke^{-3x}$
For y to be a solution to the differential equation, it is necessary that $-5ke^{-3x} = 10e^{-3x}$, so $-5k = 10$. Therefore, $y = e^{2x} + ke^{-3x}$ will be a solution to the differential equation if $k = -2$.

7-1-3

48. Of the following, which are solutions to the differential equation $y'' - 10y' + 9y = 0$?

I. $y = 2\sin(3x)$
II. $y = 5e^x$
III. $y = Ce^{9x}$, where C is a constant.

(A) I only

(B) II only

(C) III only

(D) II and III only ✓

Answer D

Correct. A solution to a differential equation can be verified by substituting the function and its derivative(s) into the equation.

For $y = 2\sin(3x)$, the derivatives are $y' = 6\cos(3x)$ and $y'' = -18\sin(3x)$.

$y'' - 10y' + 9y = -18\sin(3x) - 10(6\cos(3x)) + 9(2\sin(3x)) = -60\cos(3x) \neq 0$ for all x, so the differential equation is not satisfied.

For $y = 5e^x$, the derivatives are $y' = 5e^x$ and $y'' = 5e^x$.

$y'' - 10y' + 9y = 5e^x - 10(5e^x) + 9(5e^x) = 5e^x - 50e^x + 45e^x = 0$ for all x, so the differential equation is satisfied.

For $y = Ce^{9x}$, the derivatives are $y' = 9Ce^{9x}$ and $y'' = 81Ce^{9x}$.

$y'' - 10y' + 9y = 81Ce^{9x} - 10(9Ce^{9x}) + 9(Ce^{9x}) = 81Ce^{9x} - 90Ce^{9x} + 9Ce^{9x} = 0$ for all x, so the differential equation is satisfied for any value of the constant C.

7-1-3

49. For what value of k, if any, will $y = k\sin(5x) + 2\cos(4x)$ be a solution to the differential equation $y'' + 16y = -27\sin(5x)$?

(A) -27

(B) $-\frac{9}{5}$

(C) 3

(D) There is no such value of k.

Answer C

Correct. One way to verify that a function is a solution to a differential equation is to check that the function and its derivatives satisfy the differential equation. The differential equation in this problem involves both y and y''.

$y = k\sin(5x) + 2\cos(4x)$

$y' = 5k\cos(5x) - 8\sin(4x)$

$y'' = -25k\sin(5x) - 32\cos(4x)$

$y'' + 16y = -25k\sin(5x) - 32\cos(4x) + 16k\sin(5x) + 32\cos(4x) = -9k\sin(5x)$

For y to be a solution to the differential equation, it is necessary that $-9k\sin(5x) = -27\sin(5x)$, so $k = 3$.

7-1-3

50. Of the following, which are solutions to the differential equation $y'' - 5y' + 6y = 0$?

I. $y = e^{2t}$
II. $y = 3e^{3t}$
III. $y = 3\sin(2t)$

(A) I only

(B) II only

(C) III only

(D) I and II only ✓

Answer D

Correct. A solution to a differential equation can be verified by substituting the function and its derivatives into the equation.
For $y = e^{2t}$, the derivatives are $y' = 2e^{2t}$ and $y'' = 4e^{2t}$. It follows that
$y'' - 5y' + 6y = 4e^{2t} - 10e^{2t} + 6e^{2t} = 0$, so the differential equation is satisfied.
For $y = 3e^{3t}$, the derivatives are $y' = 9e^{3t}$ and $y'' = 27e^{3t}$. It follows that
$y'' - 5y' + 6y = 27e^{3t} - 45e^{3t} + 18e^{3t} = 0$,
so the differential equation is satisfied.
For $y = 3\sin(2t)$, the derivatives are $y' = 6\cos(2t)$ and $y'' = -12\sin(2t)$. It follows that
$y'' - 5y' + 6y = -12\sin(2t) - 30\cos(2t) + 18\sin(2t) = 6\sin(2t) - 30\cos(2t) \neq 0$,
so the differential equation is not satisfied.

51. Of the following, which is not a solution to the differential equation $y'' - 9y = 0$?

(A) $y = 0$

(B) $y = \sin(3t)$ ✓

(C) $y = e^{3t}$

(D) $y = 4e^{-3t}$

Answer B

Correct. For $y = \sin(3t)$, the derivatives are $y' = 3\cos(3t)$ and $y'' = -9\sin(3t)$. This leads to
$y'' - 9y = -9\sin(3t) - 9\sin(3t) = -18\sin(3t) \neq 0$, so the differential equation is not satisfied for this function.

7-1-3

52. For what value of k, if any, is $y = e^{-2x} + ke^{4x}$ a solution to the differential equation $y - \frac{y''}{4} = 5e^{4x}$?

(A) $-\frac{5}{3}$

(B) $\frac{20}{3}$

(C) 5

(D) There is no such value of k.

Answer A

Correct. One way to verify that a function is a solution to a differential equation is to check that the function and its derivatives satisfy the differential equation. The differential equation in this problem involves both y and y''.
$y = e^{-2x} + ke^{4x} y' = -2e^{-2x} + 4ke^{4x} y'' = 4e^{-2x} + 16ke^{4x}$
Then $y - \frac{y''}{4} = e^{-2x} + ke^{4x} - \frac{(4e^{-2x}+16ke^{4x})}{4} = -3ke^{4x}$.
If $-3ke^{4} = 5e^{4x}$, then $-3k = 5 \Rightarrow k = -\frac{5}{3}$.
Therefore, $y = e^{-2x} + ke^{4x}$ will be a solution to the differential equation if $k = -\frac{5}{3}$.

53. Of the following, which are solutions to the differential equation $y'' - 6y' + 8y = 0$?

I. $y = 2\sin(4x)$
II. $y = 3e^{2x}$
III. $y = Ce^{4x}$, where C is a constant.

(A) I only

(B) II only

(C) III only

(D) II and III only

Answer D

Correct. A solution to a differential equation can be verified by substituting the function and its derivatives into the equation.
For $y = 2\sin(4x)$, the derivatives are $y' = 8\cos(4x)$ and $y'' = -32\sin(4x)$.
Then $y'' - 6y' + 8y = -32\sin(4x) - 48\cos(4x) + 16\sin(4x) \neq 0$ for all x, so the differential equation is not satisfied.
For $y = 3e^{2x}$, the derivatives are $y = 3e^{2x}$ and $y'' = 12e^{2x}$.
Then $y'' - 6y' + 8y = 12e^{2x} - 36e^{2x} + 24e^{2x} = 0$ for all x, so the differential equation is satisfied.
For $y = Ce^{4x}$, the derivatives are $y' = 4Ce^{4x}$ and $y'' = 16Ce^{4x}$.
Then $y'' - 6y' + 8y = 16Ce^{4x} - 24e^{4x} + 8Ce^{4x} = 0$ for all x, so the differential equation is satisfied.

7-1-3

54. For what value of k, if any, will $y = ke^{-2x} + 4\cos(3x)$ be a solution to the differential equation $y'' + 9y = 26e^{-2x}$?

(A) 2

(B) $\frac{13}{5}$

(C) 26

(D) There is no such value of k.

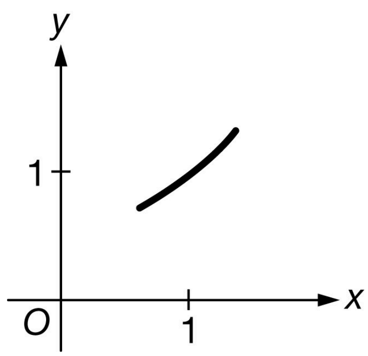
Answer A

Correct. One way to verify that a function is a solution to a differential equation is to check that the function and its derivatives satisfy the differential equation. The differential equation in this problem involves both y and y''.
$y = ke^{-2x} + 4\cos(3x) y' = -2ke^{-2x} - 12\sin(3x) y'' = 4ke^{-2x} - 36\cos(3x)$
Then $y'' + 9y = 4ke^{-2x} - 36\cos(3x) + 9ke^{-2x} + 36\cos(3x) = 13ke^{-2x}$.
If $13ke^{-2x} = 26e^{-2x}$, then $13k = 26 \Rightarrow k = 2$.
Therefore, if $k = 2$, $y = ke^{-2x} + 4\cos(3x)$ is a solution to the differential equation $y'' + 9y = 26e^{-2x}$.

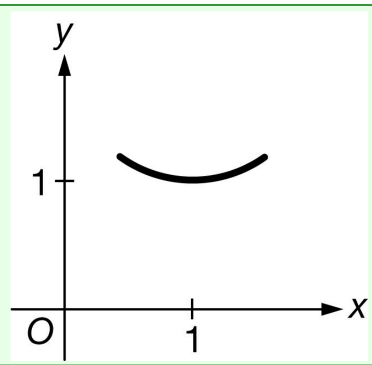
7-1-3

55. Which of the following could be the graph of a solution of the differential equation $\frac{dy}{dx} = (x-1)y^2$ near the point $(1, 1)$?

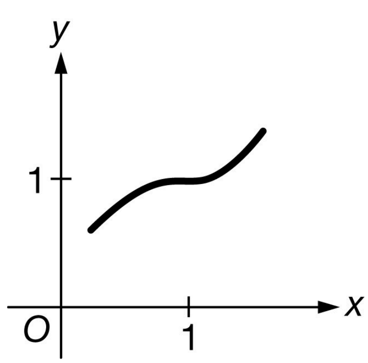
(A)



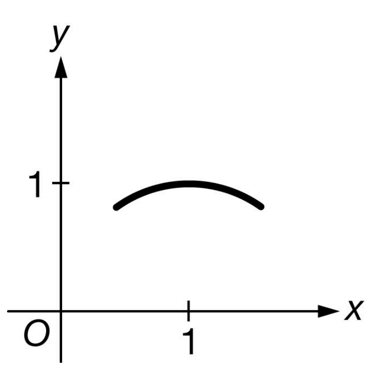
(B)



(C)



(D)



Answer B

7-1-3

Correct. For a solution y to this differential equation, $\left.\frac{dy}{dx}\right|_{(1,1)} = (1-1)(1)^2 = 0$ and
$\frac{d^2y}{dx^2} = y^2 + (x-1)2y\frac{dy}{dx} \Rightarrow \left.\frac{d^2y}{dx^2}\right|_{(1,1)} = 1^2 + 0 = 1.$
By the second derivative test, the graph of a solution to the differential equation has a local minimum at $x = 1$.
Therefore, this graph could be the graph of a solution.

56. Which of the following is a solution to the differential equation $\frac{dy}{dx} = e^{2x} + 2y$?

(A) $y = e^{2x}$

(B) $y = -e^{2x}$

(C) $y = xe^{2x}$ ✓

(D) $y = -xe^{2x}$

Answer C

Correct. One way to verify that a function is a solution to a differential equation is to check that the function and its derivative satisfy the differential equation. The differential equation in this option involves x and y . The correct derivative must be computed and the algebra correctly done to verify that the differential equation is satisfied.
$y = xe^{2x}$
$\frac{dy}{dx} = e^{2x} + 2xe^{2x} = e^{2x} + 2\left(xe^{2x}\right) = e^{2x} + 2y$
Since the differential equation is satisfied, $y = xe^{2x}$ is a solution.

7-1-3

57. Let $P(t)$ represent the number of wolves in a population at time t years, when $t \geq 0$. The population $P(t)$ is increasing at a rate directly proportional to 800 - $P(t)$, where the constant of proportionality is k.

(a) If $P(0) = 500$, find $P(t)$ in terms of t and k.

(b) If $P(2) = 700$, find k.

(c) Find $\lim_{t\to\infty} P(t)$.

Part A

1 point is earned for differential equation

1 point is earned if separates variables

2 points are earned for anti differentiation

1 point is **deducted** for errors

1 points is earned for solving for constant

1 point is earned if solves for $p(t)$

Note: If no differential equation solved, then not eligible for last two points

$$P'(t) = k(800 - P(t))$$
$$\frac{dP}{800-P} = k\,dt$$
$$-\ln|800 - P| = kt + C_0$$
$$|800 - P| = C_1 e^{-kt}$$
$$800 - 500 = C_1 e^0$$
$$C_1 = 300$$
$$\therefore\ P(t) = 800 - 300e^{-kt}$$

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned for differential equation

1 point is earned if separates variables

2 points are earned for anti differentiation

1 point is **deducted** for errors

7-1-3

1 points is earned for solving for constant

1 point is earned if solves for $p(t)$

Note: If no differential equation solved, then not eligible for last two points

$$P'(t) = k(800 - P(t))$$
$$\frac{dP}{800-P} = k\,dt$$
$$-\ln|800 - P| = kt + C_0$$
$$|800 - P| = C_1 e^{-kt}$$
$$800 - 500 = C_1 e^0$$
$$C_1 = 300$$
$$\therefore\ P(t) = 800 - 300e^{-kt}$$

Part B

1 point is earned for setting $P(2)$ equal to 700

1 point is earned for solving for k

$$P\,(2) = 700 = 800 - 300e^{-2k}$$
$$k = \frac{\ln 3}{2} \approx 0.549$$

0	1	2

The student response earns all of the following points:

1 point is earned for setting $P(2)$ equal to 700

1 point is earned for solving for k

$$P\,(2) = 700 = 800 - 300e^{-2k}$$
$$k = \frac{\ln 3}{2} \approx 0.549$$

Part C

1 point is earned for limits

7-1-3

$$\lim_{t \to \infty} \left(800 - 300e^{-\frac{\ln 3}{2}t}\right) = 800$$

0	1

The student response earns all of the following points:

1 point is earned for limits

$$\lim_{t \to \infty} \left(800 - 300e^{-\frac{\ln 3}{2}t}\right) = 800$$

7-1-3

58. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

The population of a virus in a host can be modeled by the function A that satisfies the differential equation $\frac{dA}{dt} = t - \frac{1}{5}A$, where A is measured in millions of virus cells and t is measured in days for $5 \leq t < 10$. At time $t = 5$ days, there are 10 million cells of the virus in the host.

(a) Write an equation for the line tangent to the graph of A at $t = 5$. Use the tangent line to approximate the number of virus cells in the host, in millions, at time $t = 7$ days.

(b) Show that $A(t) = -25 + 5t + 10e^{-\frac{t}{5}+1}$ satisfies the differential equation $\frac{dA}{dt} = t - \frac{1}{5}A$ with initial condition $A(5) = 10$.

(c) The host receives an antiviral medication. The amount of medication in the host is modeled by the function M that satisfies the differential equation $\frac{dM}{dt} = -\frac{1}{2}\left(\frac{M}{t+k}\right)$, where M is measured in milligrams, t is measured in days since the host received the medication, and k is a positive constant. If the amount of medication in the host is 30 milligrams at time $t = 0$ days and 15 milligrams at time $t = 3$ days, what is $M(t)$ in terms of t ?

Part A

For the second point, it is incorrect to state $A(7) = 16$ rather than $A(7) \approx 16$.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- [ ] tangent line equation
- [ ] approximation

Solution:

$A(5) = 10$

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$$\left.\frac{dA}{dt}\right|_{t=5} = 5 - \frac{1}{5} \cdot 10 = 3$$

An equation for the line tangent to the graph of A at $t = 5$ is $y = 10 + 3\,(t - 5)$.

At time $t = 7$ days, there are approximately $10 + 3\,(7 - 5) = 16$ million virus cells in the host. $A\,(7) \approx 16$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The student response accurately includes all three of the criteria below.

- ☐ verification of initial condition
- ☐ $\frac{dA}{dt} = 5 - 2e^{-\frac{t}{5}+1}$
- ☐ verification that $\frac{dA}{dt} = t - \frac{1}{5}A$

Solution:

$$A\,(5) = -25 + 5 \cdot 5 + 10e^{-1+1} = 10$$

$$\frac{dA}{dt} = \frac{d}{dt}\left(-25 + 5t + 10e^{-\frac{t}{5}+1}\right) = 5 + 10e^{-\frac{t}{5}+1}\left(-\frac{1}{5}\right) = 5 - 2e^{-\frac{t}{5}+1}$$

$$t - \frac{1}{5}A = t - \frac{1}{5}\left(-25 + 5t + 10e^{-\frac{t}{5}+1}\right) = t + 5 - t - 2e^{-\frac{t}{5}+1} = 5 - 2e^{-\frac{t}{5}+1}$$

Part C

Zero out of 4 points earned if no separation of variables.

At most 2 out of 4 points earned [1-1-0-0] if no constant of integration.

Both antiderivatives must be correct to earn the second point.

The fourth point requires an expression for $M\,(t)$. The domain of $M\,(t)$ is included with the solution; this is not a requirement to earn the fourth point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

7-1-3

The student response accurately includes all four of the criteria below.

- [ ] separation of variables
- [ ] antiderivatives
- [ ] constant of integration and uses initial conditions
- [ ] $M\ (t)$

Solution:

$$\frac{dM}{dt} = -\tfrac{1}{2}\left(\tfrac{M}{t+k}\right)$$
$$\frac{dM}{M} = -\tfrac{1}{2}\left(\tfrac{dt}{t+k}\right)$$
$$\int \frac{1}{M}\ dM = -\tfrac{1}{2}\int \left(\frac{1}{t+k}\right)\ dt$$
$$\ln|M| = -\tfrac{1}{2}\ln|t+k| + C$$

From the context, M and t are nonnegative. It is given that $k > 0$.

$$\ln\ M = -\tfrac{1}{2}\ln(t+k) + C$$
$$\ln\ M = \ln\frac{1}{\sqrt{t+k}} + C$$
$$M = \frac{D}{\sqrt{t+k}}$$
$$30 = \frac{D}{\sqrt{k}}\ \Rightarrow\ D = 30\sqrt{k}$$
$$15 = \frac{D}{\sqrt{3+k}}\ \Rightarrow\ D = 15\sqrt{3+k}$$
$$30\sqrt{k} = 15\sqrt{3+k}\ \Rightarrow\ k = 1\ \text{and}\ D = 30$$
$$M\ (t) = \frac{30}{\sqrt{t+1}}$$

Note: This is valid for $t \geq 0$.

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59. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

A large vat is initially filled with a saltwater solution. A solution with a higher concentration of salt flows into the vat, and solution flows out of the vat at the same rate. The number of pounds of salt in the vat at time t minutes is modeled by the function A that satisfies the differential equation $\frac{dA}{dt} = 6 - 0.02A$. At time $t = 10$ minutes, the vat contains 50 pounds of salt.

(a) Write an equation for the line tangent to the graph of A at $t = 10$. Use the tangent line to approximate the number of pounds of salt in the vat at time $t = 12$ minutes.

(b) Show that $A(t) = 300 - 250e^{0.2-0.02t}$ satisfies the differential equation $\frac{dA}{dt} = 6 - 0.02A$ with initial condition $A(10) = 50$.

(c) The flow of solution into the vat is stopped, and the solution is drained. The depth of solution in the vat is modeled by the function h that satisfies the differential equation $\frac{dh}{dt} = -k\sqrt{h}$, where $h(t)$ is measured in meters, t is the number of minutes since draining began, and k is a constant. If the depth of the solution is 16 meters at time $t = 0$ minutes and 4 meters at time $t = 30$ minutes, what is $h(t)$ in terms of t ?

Part A

For the second point, it is incorrect to state $A(12) = 60$ rather than $A(12) \approx 60$.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- ☐ tangent line equation
- ☐ approximation

Solution:

$A(10) = 50$

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$$\left.\frac{dA}{dt}\right|_{t=10} = 6 - 0.02 \cdot 50 = 5$$

An equation for the line tangent to the graph of A at $t = 10$ is $y = 50 + 5\,(t - 10)$.

At time $t = 12$ minutes, the vat contains approximately $50 + 5\,(12 - 10) = 60$ pounds of salt. $A\,(12) \approx 60$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The student response accurately includes all three of the criteria below.

- [ ] verification of initial condition
- [ ] $\frac{dA}{dt} = 5e^{0.2-0.02t}$
- [ ] verification that $\frac{dA}{dt} = 6 - 0.02A$

Solution:

$$A\,(10) = 300 - 250e^{0.2-0.02\cdot 10} = 300 - 250e^{0} = 50$$

$$\frac{dA}{dt} = \frac{d}{dt}\left(300 - 250e^{0.2-0.02t}\right)$$
$$= -250e^{0.2-0.02t}\,(-0.02) = 5e^{0.2-0.02t}$$

$$6 - 0.02A = 6 - 0.02\left(300 - 250e^{0.2-0.02t}\right)$$
$$= 6 - 6 + 5e^{0.2-0.02t} = 5e^{0.2-0.02t}$$

Part C

Zero out of 4 points earned if no separation of variables.

At most 2 out of 4 points earned [1-1-0-0] if no constant of integration.

Both antiderivatives must be correct to earn the second point.

The fourth point requires an expression for $h\,(t)$. The domain of $h\,(t)$ is included with the solution; this is not a requirement to earn the fourth point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response

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0	1	2	3	4

The student response accurately includes all four of the criteria below.

- ☐ separation of variables
- ☐ antiderivatives
- ☐ constant of integration and uses initial conditions
- ☐ $h\ (t)$

Solution:

$$\frac{dh}{dt} = -k\sqrt{h}$$
$$\frac{dh}{\sqrt{h}} = -k\ dt$$
$$\int \frac{1}{\sqrt{h}}\ dh = -\int k\ dt$$
$$2\sqrt{h} = -kt + C$$
$$2\sqrt{16} = 0 + C \ \Rightarrow\ C = 8$$
$$2\sqrt{4} = -30k + 8 \ \Rightarrow\ k = \frac{2}{15}$$
$$2\sqrt{h} = -\frac{2}{15}t + 8$$
$$h\ (t) = \left(-\frac{t}{15} + 4\right)^2$$

Note: This is valid for $0 \leq t \leq 60$.

7-1-3

60. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

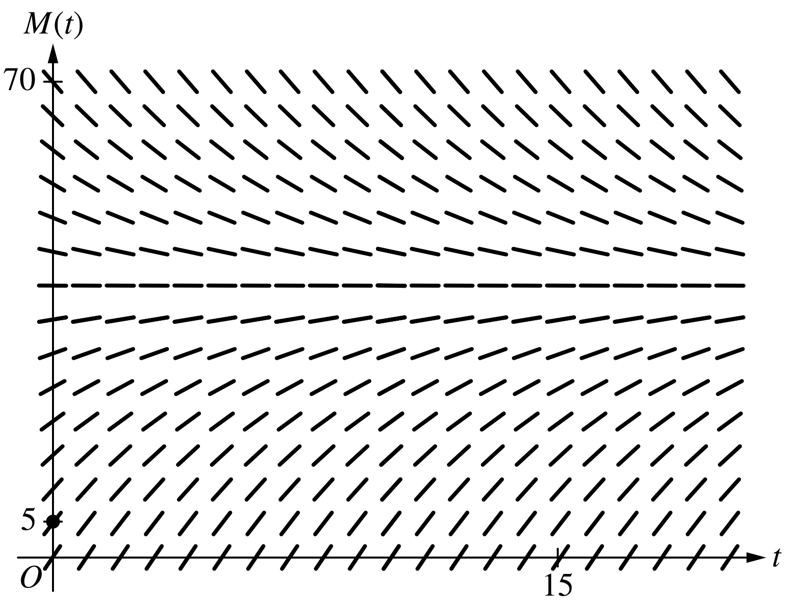
Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

A bottle of milk is taken out of a refrigerator and placed in a pan of hot water to be warmed. The increasing function M models the temperature of the milk at time t, where $M(t)$ is measured in degrees Celsius ($^\circ$C) and t is the number of minutes since the bottle was placed in the pan. M satisfies the differential equation $\frac{dM}{dt} = \frac{1}{4}(40 - M)$. At time $t = 0$, the temperature of the milk is $5^\circ\mathrm{C}$. It can be shown that $M(t) < 40$ for all values of t.

(a) A slope field for the differential equation $\dfrac{dM}{dt} = \dfrac{1}{4}(40 - M)$ is shown. Sketch the solution curve through the point $(0, 5)$.



(b) Use the line tangent to the graph of M at $t = 0$ to approximate $M(2)$, the temperature of the milk at time $t = 2$ minutes.

(c) Write an expression for $\dfrac{d^2M}{dt^2}$ in terms of M. Use $\dfrac{d^2M}{dt^2}$ to determine whether the approximation from part (b) is an underestimate or an overestimate for the actual value of $M(2)$. Give a reason for your answer.

(d) Use separation of variables to find an expression for $M(t)$, the particular solution to the differential equation $\dfrac{dM}{dt} = \dfrac{1}{4}(40 - M)$ with initial condition $M(0) = 5$.

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Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The solution curve must pass through the point $(0, 5)$, extend reasonably close to the left and right edges of the rectangle, and have no obvious conflicts with the given slope lines.

Only portions of the solution curve within the given slope field are considered.

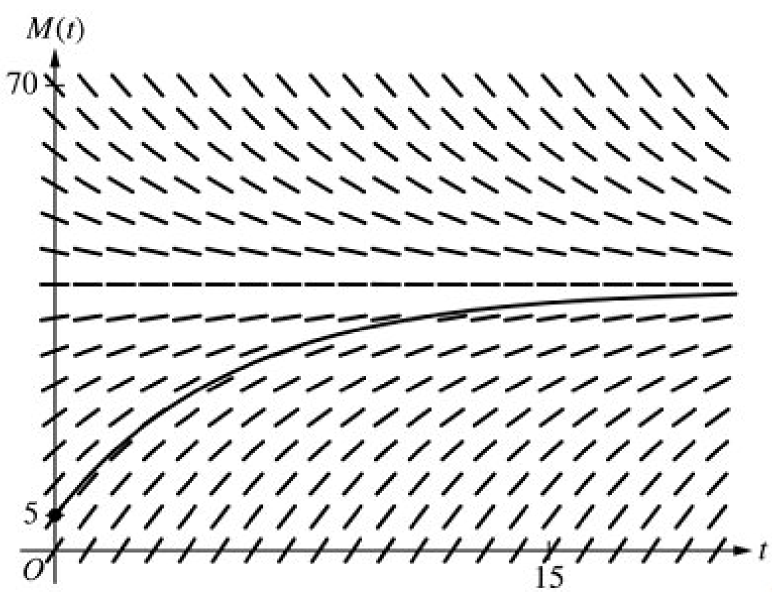
The solution curve must lie entirely below the horizontal line segments at $M = 40$.

0	1

The student response accurately includes the criteria below.

- [ ] Solution curve

Solution:



Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The value of the slope may appear in a tangent line equation or approximation.

A response of $5 + \frac{35}{4} \cdot 2$ is the minimal response to earn both points.

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A response of $\frac{1}{4}(40-5)$ earns the first point. If there are any subsequent errors in simplification, the response does not earn the second point.

In order to earn the second point the response must present an approximation found by using a tangent line that:

- passes through the point $(0,5)$ and
- has slope $\frac{35}{4}$ or a nonzero slope that is declared to be the value of $\frac{dM}{dt}$.

An unsupported approximation does not earn the second point.

The approximation need not be simplified, but the response does not earn the second point if the approximation is simplified incorrectly.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] $\left.\frac{dM}{dt}\right|_{t=0}$
- [ ] Approximation

Solution:

$$\left.\frac{dM}{dt}\right|_{t=0} = \frac{1}{4}(40-5) = \frac{35}{4}$$

The tangent line equation is $y = 5 + \frac{35}{4}(t-0)$.

$$M(2) \approx 5 + \frac{35}{4} \cdot 2 = 22.5$$

The temperature of the milk at time $t = 2$ minutes is approximately 22.5° Celsius.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for either $\frac{d^2M}{dt^2} = -\frac{1}{4}\left(\frac{1}{4}(40-M)\right)$ or $\frac{d^2M}{dt^2} = -\frac{1}{16}(40-M)$ (or equivalent). A response that presents any subsequent simplification error does not earn the second point.

A response that presents an expression for $\frac{d^2M}{dt^2}$ in terms of $\frac{dM}{dt}$ but fails to continue to an expression in terms of (i.e., $\frac{d^2M}{dt^2} = -\frac{1}{4}\frac{dM}{dt}$) does not earn the first point but is eligible for the second point.

7-1-3

If the response presents an expression for $\frac{d^2M}{dt^2}$ that is incorrect, the response is eligible for the second point only if the expression is a nonconstant linear function that is negative for $5 < M < 40$.

- Special case: A response that presents $\frac{d^2M}{dt^2} = \frac{1}{16}(40 - M)$ does not earn the first point but is eligible to earn the second point for a consistent answer and reason.

To earn the second point a response must include $\frac{d^2M}{dt^2} < 0$, or $\frac{dM}{dt}$ is decreasing, or the graph of M is concave down, as well as the conclusion that the approximation is an overestimate.

A response that presents an argument based on $\frac{d^2M}{dt^2}$ or concavity at a single point does not earn the second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] $\frac{d^2M}{dt^2}$
- [ ] Overestimate with reason

Solution:

$$\frac{d^2M}{dt^2} = -\frac{1}{4}\frac{dM}{dt} = -\frac{1}{4}\left(\frac{1}{4}(40 - M)\right) = -\frac{1}{16}(40 - M)$$

Because $M(t) < 40$, $\frac{d^2M}{dt^2} < 0$, so the graph of M is concave down. Therefore, the tangent line approximation of $M(2)$ is an overestimate.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with no separation of variables earns 0 out of 4 points.

A response that presents an antiderivative of $-\ln(40 - M)$ without absolute value symbols is eligible for all 4 points.

A response with no constant of integration can earn at most the first 2 points.

A response is eligible for the third point only if it has earned the first 2 points.

- Special Case: A response that presents $+\ln(40 - M) = \frac{t}{4} + C$ (or equivalent) does not earn the second point, is eligible for the third point, but not eligible for the fourth.

7-1-3

An eligible response earns the third point by correctly including the constant of integration in an equation and substituting 0 for t and 5 for M.

A response is eligible for the fourth point only if it has earned the first 3 points.

A response earns the fourth point only for an answer of $M = 40 - 35e^{-\frac{t}{4}}$ or equivalent.

0	1	2	3	4

The student response accurately includes **all four** of the criteria below.

- ☐ Separates variables
- ☐ Finds antiderivatives
- ☐ Constant of integration and uses initial condition
- ☐ Solves for M

Solution:

$$\frac{dM}{40-M} = \frac{1}{4}dt$$

$$\int \frac{dM}{40 - M} = \int \frac{1}{4}dt$$

$$-\ln|40 - M| = \frac{1}{4}t + C$$

$$-\ln|40 - 5| = 0 + C \Rightarrow C = -\ln 35$$

$$M(t) < 40 \Rightarrow 40 - M > 0 \Rightarrow |40 - M| = 40 - M$$

$$-\ln(40 - M) = \frac{1}{4}t - \ln 35$$

$$\ln(40 - M) = -\frac{1}{4}t + \ln 35$$

$$40 - M = 35e^{-\frac{t}{4}}$$

$$M = 40 - 35e^{-\frac{t}{4}}$$

7-1-3

61. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

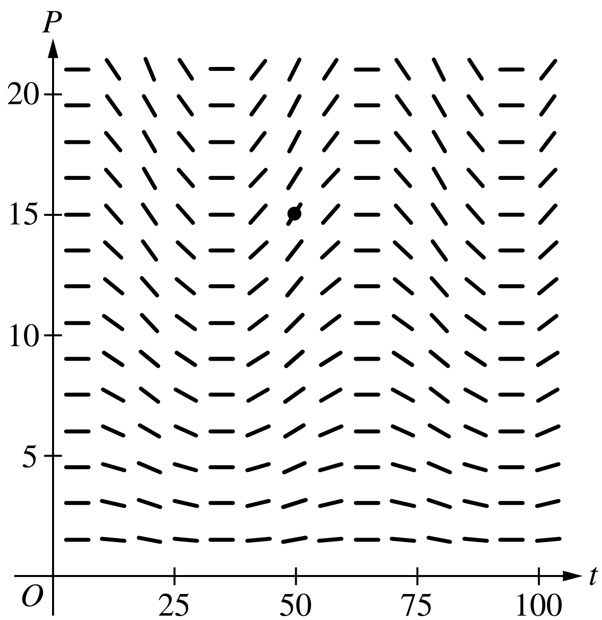
Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Scientists study a population that experiences variations in its rate of growth. The rate of growth of the population is modeled by the differential equation $\frac{dP}{dt} = P \cdot 0.02 \cos\left(\frac{\pi}{30}(t+10)\right)$, where t is measured in days since the observation of the population began. Let $P = f(t)$ be the particular solution to the given differential equation satisfying $f(50) = 15$.

(a) A slope field for the given differential equation is shown. Sketch the solution curve that passes through the point $(50, 15)$.



(b) Write an equation for the line tangent to the graph of $P = f(t)$ at time $t = 50$.

(c) It is known that $f'(t) > 0$ and $f''(t) > 0$ on the interval $47 < t < 51$. If the tangent line found in part (b) is used to approximate $f(48)$, is the approximation an underestimate or an overestimate for the actual value of $f(48)$? Give a reason for your answer.

(d) Use separation of variables to find the particular solution $P = f(t)$ to the differential equation $\frac{dP}{dt} = P \cdot 0.02 \cos\left(\frac{\pi}{30}(t+10)\right)$ that satisfies $f(50) = 15$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

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The solution curve must pass through the point $(50, 15)$, extend reasonably close to the left and right edges of the slope field, and have no obvious conflicts with the given line segments.

Only portions of the solution curve within the given slope field are considered.

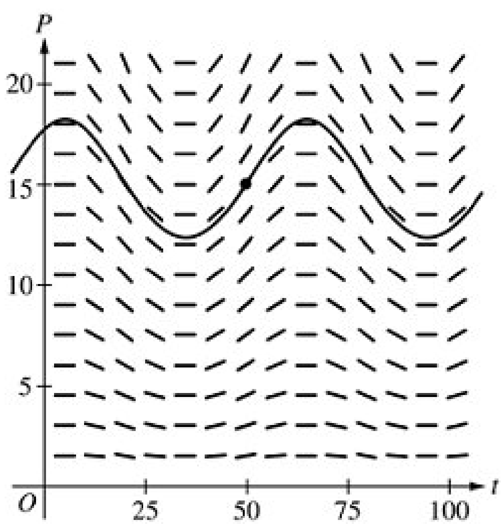
All local maximum/minimum points on the solution curve must occur at horizontal line segments in the slope field.

0	1

The student response accurately includes the criteria below.

- [ ] Solution curve through $(50, 15)$,

Solution:



Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The tangent line equation can be presented in any equivalent form and can be written in terms of any variables.

The second point can be earned with a tangent line through the point $(50, 15)$ with a slope equal to the declared value of $\left.\frac{dP}{dt}\right|_{(50,\ 15)}$.

0	1	2

The student response accurately includes **both** of the criteria below.

7-1-3

- ☐ Slope at $(50,\ 15)$
- ☐ Tangent line equation

Solution:

$f(50) = 15$

$f'(50) = \frac{dP}{dt}\Big|_{(50,15)} = 15 \cdot 0.02\cos\left(\frac{\pi}{30} \cdot 60\right) = 0.3$

An equation for the line tangent to the graph of $P = f(t)$ at time $t = 50$ is $y = 15 + 0.3(t - 50)$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The reason must include one of $f''(t) > 0$, $f'(t)$ is increasing, or $f(t)$ is concave up.

The point is not earned if the reason includes statements about the behavior of $f'(t)$ or $f''(t)$ outside the interval $47 < t < 51$.

0	1 ✓

The student response accurately includes the criteria below.

- ☐ Underestimate with reason

Solution:

The tangent line approximation of $f(48)$ is an underestimate because $f''(t) > 0$ on an interval that contains $t = 48$ and $t = 50$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with no separation of variables earns 0 out of 5 points.

A response that presents an antiderivative of without absolute value symbols is sufficient for the second point.

A response with no constant of integration can earn at most the first 3 points.

A response is eligible for the fourth point only if it has earned the first point and at least 1 of the 2 antiderivative points.

7-1-3

An eligible response earns the fourth point by correctly including the constant of integration in an equation and substituting 50 for t and P. A response is eligible for the fifth point only if it has earned the first 4 points.

0	1	2	3	4	5

The student response accurately includes **all five** of the criteria below.

- [ ] Separates variables
- [ ] Finds one antiderivative
- [ ] Finds the other antiderivative
- [ ] Constant of integration and uses initial condition
- [ ] Solves for P

Solution:

$$\int \frac{dP}{P} = \int 0.02\cos\left(\frac{\pi}{30}(t+10)\right)\, dt$$

$$\ln|P| = \frac{0.6}{\pi}\sin\left(\frac{\pi}{30}(t+10)\right) + C$$

$$\ln|15| = \frac{0.6}{\pi}\sin\left(\frac{\pi}{30}(50+10)\right) + C \Rightarrow C = \ln 15$$

Because $P(50) > 0$, $\ln|P| = \ln P$.

$$\ln P = \frac{0.6}{\pi}\sin\left(\frac{\pi}{30}(t+10)\right) + \ln 15$$

$$P = 15e^{\frac{0.6}{\pi}\sin\left(\frac{\pi}{30}(t+10)\right)}$$

7-1-3

62. A particle moves along the x-axis with position at time t given by $x(t) = e^{-t}\sin t$ for $0 \le t \le 2\pi$.

(a) Find the time t at which the particle is farthest to the left. Justify your answer.

(b) Find the value of the constant A for which $x(t)$ satisfies the equation $Ax''(t) + x'(t) + x(t) = 0$ for $0 < t < 2\pi$.

Part A

2 points are earned for $x'(t)$

1 point is earned for setting $x'(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

$x'(t) = -e^{-t}\sin t + e^{-t}\cos t = e^{-t}(\cos t - \sin t)$
$x'(t) = 0$ when $\cos t = \sin t$. Therefore, $x'(t) = 0$ on
$0 \le t \le 2\pi$ for $t = \frac{\pi}{4}$ and $t = \frac{5\pi}{4}$.

The candidates for the absolute minimum are at $t = 0, \frac{\pi}{4}, \frac{5\pi}{4}$, and 2π.

t	$x(t)$
0	$e^0\sin(0) = 0$
$\frac{\pi}{4}$	$e^{-\frac{\pi}{4}}\sin\left(\frac{\pi}{4}\right) > 0$
$\frac{5\pi}{4}$	$e^{-\frac{5\pi}{4}}\sin\left(\frac{5\pi}{4}\right) < 0$
2π	$e^{-2\pi}\sin(2\pi) = 0$

The particle is farthest to the left when $t = \frac{5\pi}{4}$.

0	1	2	3	4	5

7-1-3

The student response earns all of the following points:

2 points are earned for $x'(t)$

1 point is earned for setting $x'(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

$x'(t) = -e^{-t}\sin t + e^{-t}\cos t = e^{-t}(\cos t - \sin t)$
$x'(t) = 0$ when $\cos t = \sin t$. Therefore, $x'(t) = 0$ on
$0 \le t \le 2\pi$ for $t = \frac{\pi}{4}$ and $t = \frac{5\pi}{4}$.

The candidates for the absolute minimum are at $t = 0, \frac{\pi}{4}, \frac{5\pi}{4}$, and 2π.

t	$x(t)$
0	$e^0 \sin(0) = 0$
$\frac{\pi}{4}$	$e^{-\frac{\pi}{4}} \sin\left(\frac{\pi}{4}\right) > 0$
$\frac{5\pi}{4}$	$e^{-\frac{5\pi}{4}} \sin\left(\frac{5\pi}{4}\right) < 0$
2π	$e^{-2\pi} \sin(2\pi) = 0$

The particle is farthest to the left when $t = \frac{5\pi}{4}$.

Part B

2 points are earned for $x''(t)$

1 point is earned for substituting $x''(t)$, $x'(t)$, and $x(t)$ into $Ax''(t) + x'(t) + x(t)$

1 point is earned for the answer

7-1-3

$$
\begin{aligned}
x''(t) &= -e^{-t}(\cos t - \sin t) + e^{-t}(\sin t - \cos t) \\
&= -2e^{-t}\cos t
\end{aligned}
$$

$$
\begin{aligned}
&Ax''(t) + x'(t) + x(t) \\
&\quad = A\left(-2e^{-t}\cos t\right) + e^{-t}(\cos t - \sin t) + e^{-t}\sin t \\
&\quad = (-2A+1)e^{-t}\cos t \\
&\quad = 0
\end{aligned}
$$

Therefore, $A = \frac{1}{2}$.

0	1	2	3	4

The student response earns all of the following points:

2 points are earned for $x''(t)$

1 point is earned for substituting $x''(t)$, $x'(t)$, and $x(t)$ into $Ax''(t) + x'(t) + x(t)$

1 point is earned for the answer

$$
\begin{aligned}
x''(t) &= -e^{-t}(\cos t - \sin t) + e^{-t}(\sin t - \cos t) \\
&= -2e^{-t}\cos t
\end{aligned}
$$

$$
\begin{aligned}
&Ax''(t) + x'(t) + x(t) \\
&\quad = A\left(-2e^{-t}\cos t\right) + e^{-t}(\cos t - \sin t) + e^{-t}\sin t \\
&\quad = (-2A+1)e^{-t}\cos t \\
&\quad = 0
\end{aligned}
$$

Therefore, $A = \frac{1}{2}$.

7-1-3

63. The height of an object at time $t \geq 1$ is given by $h(t) = t^2 - \frac{16}{t} + 15$. What is the velocity of the object at time $t = 3$?

(A) 0.815

(B) 7.778 ✓

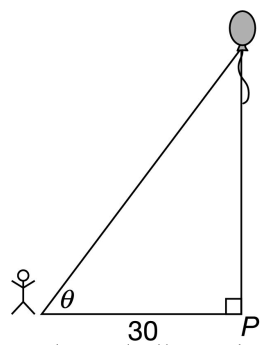
(C) 18.667

(D) 21.089

Answer B

Correct. The velocity is the derivative of the height. Using the calculator, $v(3) = h'(3) = 7.778$.

64.



A person stands 30 feet from point P and watches a balloon rise vertically from the point, as shown in the figure above. The balloon is rising at a constant rate of 2 feet per second.

What is the rate of change, in radians per second, of angle θ at the instant when the balloon is 40 feet above point P ?

(A) $\frac{3}{100}$

(B) $\frac{3}{125}$ ✓

(C) $\frac{1}{12}$

(D) $\frac{5}{27}$

Answer B

Correct. If $h(t)$ is the height of the balloon above the point P at time t, then $\tan\theta = \frac{h}{30}$. Using implicit differentiation with respect to t shows that $\sec^2\theta \frac{d\theta}{dt} = \frac{1}{30}\frac{dh}{dt}$. At the instant when $h = 40$, the distance from the person to the balloon is $\sqrt{30^2 + 40^2} = \sqrt{2500} = 50$. At this instant, $\sec\theta = \frac{50}{30} = \frac{5}{3}$ and therefore $\left(\frac{5}{3}\right)^2 \frac{d\theta}{dt} = \frac{1}{30} \cdot 2 = \frac{1}{15} \Rightarrow \frac{d\theta}{dt} = \frac{1}{15} \cdot \frac{9}{25} = \frac{3}{125}$.

7-1-3

65. A rectangle has length x, width y, and area A. The instantaneous rate of change of the area with respect to the width is equal to the sum of the length and twice the width. Which of the following equations describes this relationship?

(A) $A = x + 2y$

(B) $\frac{dy}{dx} = x + 2y$

(C) $\frac{dA}{dx} = x + 2y$

(D) $\frac{dA}{dy} = x + 2y$ ✓

Answer D

Correct. The instantaneous rate of change of the area A with respect to the width y is given by the derivative $\frac{dA}{dy}$. The sum of the length x and twice the width y is $x + 2y$. Therefore the relationship is described by the equation $\frac{dA}{dy} = x + 2y$.

66. A population described by a function P at time t decreases at a rate proportional to P. Which of the following differential equations could describe the rate of change of the population?

(A) $\frac{dP}{dt} = -0.015P$ ✓

(B) $\frac{dP}{dt} = -\frac{0.015}{P}$

(C) $\frac{dP}{dt} = -0.02t$

(D) $\frac{dP}{dt} = 6e^{-0.02t}$

Answer A

Correct. The model for exponential growth and decay that arises from the statement “The rate of change of a quantity is proportional to the size of the quantity” is $\frac{dy}{dt} = ky$. Here the quantity is the size P of the population. Since the population is decreasing, the constant k would be negative. This differential equation matches the conditions for exponential decay with $k = -0.015$.

7-1-3

67. The rate of change of the volume V with respect to time t of water leaking from a tank is proportional to the cube of the volume. Which of the following is a differential equation that could describe this relationship?

(A) $\frac{dV}{dt} = -1.4t^3$

(B) $\frac{dV}{dt} = -0.2V^3$ ✓

(C) $\frac{dV}{dt} = 0.75t^3$

(D) $\frac{dV}{dt} = 0.15V^3$

Answer B

Correct. The rate of change of the volume with respect to time is given by $\frac{dV}{dt}$. Since the rate of change is proportional to the cube of the volume, $\frac{dV}{dt} = kV^3$ where k is a constant. The tank is leaking so the volume of water in the tank will be decreasing. Therefore $\frac{dV}{dt} < 0$, which means that the proportionality constant k should be negative.

68. A scientist is studying the relationship of two quantities S and T in an experiment. The scientist finds that under certain conditions, as the quantity of S increases, the quantity of T decreases. After taking measurements, the scientist determines that the rate of change of the quantity of S with respect to the quantity of T present is inversely proportional to the natural logarithm of the quantity of T. Which of the following is a differential equation that could describe this relationship?

(A) $S = \frac{k}{\ln T}$, where k is a nonzero constant.

(B) $\frac{dS}{dT} = \frac{\ln T}{k}$, where k is a nonzero constant.

(C) $\frac{dS}{dT} = \frac{k}{\ln T}$, where k is a nonzero constant. ✓

(D) $\frac{dT}{dS} = \frac{k}{\ln S}$, where k is a nonzero constant.

Answer C

Correct. The rate of change of the quantity of S with respect to the quantity of T is given by the derivative $\frac{dS}{dT}$. Since the rate of change of the quantity of S with respect to the quantity of T present is inversely proportional to the natural logarithm of the quantity of T, $\frac{dS}{dT} = \frac{k}{\ln T}$, where k is a nonzero constant.

7-1-3

69. A vehicle moves along a straight road. The vehicle's position is given by $f(t)$, where t is measured in seconds since the vehicle starts moving. During the first 10 seconds of the motion, the vehicle's acceleration is proportional to the cube root of the time since the start. Which of the following differential equations describes this relationship, where k is a positive constant?

(A) $\frac{df}{dt} = k\sqrt[3]{t}$

(B) $\frac{df}{dt} = k\sqrt[3]{f}$

(C) $\frac{d^2f}{dt^2} = k\sqrt[3]{t}$ ✓

(D) $\frac{d^2f}{dt^2} = k\sqrt[3]{f}$

Answer C

Correct. The vehicle's acceleration is given by the second derivative of position, $\frac{d^2f}{dt^2}$.

70. A hard-boiled egg is removed from a pot of hot water and set on the table to cool. The rate of change of the egg's temperature T with respect to time t is proportional to the difference between the egg's temperature in degrees Fahrenheit ($^\circ\mathrm{F}$) and the room temperature of $75^\circ\mathrm{F}$. Which of the following is a differential equation that describes this relationship, where k is a constant?

(A) $\frac{dT}{dt} = kT$

(B) $\frac{dT}{dt} = kT - 75$

(C) $\frac{dT}{dt} = k(t - 75)$

(D) $\frac{dT}{dt} = k(T - 75)$ ✓

Answer D

Correct. The rate of change of the egg's temperature with respect to time is given by the derivative $\frac{dT}{dt}$. The difference between the egg's temperature and the room temperature of 75 is $T - 75$. Therefore, $\frac{dT}{dt} = k(T - 75)$, where k is a constant. (The constant k would be negative because when the egg's temperature is greater than the room temperature, the egg will be cooling so $\frac{dT}{dt}$ will be negative.)

7-1-3

71. A jogger runs along a straight track. The jogger's position is given by the function $p(t)$, where t is measured in minutes since the start of the run. During the first minute of the run, the jogger's acceleration is proportional to the square root of the time since the start of the run. Which of the following is a differential equation that describes this relationship, where k is a positive constant?

(A) $\frac{dp}{dt} = k\sqrt{t}$

(B) $\frac{dp}{dt} = k\sqrt{p}$

(C) $\frac{d^2p}{dt^2} = k\sqrt{t}$ ✓

(D) $\frac{d^2p}{dt^2} = k\sqrt{p}$

Answer C

Correct. The jogger's acceleration is given by the second derivative of position, $\frac{d^2p}{dt^2}$.

72. The weight of a population of yeast is given by a differentiable function, W, where $W(t)$ is measured in grams and t is measured in hours. The weight of the yeast population increases with respect to t at a rate that is directly proportional to the weight. At time $t = 10$ hours, the weight of the yeast is 200 grams and is increasing at the rate of 5 grams per hour. Which of the following is a differential equation that models this situation?

(A) $W = 5(t - 10) + 200$

(B) $\frac{dW}{dt} = \frac{1}{2}t$

(C) $\frac{dW}{dt} = \frac{1}{40}W$ ✓

(D) $\frac{dW}{dt} = \frac{1}{2}W$

Answer C

Correct. The rate of change of W with respect to t is proportional to W, so the differential equation is $\frac{dW}{dt} = kW$, where k is a constant.
$\frac{dW}{dt} = 5$ when $W = 200$ implies that $5 = 200k$, so $k = \frac{5}{200} = \frac{1}{40}$.

7-1-3

73. If the pressure P applied to a gas is increased while the gas is held at a constant temperature, then the volume V of the gas will decrease. The rate of change of the volume of gas with respect to the pressure is proportional to the reciprocal of the square of the pressure. Which of the following is a differential equation that could describe this relationship?

(A) $V = \frac{k}{P^2}$, where k is a positive constant.

(B) $\frac{dV}{dP} = \frac{k}{P^2}$, where k is a positive constant.

(C) $\frac{dV}{dP} = \frac{k}{P^2}$, where k is a negative constant. ✓

(D) $\frac{dP}{dV} = \frac{k}{P^2}$, where k is a negative constant.

Answer C

Correct. The rate of change of the volume of the gas with respect to pressure is given by the derivative $\frac{dV}{dP}$. Since the volume will decrease if the pressure is increased, the derivative should be negative. Therefore, k should be a negative constant.

74. The population of a certain species of insect is given by a differentiable function P, where $P(t)$ is the number of insects in the population, in millions, at time t, where t is measured in days. When the environmental conditions are right, the population increases with respect to time at a rate that is directly proportional to the population. Starting August 15, the conditions were favorable and the population began increasing. On August 20, five days later, there were an estimated 10 million insects and the population was increasing at a rate of 2 million insects per day. Which of the following is a differential equation that models this situation?

(A) $P = 2(t-5) + 10$

(B) $\frac{dP}{dt} = \frac{2}{5}t$

(C) $\frac{dP}{dt} = \frac{1}{5}P$ ✓

(D) $\frac{dP}{dt} = 5P$

Answer C

Correct. The rate of change of P with respect to t is proportional to P, so the differential equation is $\frac{dP}{dt} = kP$, where k is a constant. $\frac{dP}{dt} = 2$ when $P = 10$, implies $2 = 10k$, so $k = \frac{2}{10} = \frac{1}{5}$.

7-1-3

75. The rate at which a quantity M of a certain radioactive substance decays is proportional to the amount of the substance present at a given time. Which of the following is a differential equation that could describe this relationship?

(A) $\frac{dM}{dt} = -3.7t^2$

(B) $\frac{dM}{dt} = -0.11M$ ✓

(C) $\frac{dM}{dt} = 0.08t^2$

(D) $\frac{dM}{dt} = 1.2M$

Answer B

Correct. The rate of change of the quantity of the substance is given by $\frac{dM}{dt}$. Since the rate of change is proportional to the amount, $\frac{dM}{dt} = kM$, where k is a constant. The substance is decaying, so the amount will be decreasing. Therefore $\frac{dM}{dt} < 0$, which means that the proportionality constant k should be negative.

76. Students are asked to memorize a list of 100 words. The students are given periodic quizzes to see how many words they have memorized. The function L gives the number of words memorized at time t. The rate of change of the number of words memorized is proportional to the number of words left to be memorized. Which of the following differential equations could be used to model this situation, where k is a positive constant?

(A) $\frac{dL}{dt} = kL$

(B) $\frac{dL}{dt} = 100 - kL$

(C) $\frac{dL}{dt} = k(100 - L)$ ✓

(D) $\frac{dL}{dt} = kL - 100$

Answer C

Correct. The rate of change of the number of words memorized is $\frac{dL}{dt}$. The number of words left to be memorized at time t is $100 - L$. According to the model, the rate of change of the number of words memorized is positive and is proportional to the number of words left to be memorized, which is nonnegative. Therefore $\frac{dL}{dt} = k(100 - L)$, where k is a positive constant, is a model of the situation.

7-1-3

77. Ana opened a bank account with $\$1000$ that earns interest, compounded continuously, at an annual rate of $r = 0.02$. Money can be withdrawn from the account at regular intervals, and no additions to the account can be made. The function P models the balance of the account, in dollars, at time t. Assume that Ana withdraws money from the account continuously at a rate of N dollars per year. Which of the following differential equations best describes the relationship between P and t ?

(A) $\frac{dP}{dt} = 0.02P$

(B) $\frac{dP}{dt} = 0.02P + 1000$

(C) $\frac{dP}{dt} = 0.02\left(P - N\right)$

(D) $\frac{dP}{dt} = 0.02P - N$ ✓

Answer D

Correct. The rate of change of the balance of the account is given by the derivative $\frac{dP}{dt}$. If there are no additions or withdrawals, the rate of change of the balance is $\frac{dP}{dt} = 0.02P$. Continuously withdrawing N dollars per year gives $\frac{dP}{dt} = 0.02P - N$.

78. Out of a total of N students at a school, the number of students who have seen a new television program increases at a rate proportional to the product of the number of students who have seen the program and the number of students who have not seen the program. If S denotes the number of students who have seen the program at time t, which of the following differential equations could be used to model this situation, where k is a positive constant?

(A) $\frac{dS}{dt} = kS$

(B) $\frac{dS}{dt} = kt\left(N - t\right)$

(C) $\frac{dS}{dt} = kS\left(N - S\right)$ ✓

(D) $\frac{dS}{dt} = kS\left(S - N\right)$

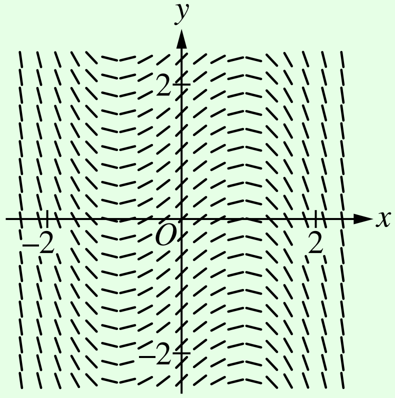
Answer C

Correct. The number of students who have seen the program is S, and the number of students who have not seen the program is $N - S$. Since the rate of increase of S is proportional to the product of S and $N - S$, the differential equation $\frac{dS}{dt} = kS\left(N - S\right)$ models this situation.

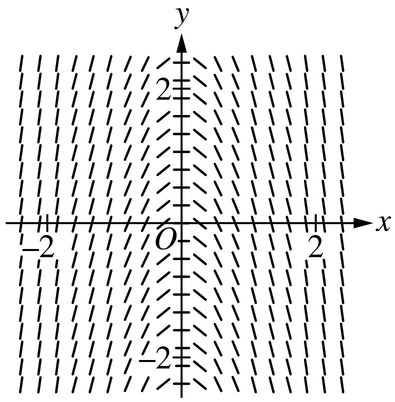
7-1-3

79. Which of the following is a slope field for the differential equation $\frac{dy}{dx} = 1 - x^2$?

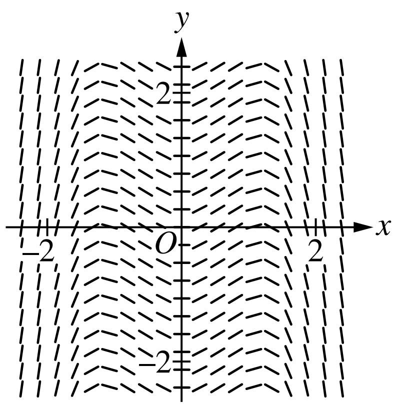
(A)



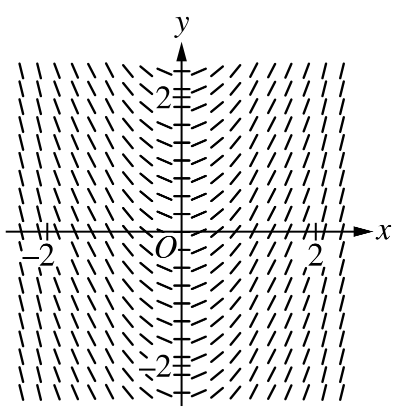
(B)



(C)



(D)

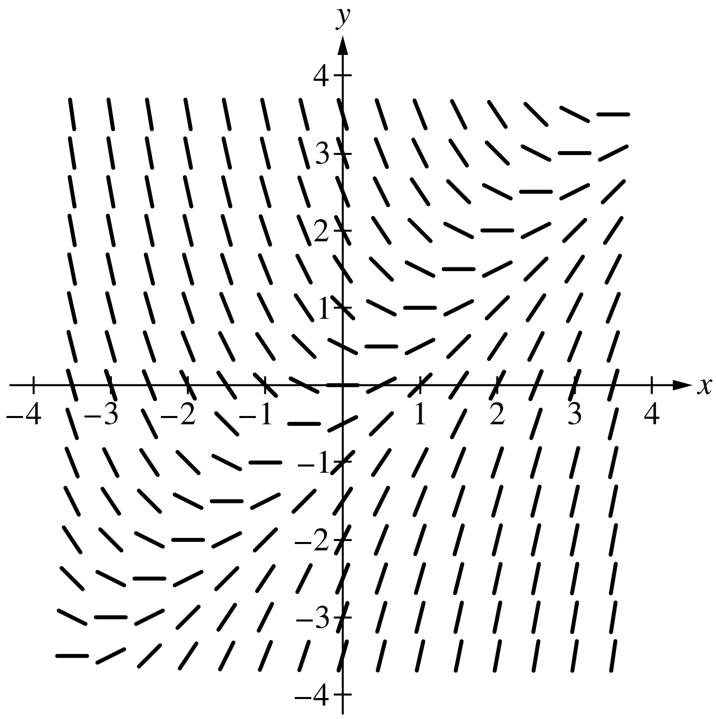


7-1-3

Answer A

Correct. The line segments in the slope field have slopes given by $\frac{dy}{dx} = 1 - x^2$ at the point (x, y). The slopes should be positive for all points (x, y) with $-1 < x < 1$. In particular, the line segments along the y-axis should all have slope 1. This is the only option in which all those conditions are true.

80.



A portion of the slope field for the differential equation $\frac{dy}{dx} = x - y$ is shown in the figure. If the linear function $y = f(x)$ is the solution to the differential equation such that $f(-1) = -2$, then $f(3) =$

(A) 0

(B) 1

(C) 2

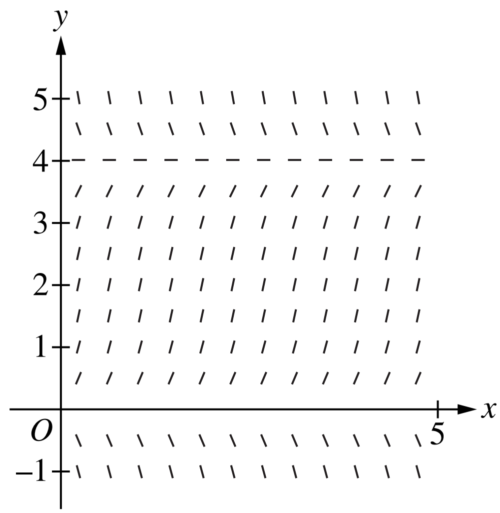
(D) 3

Answer C

Correct. To estimate the value of $f(3)$ graphically, sketch the graph of the solution in the slope field starting at $(-1, -2)$. Since the solution is linear, it follows the slope lines that have the same slope as the slope line at $(-1, -2)$. At $x = 3$ the solution curve would be at $y = 2$.

7-1-3

81.



The slope field shown is for the differential equation $\frac{dy}{dx} = ky - 2y^2$, where k is a constant. What is the value of k ?

(A) 2

(B) 4

(C) 6

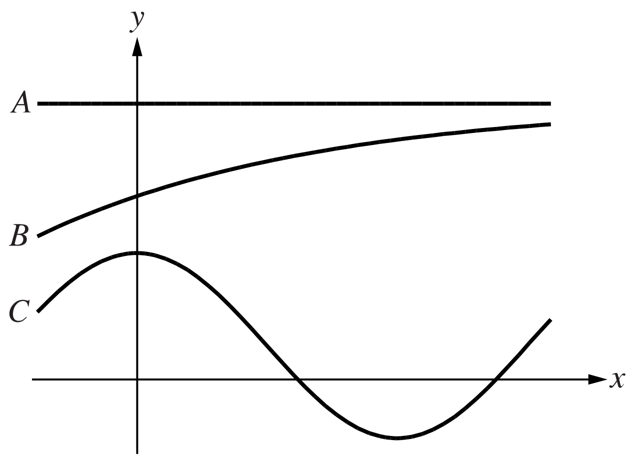
(D) 8

Answer D

Correct. The slope field indicates that $\frac{dy}{dx} = 0$ for points (x, y) on the horizontal line $y = 4$ for $0 \leq x \leq 5$. Therefore, when $y = 4$, $\frac{dy}{dx} = 4k - 32 = 0$ and so $k = 8$.

7-1-3

82.



Of the graphs shown, which could be the graph of a solution to a differential equation whose slope field has slopes that depend only on the y-coordinate at each point?

(A) Graph A only

(B) Graph B only

(C) Graphs A and B only

(D) Graphs B and C only

Answer C

Correct. If a horizontal line intersects the graph of a solution to such a differential equation in two different points, then the graph must have the same slope at each intersection point.

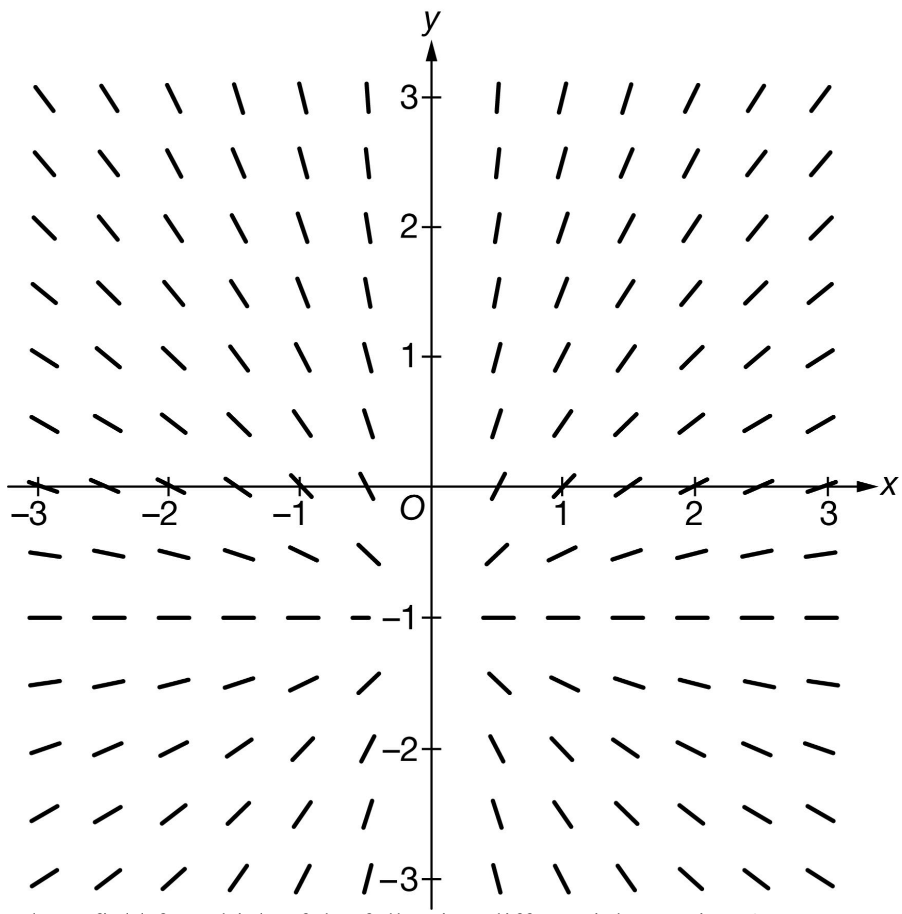
This is true for Graph A, where the slope at every point on the graph is 0, so graph A could be the solution of such a differential equation. Graph A could, for example, be a solution to a logistic differential equation of the form $\frac{dy}{dx} = ky(A - y)$ with initial condition $y = A$ at $x = 0$, where A is the carrying capacity.

Graph B is the graph of an increasing function, so a horizontal line cannot intersect the graph in more than one point. Therefore, Graph B could be the solution of such a differential equation. This could, for example, be the graph of a solution to a logistic differential equation $\frac{dy}{dx} = ky(A - y)$ with an initial condition at $x = 0$ between $y = \frac{A}{2}$ and $y = A$, where A is the carrying capacity.

There are horizontal lines that intersect Graph C in two points, where one slope is positive and the other slope is negative. Therefore Graph C could not be the solution to a differential equation whose slope field has slopes that depend only on the y-coordinate at each point.

7-1-3

83.



Shown above is a slope field for which of the following differential equations?

(A) $\frac{dy}{dx} = \frac{y+1}{x}$

(B) $\frac{dy}{dx} = \frac{-(y+1)}{x}$

(C) $\frac{dy}{dx} = \frac{y-1}{x}$

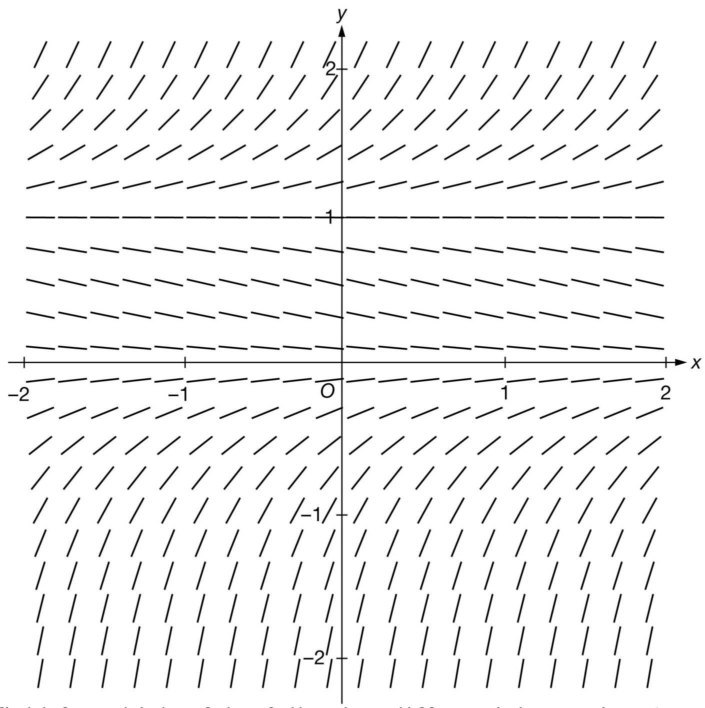
(D) $\frac{dy}{dx} = \frac{-(y-1)}{x}$

Answer A

Correct. For this differential equation, the line segments at points (x, y) will have positive slopes when $x > 0$ and $y > -1$, or when $x < 0$ and $y < -1$. The line segments at points (x, y) will have negative slopes when $x > 0$ and $y < -1$, or when $x < 0$ and $y > -1$. The line segments for points (x, y) on the line $y = -1$ will be horizontal. The given slope field displays this behavior.

7-1-3

84.



Shown above is a slope field for which of the following differential equations?

(A) $\frac{dy}{dx} = y^2$

(B) $\frac{dy}{dx} = y - 1$

(C) $\frac{dy}{dx} = y^2 + y$

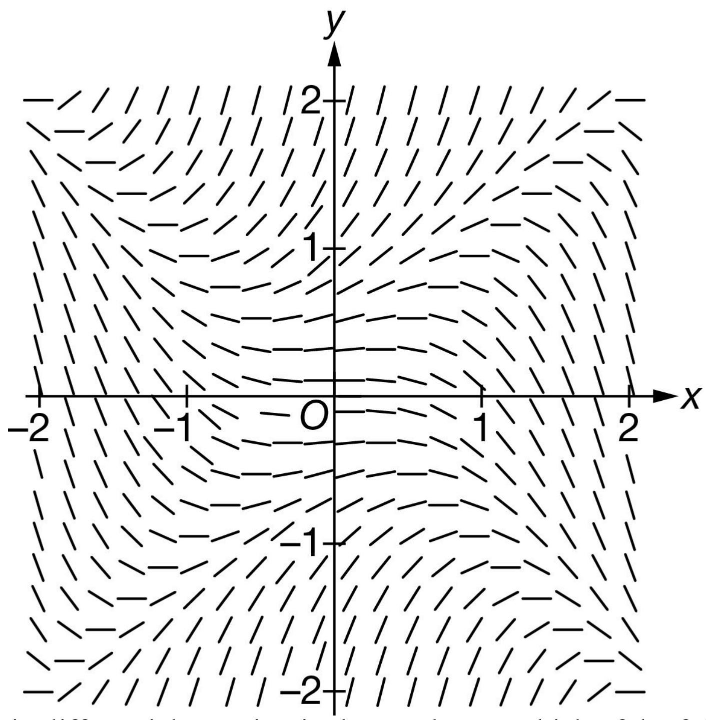
(D) $\frac{dy}{dx} = y^2 - y$

Answer D

Correct. For this differential equation, the slopes for the line segments at points on the x-axis and the line $y = 1$ will be zero. At points where $0 < y < 1$, the slopes of the line segments will be negative. At points where $y < 0$ or $y > 1$, the slopes of the line segments will be positive. The given slope field represents this behavior.

7-1-3

85.



The slope field for a certain differential equation is shown above. Which of the following statements about a solution $y = f(x)$ to the differential equation must be false?

(A) The graph of the particular solution that satisfies $f(1) = -1$ is concave down on the interval $0 < x < 2$.

(B) The graph of the particular solution that satisfies $f(-1) = 1$ is concave up on the interval $-1 < x < 0$.

(C) The graph of the particular solution that satisfies $f(0) = 0$ is concave up on the interval $-1 < x < 1$.

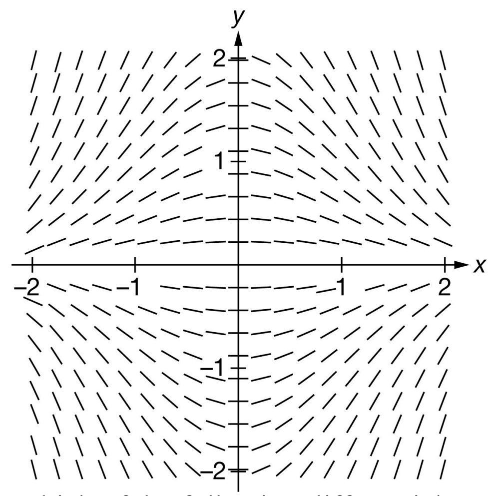
(D) The graph of the particular solution that satisfies $f(0) = 0$ has a point of inflection at $x = 0$.

Answer C

Correct. Using the slope field to estimate the particular solution to the differential equation through the point $(0, 0)$ indicates that the graph there is decreasing for $x < 0$ and decreasing for $x > 0$, where $-1 < x < 1$, since the slopes of the line segments are negative for $x < 0$ and negative for $x > 0$. If the graph of the particular solution was concave up on the interval, the slopes of the line segments would change from negative to positive near 0. Therefore, the graph of this particular solution cannot be concave up on the interval.

7-1-3

86.



Shown above is the slope field for which of the following differential equations?

(A) $\frac{dy}{dx} = -(x + y)$

(B) $\frac{dy}{dx} = -xy$

(C) $\frac{dy}{dx} = -x^2y$

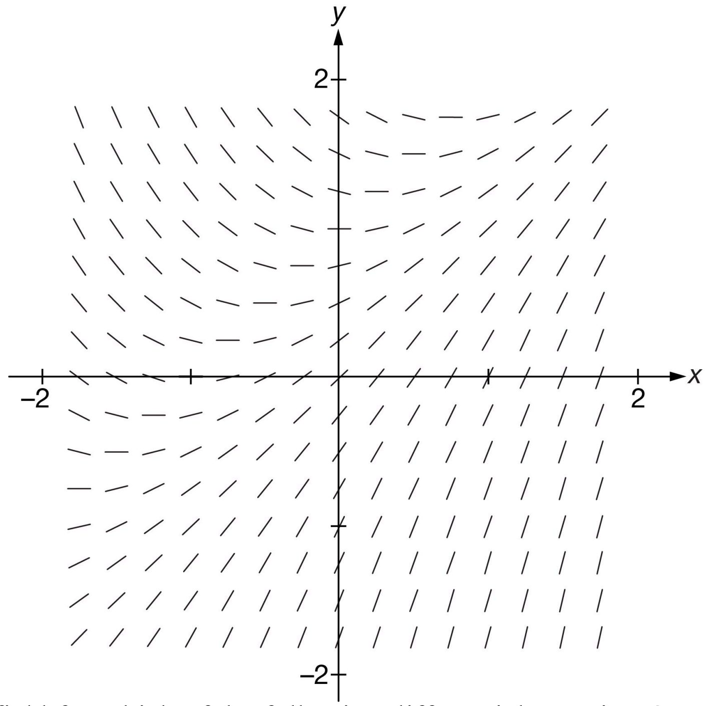
(D) $\frac{dy}{dx} = -\frac{x}{y}$

Answer B

Correct. For this differential equation, the line segments at points (x, y) in Quadrants I and III will have negative slopes, and the line segments at points (x, y) in Quadrants II and IV will have positive slopes. The slopes for points on the x-axis and for points on the y-axis will be 0. The given slope field represents this behavior.

7-1-3

87.



Shown above is a slope field for which of the following differential equations?

(A) $\frac{dy}{dx} = x + y$

(B) $\frac{dy}{dx} = xy - 1$

(C) $\frac{dy}{dx} = 1 - x - y$

(D) $\frac{dy}{dx} = x - y + 1$

Answer D

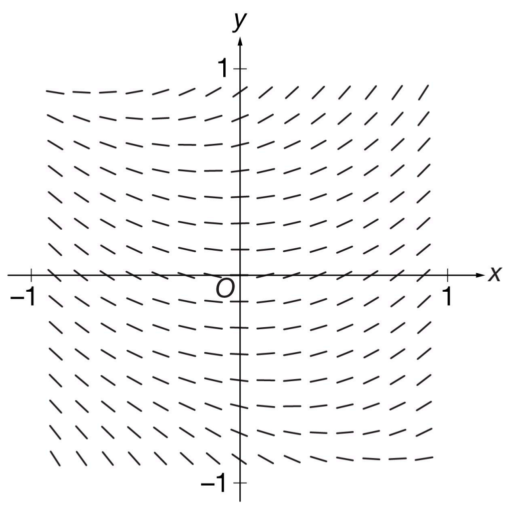
Correct. In a slope field for $\frac{dy}{dx} = x - y + 1$, all line segments at points on the diagonal line $y = x$ would have slope 1. Also, all line segments at points on the line $y = x + 1$ would have slope 0. All line segments at points below the line $y = x + 1$ would have positive slopes, and all line segments at points above the line $y = x + 1$ would have negative slopes. All of these conditions are satisfied by the line segments shown in this slope field. No inconsistencies can be found as with the differential equations in the other options. Therefore, this is the best of the options given.

7-1-3

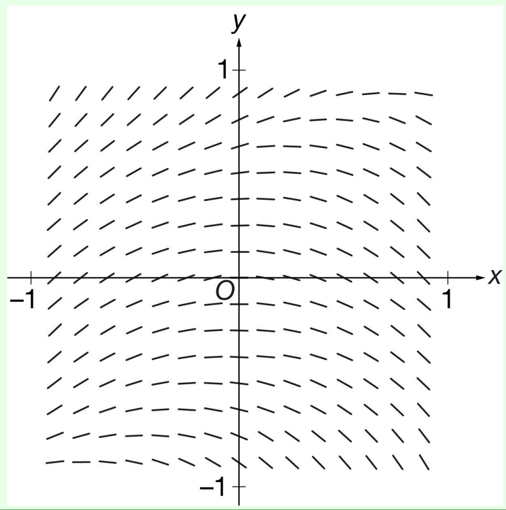
88. Which of the following could be a slope field for the differential equation $\frac{dy}{dx} = y^3 - x$?

7-1-3

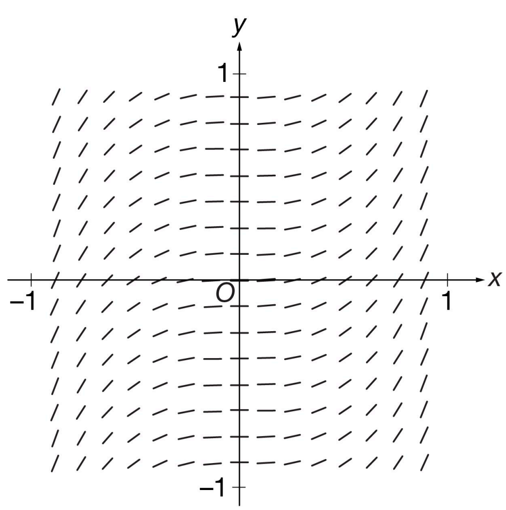
(A)



(B)

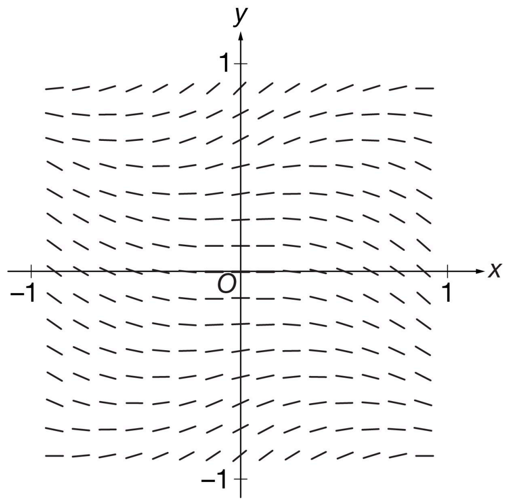


(C)



7-1-3

(D)

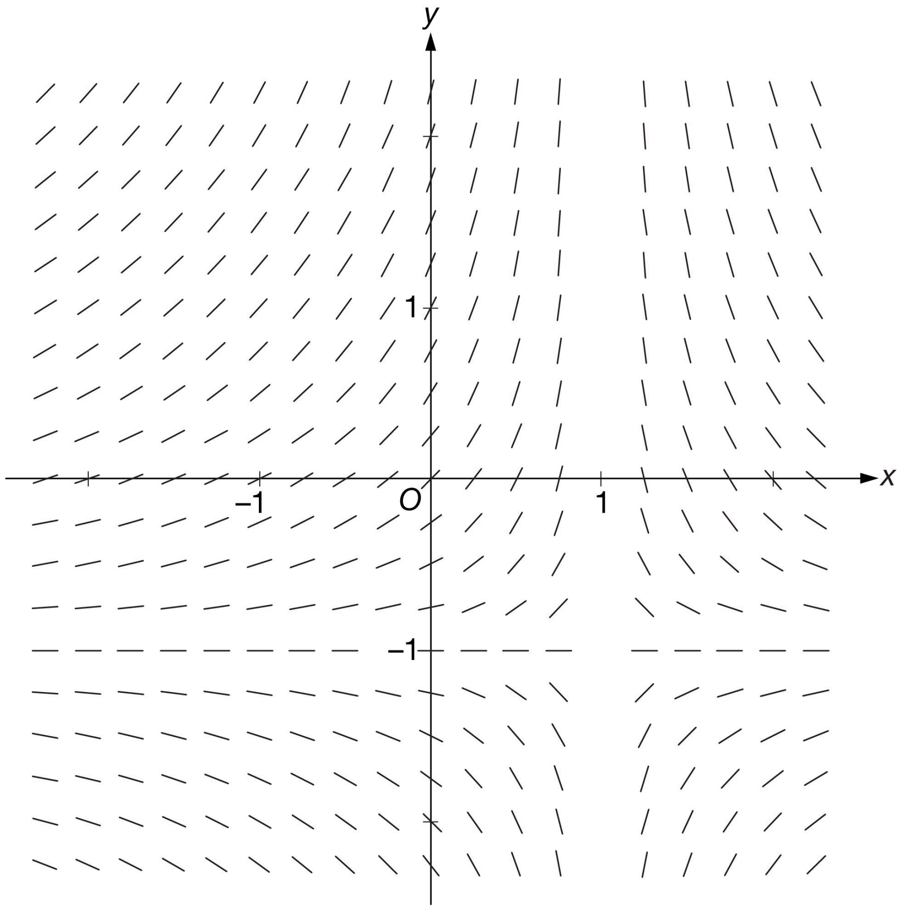


Answer B

Correct. The line segments in the slope field have slopes given by $\frac{dy}{dx} = y^3 - x$ at the point (x, y). In Quadrant II, all slopes must be positive, since $y^3 > 0$ and $x < 0$. In Quadrant IV, all slopes must be negative, since $y^3 < 0$ and $x > 0$. All line segments along the negative x-axis should have positive slope, and all line segments along the positive x-axis should have negative slope. This is the only option in which all of those conditions are true.

7-1-3

89.



The slope field for a certain differential equation is shown above. Which of the following could be a solution to the differential equation with initial condition $y(0) = 0$?

(A) $y = \ln|1 - x|$

(B) $y = e^{-x^2} - 1$

(C) $y = \frac{x}{1-x}$

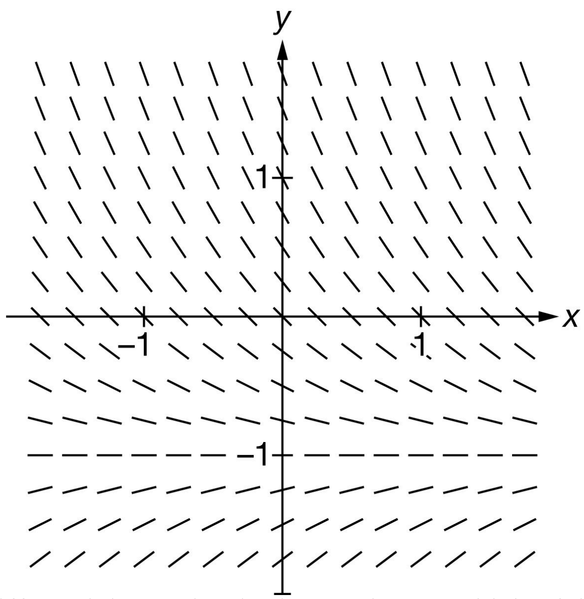
(D) $y = \frac{x^2}{1-x^2}$

Answer C

Correct. The slope field indicates that the slopes are undefined at $x = 1$. A solution with initial condition $y(0) = 0$ appears to approach a vertical asymptote as x approaches 1 from the left. The solution also appears to approach a horizontal asymptote at $y = -1$ as x goes towards $-\infty$. The graph of this function has both of those properties.

7-1-3

90.



The slope field for a certain differential equation is shown above. Which of the following could be the particular solution to the differential equation satisfying $y(0) = 0$?

(A) $y = x^2 - x$

(B) $y = -\sin x$

(C) $y = e^{-x} - 1$

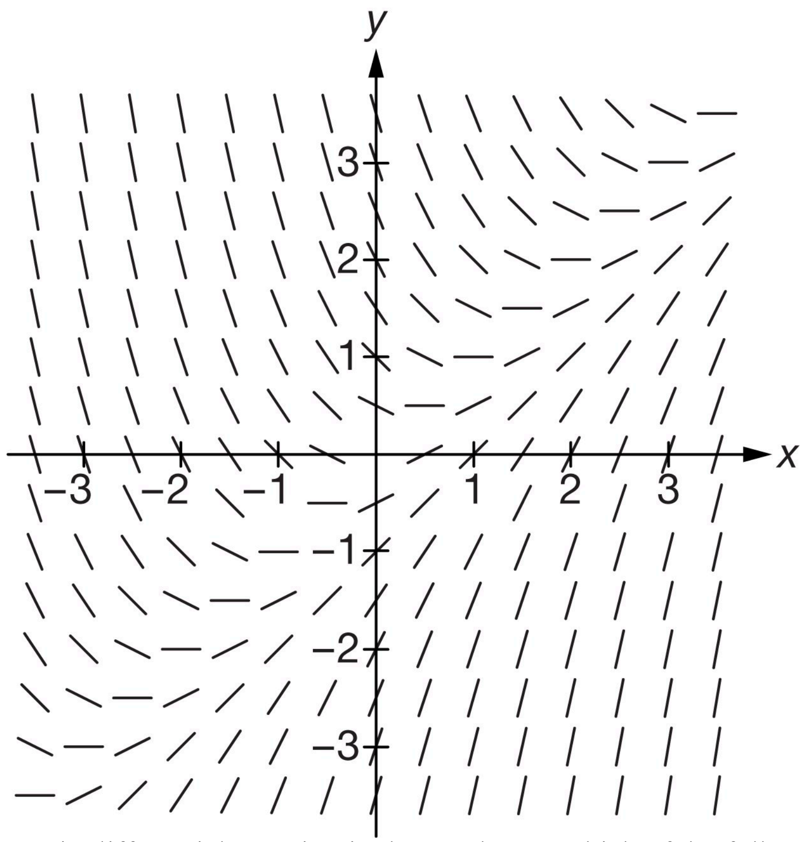
(D) $y = e^{-x^2} - 1$

Answer C

Correct. The slope field suggests that the particular solution through the origin approaches a horizontal asymptote at $y = -1$ as x increases and that the solution continues to increase toward infinity as x decreases in Quadrant II. That behavior is consistent with the graph of $y = e^{-x} - 1$.

7-1-3

91.



The slope field for a certain differential equation is shown above. Which of the following statements about a solution $y = f(x)$ to the differential equation must be false?

(A) The graph of the particular solution that satisfies $f(-3) = 2$ is concave up on the interval $-3 < x < 3$.

(B) The graph of the particular solution with $f(-2) = -2$ has a relative minimum at $x = -2$.

(C) The graph of the particular solution that satisfies $f(0) = -2$ is concave up on the interval $-1 < x < 3$.

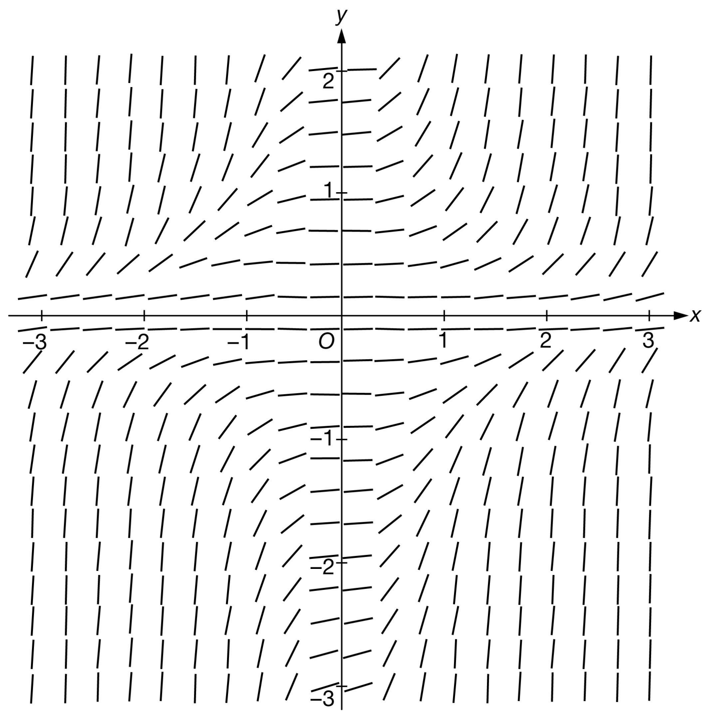
(D) The graph of the particular solution that satisfies $f(1) = 0$ is linear.

Answer C

Correct. Using the slope field to estimate the particular solution to the differential equation through the point $(0, -2)$ indicates a graph that is concave down. The line segments in this portion of the slope field have positive slopes that are decreasing as the line segments get less steep along the estimated graph of the particular solution. If the graph of the particular solution was concave up on the interval $-1 < x < 3$, the slopes would have to be increasing, not decreasing. Therefore the graph of this particular solution cannot be concave up on that interval.

7-1-3

92.



Shown above is a slope field for which of the following differential equations?

(A) $\frac{dy}{dx} = x^2 + y^2$

(B) $\frac{dy}{dx} = x^2y^2$

(C) $\frac{dy}{dx} = x^2 - y^2$

(D) $\frac{dy}{dx} = y^2$

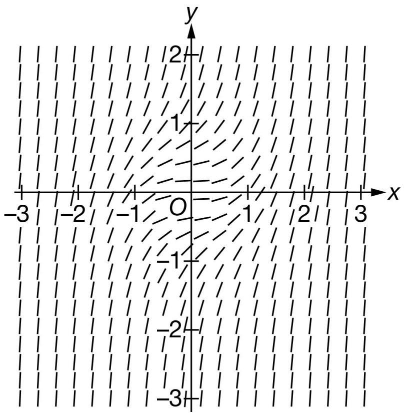
Answer B

Correct. For this differential equation, the slopes for line segments at points on the x-axis and y-axis will be zero. The slopes of line segments at points in each quadrant will be positive. The given slope field represents this behavior.

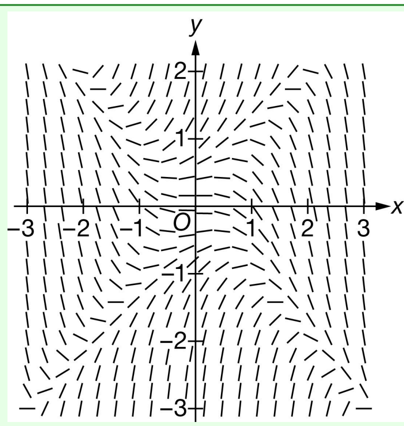
7-1-3

93. Which of the following could be a slope field for the differential equation $\frac{dy}{dx} = y^2 - x^2$?

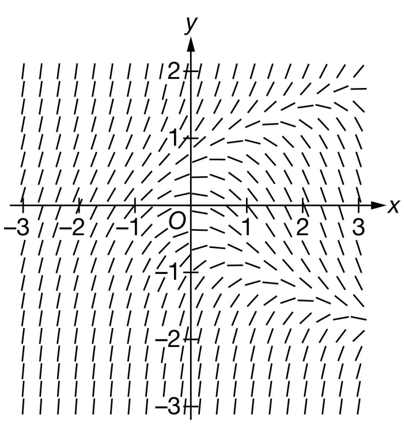
(A)



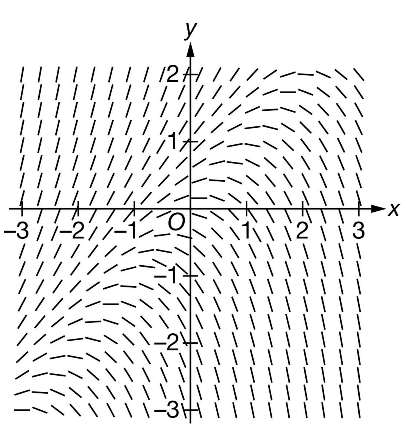
(B)



(C)



(D)

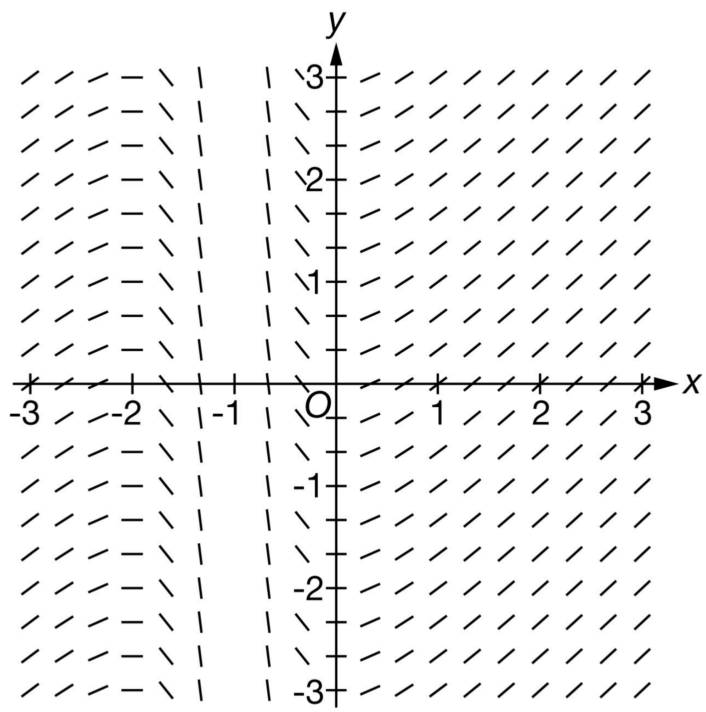


7-1-3

Answer B

Correct. The line segments in the slope field have slopes given by $\frac{dy}{dx} = y^2 - x^2$ at the points (x, y). Since $y^2 - x^2 = (y + x)(y - x)$, at points on the lines $y = x$ and $y = -x$, the slopes of the line segments will be zero. At points on the y-axis, where $x = 0$, the slopes of the line segments will be positive. At points on the x-axis, where $y = 0$, the slopes of the line segments will be negative. This is the only option in which all of these conditions are true.

94.



The slope field for a certain differential equation is shown above. Which of the following could be a solution to the differential equation with the initial condition $y(0) = 0$?

(A) $y = -e^{x+1}$

(B) $y = \frac{1}{x^2-1}$

(C) $y = \frac{x^2}{1+x}$

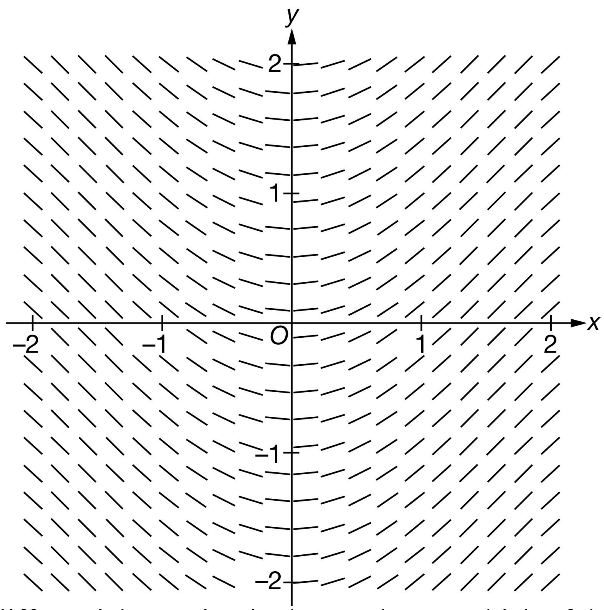
(D) $y = \frac{\tan x}{1+x}$

Answer C

Correct. Taking the derivative, the slope is $\frac{dy}{dx} = \frac{2x+x^2}{(1+x)^2} = \frac{(1+x)^2-1}{(1+x)^2} = 1 - \frac{1}{(1+x)^2}$. The slope field indicates that the slopes are undefined at $x = -1$. A solution with initial condition $y(0) = 0$ approaches a vertical asymptote as x approaches -1. As $x \to \infty$ the slopes of the line segments are positive, and as $x \to -\infty$ the slopes are also positive.

7-1-3

95.



The slope field for a certain differential equation is shown above. Which of the following could be the particular solution to the differential equation satisfying $y(0) = 0$?

(A) $y = -\cos x + 1$

(B) $y = e^{x^2} - 1$

(C) $y = \sin x$

(D) $y = \tan x$

Answer A

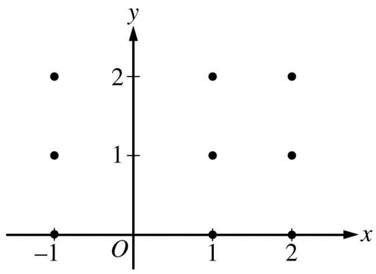
Correct. The slope field suggests that the particular solution through the origin has negative slopes for the line segments when $-2 < x < 0$, zero slope for the line segments when $x = 0$, and positive slopes for the line segments when $0 < x < 2$. That behavior is consistent with the solution to $y' = \sin x$, which is $y = -\cos x + C$ when $-2 < x < 2$. Using the initial condition $(0, 0)$ gives $y = -\cos x + 1$.

7-1-3

96. Consider the differential equation $\frac{dy}{dx} = \left(1 - \frac{2}{x^2}\right)(y - 1)$, where $x \neq 0$. Let $y = f(x)$ be the particular solution to the differential equation with initial condition $f(1) = 2$.

(a) Find the slope of the line tangent to the graph of f at the point $(1, 2)$.

(b) On the axes provided, sketch a slope field for the given differential equation at the nine points indicated.



(c) Find the particular solution $y = f(x)$ to the differential equation $\frac{dy}{dx} = \left(1 - \frac{2}{x^2}\right)(y - 1)$ with initial condition $f(1) = 2$.

Part A

1 point is earned for the correct answer

$$\left.\frac{dy}{dx}\right|_{(x,y)=(1,2)} = \left(1 - \frac{2}{1}\right)(2 - 1) = -1$$

0	1

The student earns all of the following point:

1 point is earned for the correct answer

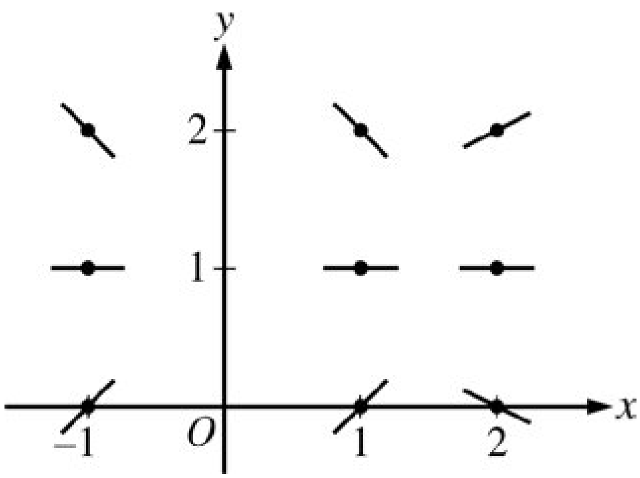
$$\left.\frac{dy}{dx}\right|_{(x,y)=(1,2)} = \left(1 - \frac{2}{1}\right)(2 - 1) = -1$$

Part B

1 point is earned for zero slope at each point (x, y) where $y = 1$

1 point is earned for the remaining slopes

7-1-3

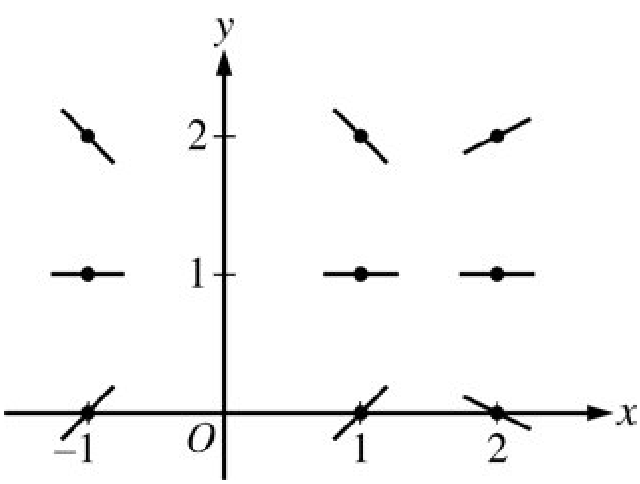


0	1	2

The student earns all of the following points:

1 point is earned for zero slope at each point (x, y) where $y = 1$

1 point is earned for the remaining slopes



Part C

1 point is earned for the separation of variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for uses of the initial condition

1 point is earned for the solution for y

7-1-3

$$\frac{dy}{dx} = \left(1 - \frac{2}{x^2}\right)(y-1)$$
$$\int \frac{dy}{y-1} = \int \left(1 - \frac{2}{x^2}\right) dx$$
$$\ln|y-1| = x + \frac{2}{x} + C$$
$$\ln|2-1| = 1 + \frac{2}{1} + C \Rightarrow C = -3$$
$$\ln|y-1| = x + \frac{2}{x} - 3$$

Note that $y - 1 > 0$ since the solution curve includes the point $(1, 2)$.

$$\ln(y-1) = x + \frac{2}{x} - 3$$
$$y = f(x) = e^{\left(x + \frac{2}{x} - 3\right)} + 1$$

Note: This solution is valid for $x > 0$.

Note: max 3/6 [1-2-0-0-0] if no constant of integration.

Note: 0/6 if no separation of variables

0	1	2	3	4	5	6

The student earns all of the following points:

1 point is earned for the separation of variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for uses of the initial condition

1 point is earned for the solution for y

$$\frac{dy}{dx} = \left(1 - \frac{2}{x^2}\right)(y-1)$$
$$\int \frac{dy}{y-1} = \int \left(1 - \frac{2}{x^2}\right) dx$$
$$\ln|y-1| = x + \frac{2}{x} + C$$
$$\ln|2-1| = 1 + \frac{2}{1} + C \Rightarrow C = -3$$
$$\ln|y-1| = x + \frac{2}{x} - 3$$

Note that $y - 1 > 0$ since the solution curve includes the point $(1, 2)$.

7-1-3

$$\ln(y-1) = x + \frac{2}{x} - 3$$
$$y = f(x) = e^{\left(x+\frac{2}{x}-3\right)} + 1$$

Note: This solution is valid for $x > 0$.

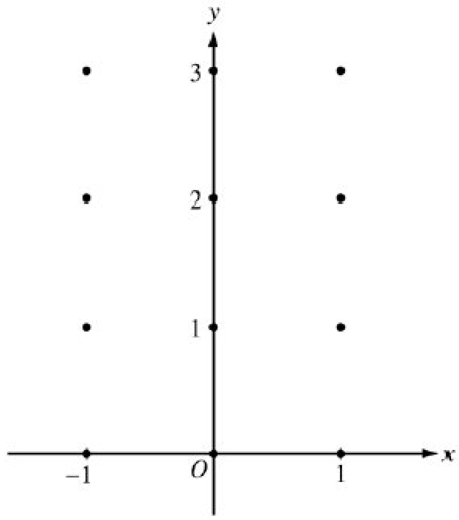
Note: max 3/6 [1-2-0-0-0] if no constant of integration.

Note: 0/6 if no separation of variables

7-1-3

97. Consider the differential equation $\frac{dy}{dx} = x^2(y-1)$.

(a) On the axes provided, sketch a slope field for the given differential equation at the twelve points indicated.



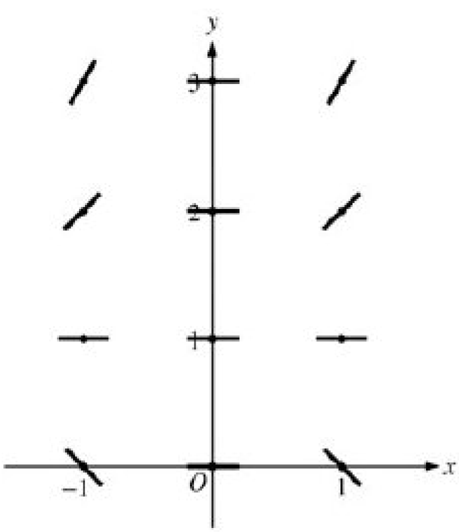
(b) While the slope field in part (a) is drawn at only twelve points, it is defined at every point in the *xy*-plane. Describe all points in the *xy*-plane for which the slopes are positive.

(c) Find the particular solution $y = f(x)$ to the given differential equation with the initial condition $f(0) = 3$.

Part A

1 point is earned for zero slope at each point (x, y) where $x = 0$ or $y = 1$

1 point is earned for positive slope at each point (x, y) where $x \neq 0$ or $y > 1$ and for negative slope at each point (x, y) where $x \neq 0$ or $y < 1$



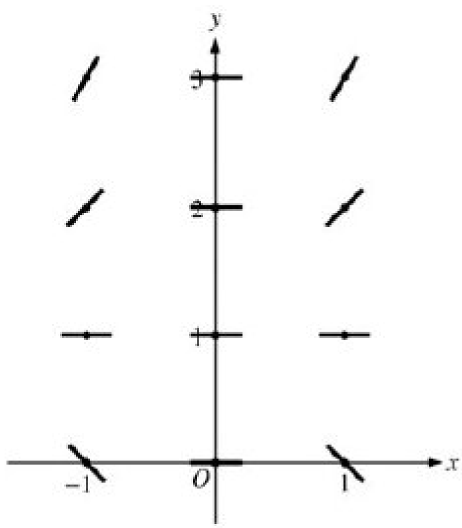
7-1-3

0	1	2

The student response earns all of the following points:

1 point is earned for zero slope at each point (x, y) where $x = 0$ or $y = 1$

1 point is earned for positive slope at each point (x, y) where $x \neq 0$ or $y > 1$ and for negative slope at each point (x, y) where $x \neq 0$ or $y < 1$



Part B

1 point is earned for the description

Slopes are positive at points (x, y) where $x \neq 0$ and $y > 1$.

0	1

The student response earns all of the following points:

1 point is earned for the description

Slopes are positive at points (x, y) where $x \neq 0$ and $y > 1$.

Part C

1 point is earned for the separates variables

2 points are earned for the antiderivatives

7-1-3

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

0/1 if y is not exponential

$\frac{1}{y-1}dy = x^2dx$
$\ln|y-1| = \frac{1}{3}x^3 + C$
$|y-1| = e^C e^{\frac{1}{3}x^3}$
$y-1 = Ke^{\frac{1}{3}x^3}, K = \pm e^C$
$2 = Ke^0 = K$
$y = 1 + 2e^{\frac{1}{3}x^3}$

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned for the separates variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

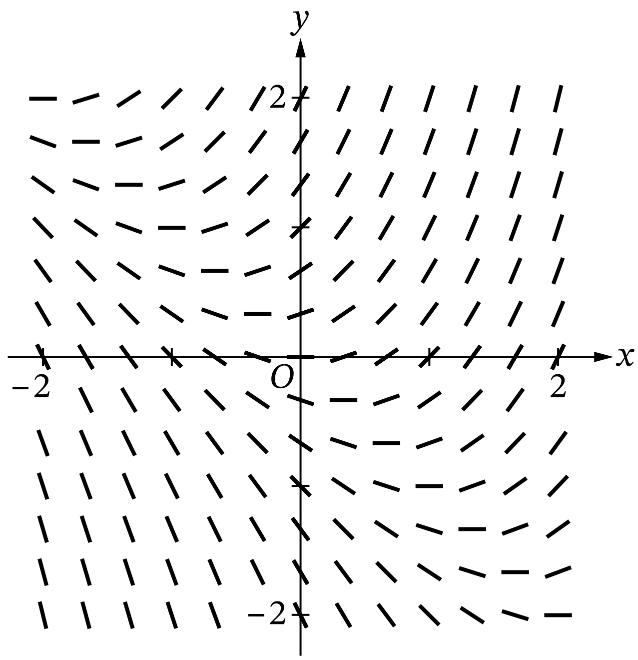
1 point is earned for solving for y

0/1 if y is not exponential

$\frac{1}{y-1}dy = x^2dx$
$\ln|y-1| = \frac{1}{3}x^3 + C$
$|y-1| = e^C e^{\frac{1}{3}x^3}$
$y-1 = Ke^{\frac{1}{3}x^3}, K = \pm e^C$
$2 = Ke^0 = K$
$y = 1 + 2e^{\frac{1}{3}x^3}$

7-1-3

98.



The figure shown is a slope field for which of the following differential equations?

(A) $\frac{dy}{dx} = \frac{x}{y}$

(B) $\frac{dy}{dx} = x - y$

(C) $\frac{dy}{dx} = x + y$

(D) $\frac{dy}{dx} = -x + y$

Answer C

Correct. For this differential equation, the slope segments at all points along the line $y = -x$ should be horizontal. For points above the line $y = -x$, the slope segments should have positive slopes, and for points below the line $y = -x$, the slope segments should have negative slopes. The given slope field demonstrates all these behaviors.